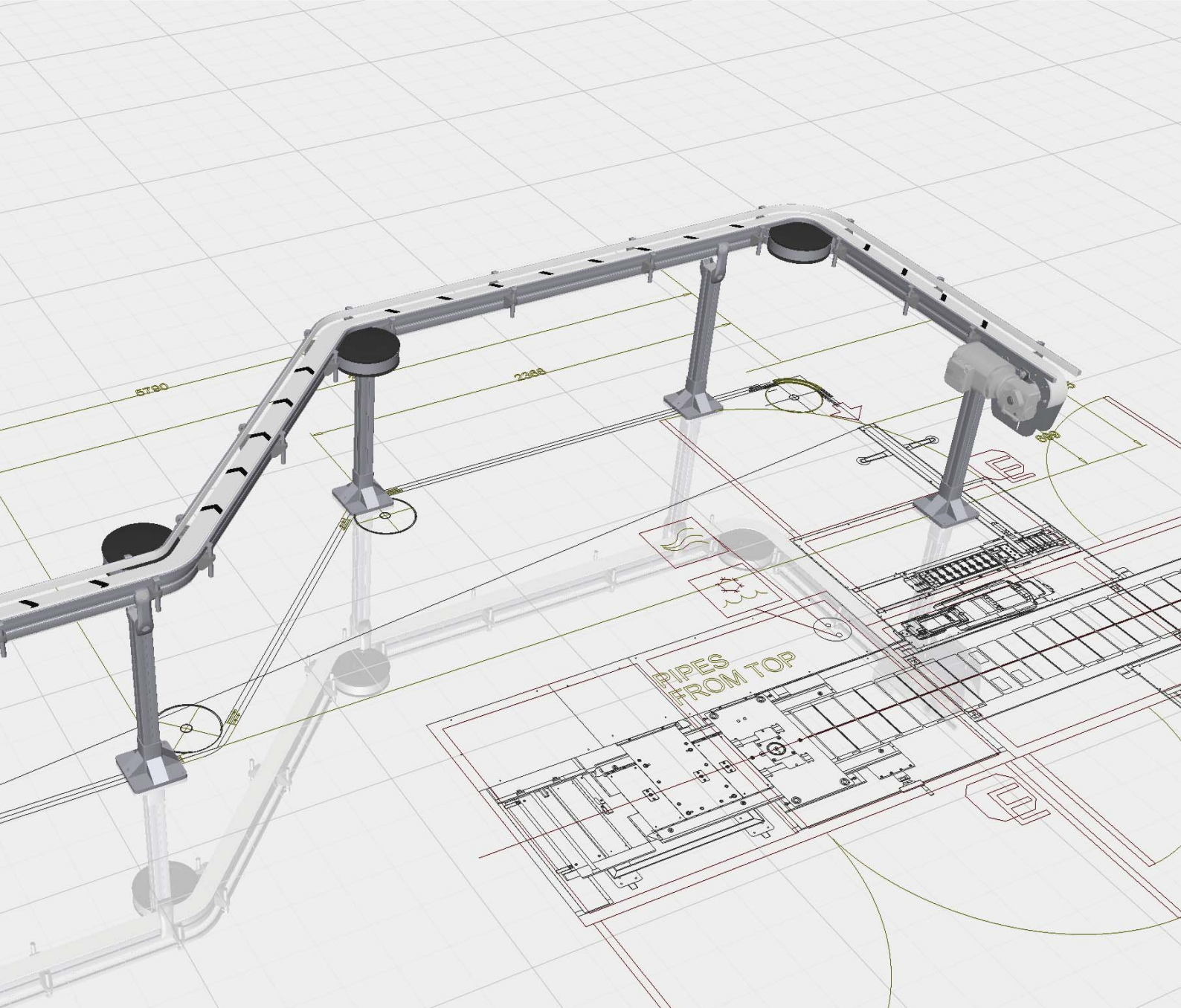




FLEXLINK DESIGN TOOL

User documentation

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Background

FlexLink has a number of different software's to help both internal and external customers to design and order conveyor systems. This software package is a suite of three main software's which focus on different parts of the design and ordering process.

FlexLink Design Tool

This is the software described in this manual and is later in this document referred to as the Design Tool. The main feature provided by the software is the ability to quickly design a conveyor line in 3D with a number of different outputs as drawings, BOM and simulations.

FlexLink Calculation Tool

Next software in the suite is the Calculation Tool. The objective with this tool is to verify if the conveyor design is valid in terms of chain tension calculation. This tool provides the feature of creating the conveyor configuration and supports the import from the Design tool.

FlexLink Quotation Tool

The result of the Design tool could also be imported into the Quotation Tool in order to help the application engineers and sale engineers in the quotation process.

My FlexLink

There is also a portal on the www.flexlink.com webpage where all the software above can be downloaded. For entering My FlexLink a user account is mandatory. There is also a possibility to apply for an account directly from this page. In the My FlexLink portal there is a Download Area containing useful software. Also this is the path to access the Online store, FlexLink's web shop.

A full list of abbreviations and terms used in this document is listed in the Appendix.

Software Installation

Software prerequisites

The Design Tool has the following prerequisites:

Windows 7

Microsoft .NET Framework 4.5

- <http://www.microsoft.com/sv-se/download/details.aspx?id=30653>

DirectX End-User Runtime

- <http://www.microsoft.com/sv-se/download/details.aspx?id=35>

Microsoft SQL Server Compact 4.0

- <http://www.microsoft.com/en-us/download/details.aspx?id=17876>

Exists as X86 or X64 version, user must install correct one

Visual C++ 2008 SP1

- <http://www.microsoft.com/en-us/download/details.aspx?id=11895>

Exists as X86 or X64 version, user must install correct one

Adobe Acrobat Reader

- <http://get.adobe.com/se/reader/>

Hardware requirements

To be added later

Installation procedure and settings

In the installation process the software is requesting a lease. See separate chapter about the Lease Management System.

Files created in the Design Tool will have the extension vcmx and will be associated with the software.



filename.vcmx

Picture: File extension

Startup of software

The Design Tool could be started from the start menu in Windows. As default it is located in a folder called FlexLink Design Tool.

After the start up screen the software will check if a new version of the Design Tool is available. If so, a screen will appear with the possibility to accept the downloading of the latest version. After a successfully executed update there will be a notification listing all new features, changes and bug fixes in this version.

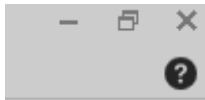
Besides the software version all of the component libraries will be checked and updated. These updates are only done if the PC is connected to internet.

Software overview

The Design Tool is built upon an existing 3D manufacturing, simulation and visualization software from Visual Components called 3D Realize.

The Design Tool has modified the software to fulfill all our special requirements. This document focuses mainly on the Design Tool features but also to some extent the original software. A continuously updated version of this document can be found in the **Help** menu and the **Documentation** ribbon group.

If some topic is not covered or described too briefly in this document there might be more information available in documents (manuals, tutorials, etc.) from Visual Components (www.visualcomponents.com). Also the question mark symbol in the top right corner of the screen (or F1) opens up a FlexLink Design Tool **Help** for further reading.



Picture: FlexLink Design Tool Help (question mark symbol)

FL Configurator

The standard version of the Design Tool is called **FL Configurator** and is the suitable for most users. This contains all the FlexLink features but only basic functionality regarding other parts of the Visual Component software suites. The license for **FL Configurator** covers both the FlexLink Design Tool and the basic features of the software from Visual Components.

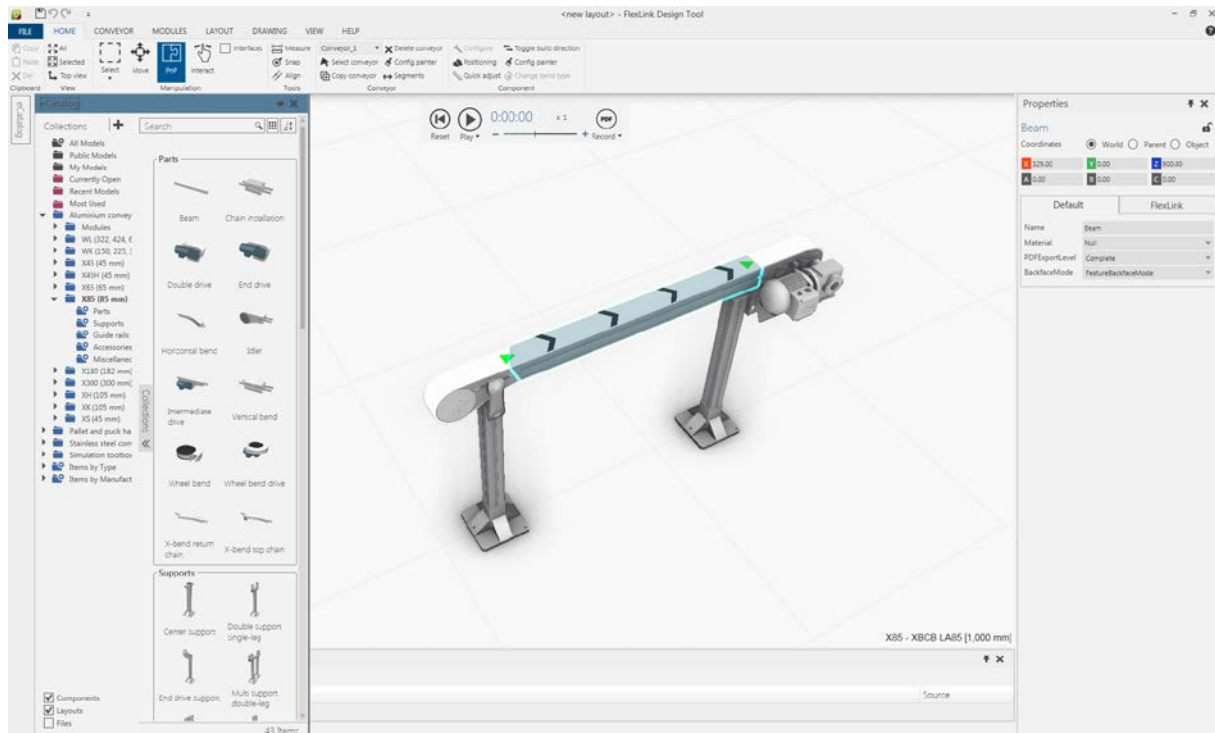
FL Analyzer

User that require more features from the software suite from Visual Components have two more options. The first is the **FL Analyzer**. This includes more features regarding robotics and statics. This software version requires a special **FL Analyzer** license which has to be purchased directly from Visual Components. This version is not yet available.

FL Creator

Even more advanced users may require the possibility to model their own components completely. Then the **FL Creator** version is needed. This also includes all **FL Analyzer** features. This software version requires a special **FL Creator** license which also has to be purchased from Visual Components. This version is not yet available.

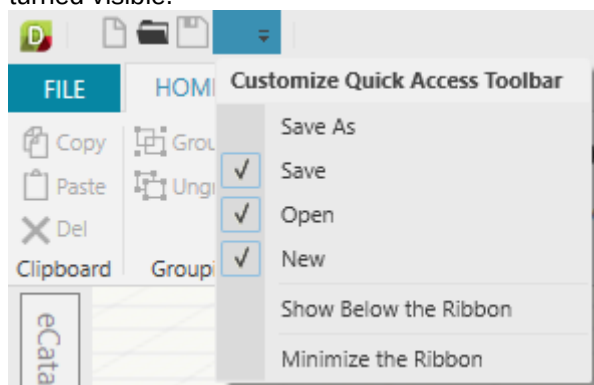
Screen layout



Picture: Screen layout

Quick Access Toolbar

At the absolute top part of the screen there are some symbols put there for easy access. This toolbar can be adjusted to fit the users need. The most common commands from the FILE menu can be set to visible in this toolbar (see picture below). As default the three commands shown in the picture are turned visible.



Picture: Quick Access toolbar - Customize

The Quick Access Toolbar is possible to move below the ribbon groups. The ribbon group toolbar can also be minimized so only the menu bar is shown. Then when any of these menus are selected the ribbon group shows, but only until a command is chosen.

Top menu bar

At the top part of the screen there is the top menu where the user could switch between main

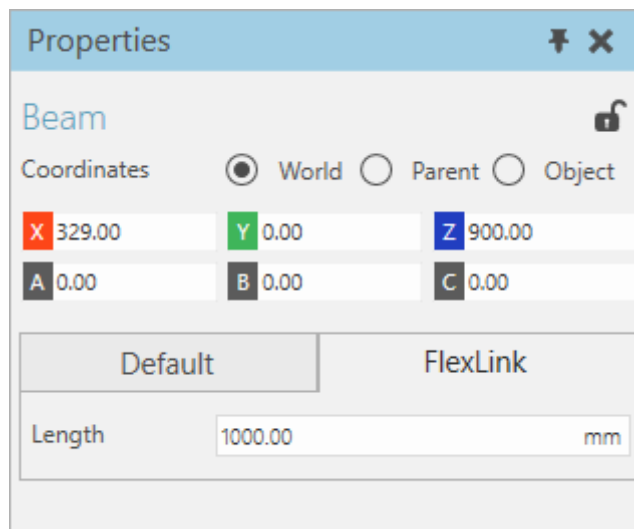
command alternatives in the software. The tabs in this menu are ordered in a work process order. The most left menu is the File menu which contains ordinary commands regarding the application file (e.g. new, open, close). This menu is also containing lot of setting for the software. The **Home** menu is containing the most frequently used commands during the build and simulation process. The **Conveyor** menu and the **Layout** menu are for more detailed commands for each level. The **View** menu contains settings for the appearance of the 3D view. The **Drawing** menu is dedicated for all commands concerning the 2D drawing feature. Each menu is built up by a number of ribbon groups containing a cluster of more commands. These differ depending on the alternative selected on the top menu.

eCatalog

On the left side of the screen there is an **eCatalog** window. This contains all components, layouts and geometry files that can be used in the applications. Details about this is described in a separate chapter.

Property window

On the right side of the screen the **Properties** windows shows settings for the selected component. This window has different number of tabs depending on the selected component. At least it has both the **Default** tab and the FlexLink tab. A lot more info about this window and the properties can be read in the **Component** chapter.

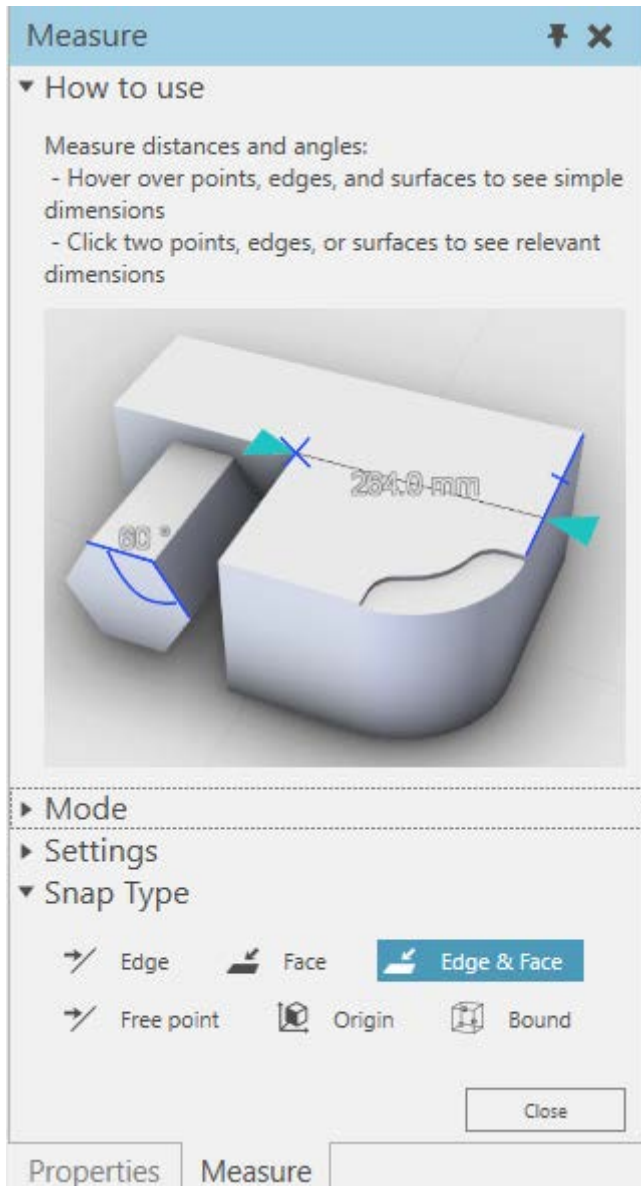


Picture: Property window

Command window

On the right side of the screen the **Command** window will be shown for many of the commands. This window shows settings and information related to the active command. The window will disappear when the command completes or is cancelled.

Each command has one or two buttons at the bottom of the window to either **Close** or **Cancel** the command.



Picture: Command window: Measure command

Messages / Outputs window

In the bottom of the screen there is one window divided into two areas for user information. In the **Messages** window there are three types of information to the user. **Errors** has a red stop symbol and indicates issues with the current layout that has to be fixed in order to get a valid conveyor line. The source describes which area the issue is concerning.



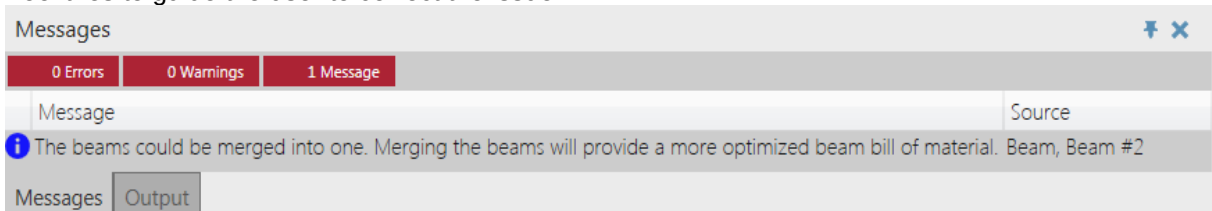
Picture: Message - Error example

The next type of messages is the **Warning** message. It is an issue which could be a potential problem but not necessary. The user is notified and can do changes but it is not mandatory to solve it to continue working with the layout (e.g. export to other tools).



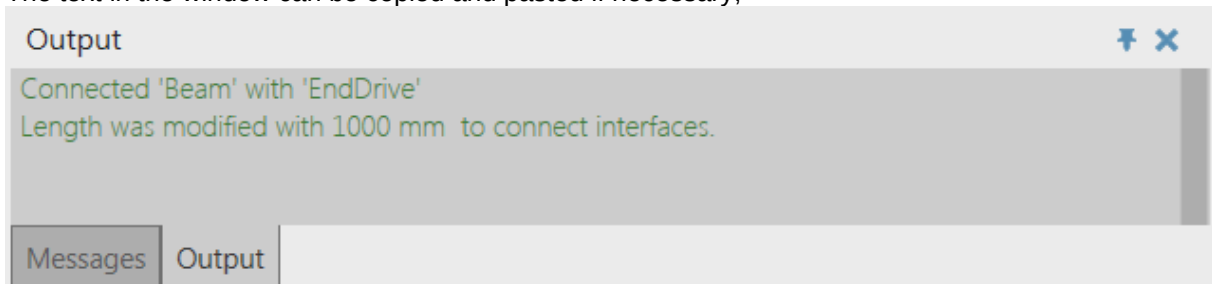
Picture: Message – Warning example

Last there is a third type of messages called only **Messages**. This is good-to-know information for the user. In all these types of messages it is possible to double click on the message and Design Tool tries to guide the user to correct the issue.



Picture: Message – Message example

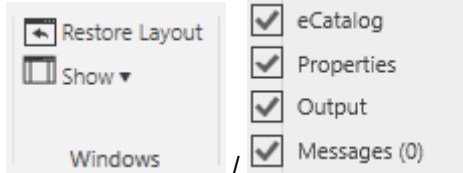
The **Output** message windows informs the user about more detailed actions done by the software. The text in the window can be copied and pasted if necessary,



Picture: Output example

Windows settings

The screen appearance described above has the default settings. The locating of the windows could easily be changed by drag and drop on the window top bar. Also they could individually be turned off and on if required. This is from the **Show** command (the **Windows** ribbon group in the **View** menu).



Picture: Panels (Windows ribbon group & Show settings)

Each window could be set to **Auto Hide**. In that case the windows hides to only a single button when it isn't in use. This is the default for the **eCatalog** window. If not the windows is constantly shown on the screen, default setting for the **Properties** window.



Picture: Auto Hide (turned off) and Close symbols

eCatalog

The **eCatalog** is a library containing components, layouts and geometry files that can be used in the application. Both default items from Visual Component and FlexLink are available and also earlier created user items. Both components and layouts have the file extension vcm or vcmx.

Collections

When the **eCatalog** window is opened the structure of collections are visible. There are different types of collections with different colors and symbols. Two important colors are:

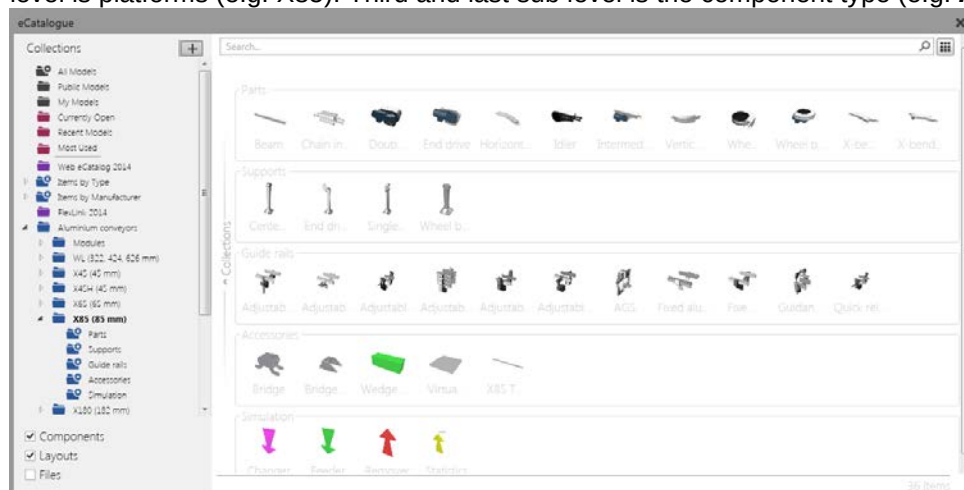


The purple folders are default source folders which contains a lot of components



The blue folders are only collections of components actually located in another source folder. This can be considered as filter view of the source folder.

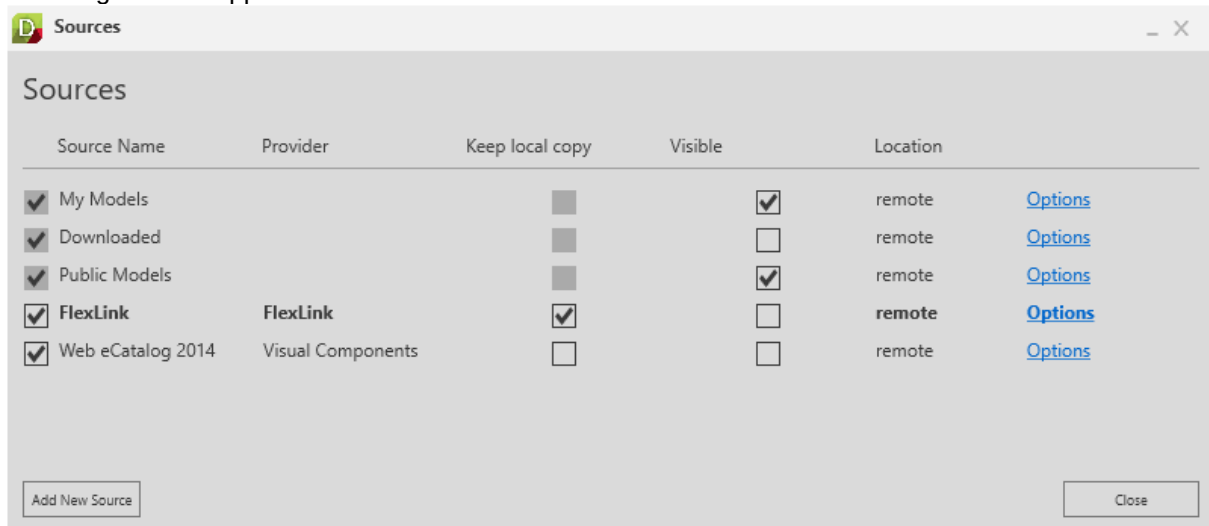
When the software is installed the structure of the FlexLink components are automatically created. All these components are located in the FlexLink collection. Then there is a sub structure with more collections. The first sub level is platform category (e.g. **Aluminum conveyors**). Then second sub level is platforms (e.g. X85). Third and last sub level is the component type (e.g. **Parts, Supports**).



Picture: eCatalog window

Sources

In the **eCatalog** window there is a plus sign after the **Collections** (see picture above). This is used for editing the sources and adding new collections. If the **Edit sources** command is selected the following window appears.



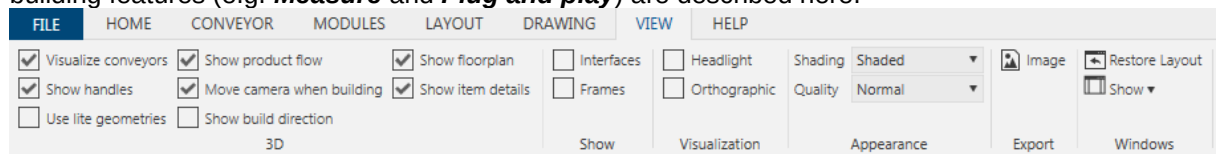
Picture: eCatalog – Edit sources

All of the components for FlexLink is located in the FlexLink source. It is recommended to have this folder as **Keep local copy** in order to be able to use these components when working with the tool offline. The other source, **Web eCatalog 2014** is from VC and is not recommended to have as a local copy because of the required storage size. More information about the **eCatalog** and all features can be read in the FlexLink Design Tool **Help**.

Building concept

The Design Tool is designed to make it as easy as possible for the user to build complete conveyor systems and at the same time having maximal flexibility. A lot of limitations are also added into the software as building rules which actually is an assistance to the designer. If the engineering guidelines in this documentation is not sufficient for being able to complete the layouts it is possible to find more detailed information in the FlexLink Product Catalog. For external users the local FlexLink operating unit will be available for support.

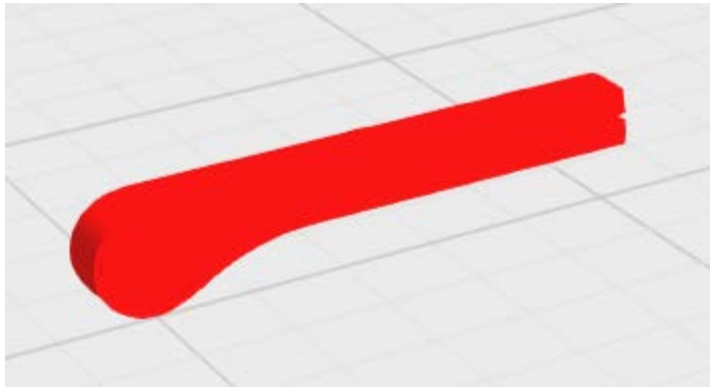
The majority of the settings for this chapter is located in the **View** menu. But also some other building features (e.g. **Measure** and **Plug and play**) are described here.



Picture: Building concept – View menu

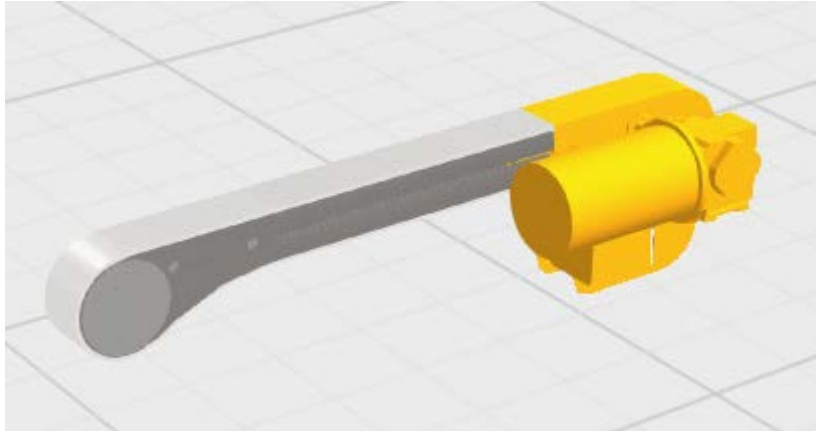
Visualize conveyors (View menu, 3D Ribbon group)

To assist the user in the building process there are some color indication of the conveyor components. This is an option called **Visualize conveyors** and could be turned on and off from the **View** menu. For any uncompleted conveyor all components will be colored red. A typical uncompleted conveyor is where either the end drive or the idler end are missing. This red color coding will also be presented as an **Error** message in the **Messages** window. This message gives more details about the cause.



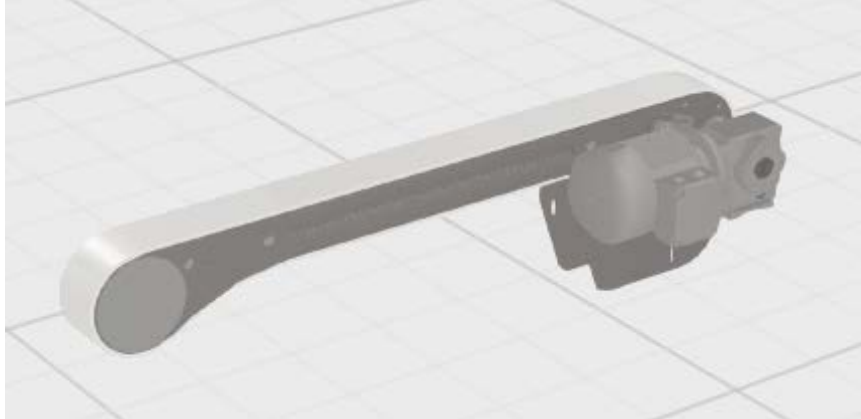
Picture: Red color – Error, Not completed conveyor

A conveyor is complete when there are no unconnected conveyor ends and there is a drive unit included. When the conveyor is completed there might still be components with color coding. These are colored yellow and indicates a **Warning** message. One common reason for this is that the component needs to be configured. But in some case the complete conveyor is colored yellow. In any case the reason for the coloring should appear in the **Messages** window as a **Warning** message.



Picture: Yellow color - Completed conveyor, End drive not configured

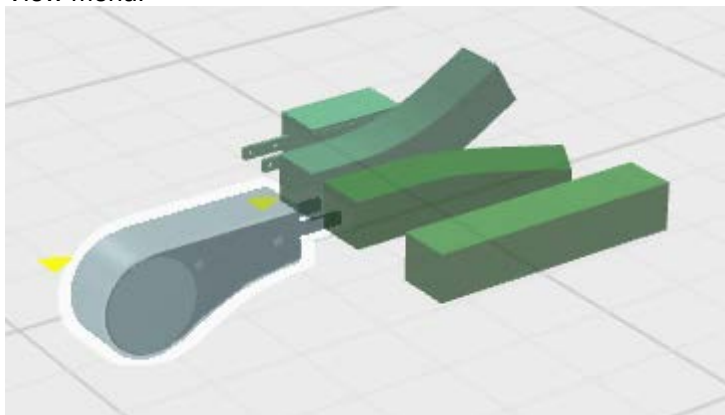
If the conveyor doesn't have any errors or warnings all color indicators disappears and the conveyor is prepared to proceed further to other features of the software (e.g. exporting to BOM).



Picture: No color indications - Completed conveyor, no warnings

Show handles (View menu, 3D Ribbon group)

After a component has been added into the layout a number of building handles appears. These represent the most frequent use component that is compatible to be connected as the next component. This visualization help could be turned on and off with the show handles tick box in the View menu.



Picture: Building handles – Example

If one of them corresponds to the required layout it can be automatically added by clicking on that

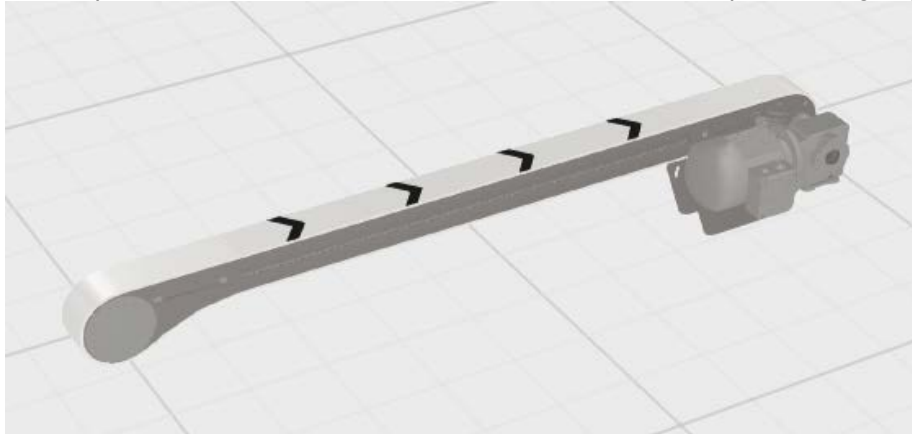
build handle. Then the component is automatically added and connected to the existing conveyor. This new component will be automatically selected and a new set of building handles will appear. In this manner it is easy to quickly draw an easy standard conveyor.

Use lite geometries (View menu, 3D Ribbon group)

A lot of the standard FlexLink components are created with two geometry sets. If this option is selected then all components will look slightly less detailed but will also be much less demanding for the PC to handle. Especially large layouts with a lot of components benefit from this. No other feature (e.g. BOM, simulation) is affected by this choice.

Show product flow (View menu, 3D Ribbon group)

Idler ends and end drive units has a default product flow that goes from the idler end to the end drive unit. This is regardless of the building direction (in which order the conveyor expands in the building process). It is caused by the end drive ability to pull the chain instead of pushing it. These indicator arrow option could be turned on and off from the **View** menu (**3D** ribbon group).



Picture: Indication product flow

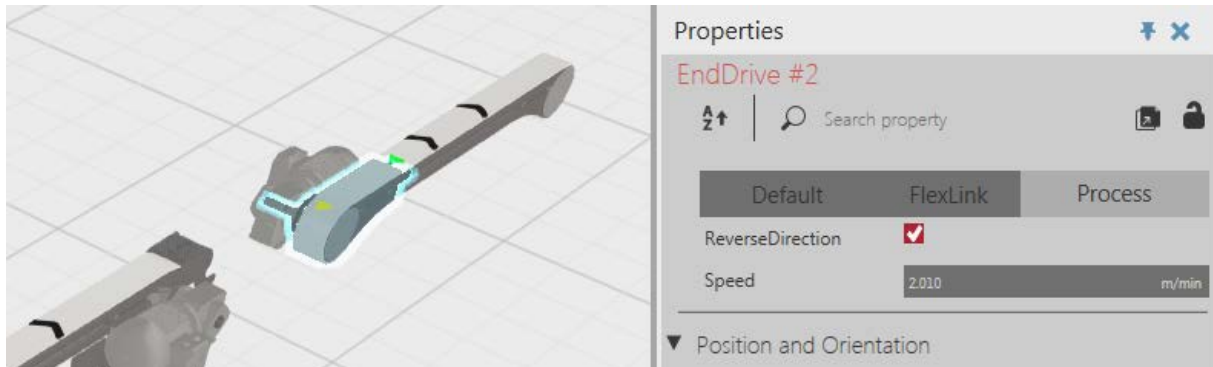
If a conveyor is built up starting with an idler end the building direction will correlate with the default product flow. On the contrary, if a conveyor is built started with an end drive unit, the building direction will be the opposite as the default product flow.

Show item details (View menu, 3D Ribbon group)

When a component in the 3D-world is selected a description of which component it is will be shown in the lower right corner. This information can be turned on and off by toggling the **Show item details** setting.

Reverse direction

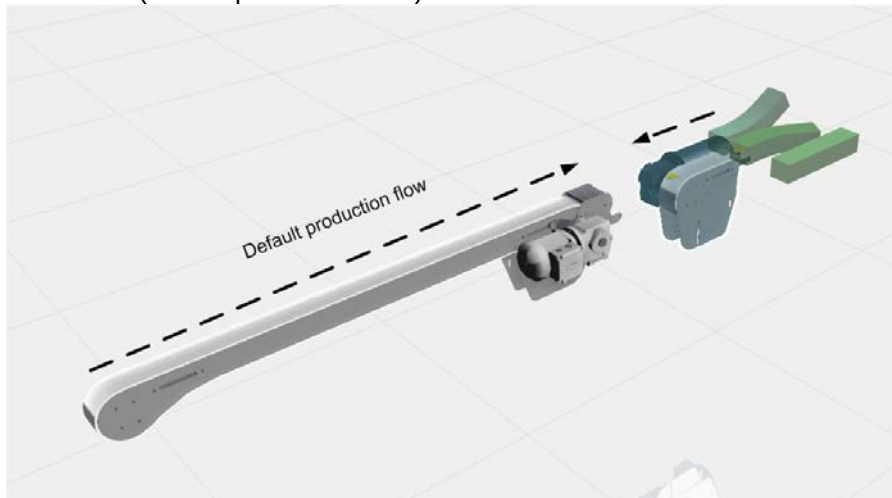
To change the product flow it is necessary to have a drive unit that is allowed to run backwards. Then there is a **Properties** setting on the drive unit component under the **Process** tab, called **ReverseDirection**. Setting this will force the product flow on that complete conveyor to the opposite compared to the default product flow. It is only possible to set reverse direction if the drive unit is guided. The **ReverseDirection** property is not even visible if it is not possible to change it.



Picture: Reverse direction

Inverted Insert

If a component is supposed to be added using **Plug and play** it is not certain that it has the same default product flow as the rest of the conveyor. In this case the new component will not be possible to connect (snap) to the rest of the conveyor. Then the property **Inverted insert** has to be set on that component. It makes the component connect in the opposite way compared to the default production flow. Basically it means idler ends can connect where normally end drives connect and vice versa. As long as the building handles are used for adding the components, the new components will automatically set the inverted insert if needed. In the case below the drive unit can't be connected to the bridge component because the building orientation (default production flow) is different.



Picture: Different default production flow, example

To be able to connect the end drive unit to the bridge the **Inverted insert** property has to be turned on. This doesn't change the default product flow orientation but opens up the possibility to connect the components. If the drive unit was added through the building handle of the bridge it would have set this **Inverted insert** property automatically. Then the **ReverseDirection** on the end drive has to be changed too, at least if simulation should be performed.

Note:

Both the **Inverted insert** and **ReversedDirection** acts compared to the default production flow. The **Inverted insert** is only affecting how the component is possible to connect while the **ReverseDirection** changes the production flow in simulation mode.

Move camera when building (View menu, 3D Ribbon group)

When adding components to the layout view using build handles it could expand beyond the limits of the 3D view. If the move camera when building option is turned on (from the **View** menu) the camera view is optimized to fit the new conveyor system.

Headlight (View menu, 3D Ribbon group)

When switched on the conveyors are lit with a light making them brighter.

Orthographic (View menu, 3D Ribbon group)

When switched on the conveyors are seen in an orthographic view.

Shading (View menu, 3D Ribbon group)

There are three different shading modes available:

- Wireframe
- Shaded hidden Line
- Shaded

This setting will affect the appearance of the conveyor.

Quality (View menu, 3D Ribbon group)

The quality can be set to four different settings:

- Fast
- Normal
- High
- Very high

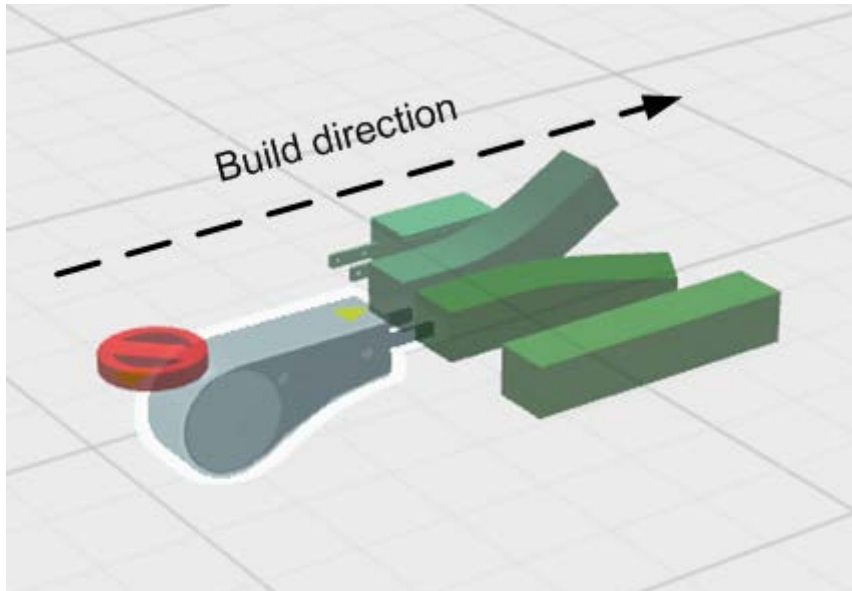
If the design tool performs slow it is recommended to lower the quality setting.

Build direction

The build direction means in which direction the conveyor will expand using the building handles. Also it indicates which part of the conveyor that is static and not affected by any change on any settings or adding any new component. The build direction is visualized by the position of the stop symbol but this is actually the direction which is opposite of the build direction. If the build handles is turned on this is the same as the build direction.

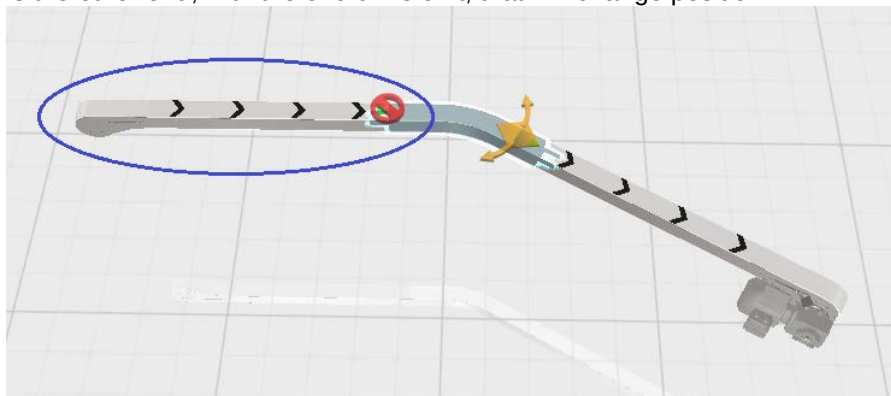
Build direction (View menu, 3D Ribbon group)

The stop symbol of the build direction is turned on and off with the show build direction option in the **View** menu in the **3D** ribbon group.



Picture: Build direction – stop symbol

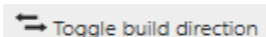
As mentioned above the stop symbol indicates the fixed part of the component. Even all components connected to this part of the selected component is included in this fixed state. The position of all these components will not change if the selected component is changed. In the example below the components marked with the blue oval is not affected on an angle change on the horizontal bend. It is the other end, with the end drive unit, that will change position.



Picture: Static part of the conveyor

Toggle build direction (Home menu/Component ribbon group)

Short key: Ctrl + H

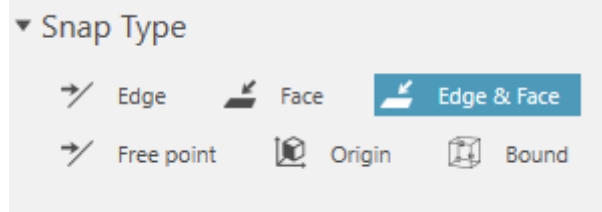


The build direction has nothing to do with the flow of the products on the conveyor. It is possible to build a conveyor completely from one end to the other using the building handles and then the building direction will be the same through the whole conveyor. But it is also possible to start from a middle component (e.g. a beam) and then build in two directions (but one at a time). When a component is selected it will show the building handles for the first open component end on that component. If it has more than one open end the building handles could be changed to the next open end using the **Toggle build direction** command, from the **Component** ribbon group (**Home**

menu) or with the short key (Ctrl +H). It is even a third option to toggle the build direction and that is simply to click on the stop symbol. Changing the build direction is especially useful for components that has more than two component ends (interfaces for connecting to other components), e.g. X-bends and diverters. This is described more in the **Component** chapter.

Snapping tools

When measuring, aligning or working with a drawing, there are different types of snapping modes. This is displayed in a Command window that is displayed for some menu commands.



Guide frames are certain points in the components that can simplify measuring, for instance they are displayed on the top of chain at the start and end of a component. Using these points it is easy to measure the L-distance in a conveyor.

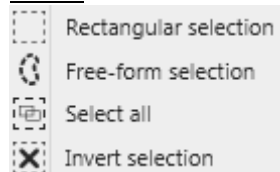
Manipulation (Home menu)



Picture: Grouping ribbon group

The **Manipulation** ribbon group contains of four manipulation symbols and a selection box. Exactly one of these symbols is always selected. The Interface selection box enables interface connection and is for future use.

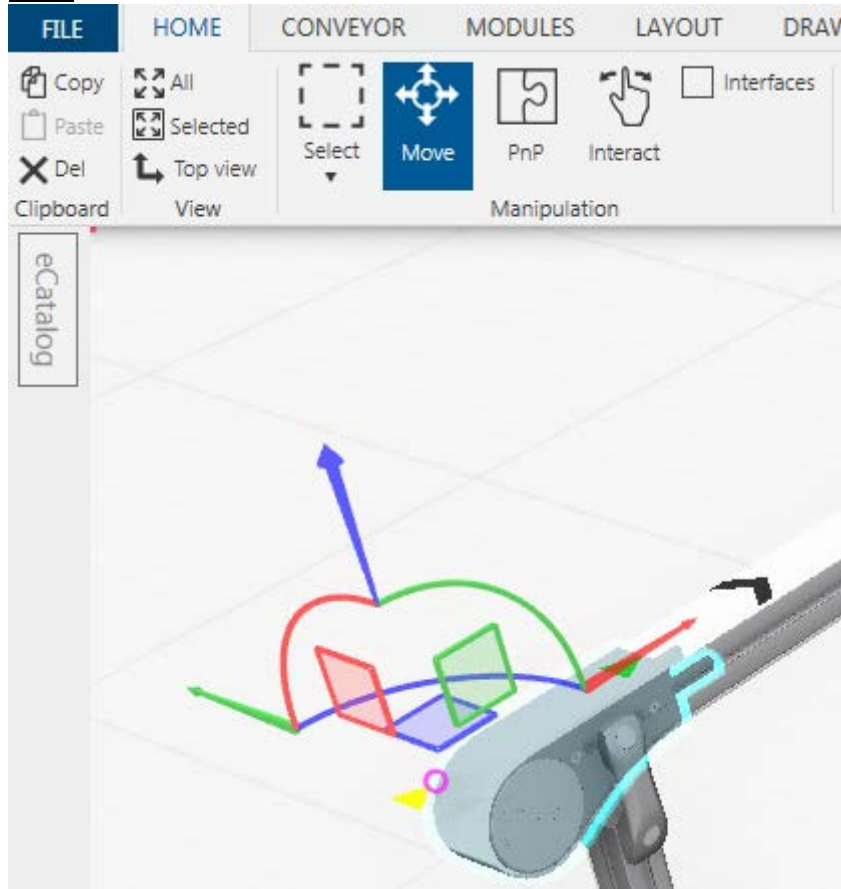
Select



Picture: Select options (Grouping ribbon group)

There is a number of different ways to select multiple components. The first method is to press down the CTRL key while selecting several components, either by clicking on them or selection with a rectangular selection. Then there are three more ways under the **Select** command in the **Manipulation** ribbon group. The **Rectangular selection** selects all components that is covered by a user defined rectangle window. The components doesn't have to be completely inside the rectangle. The next command is the Free-form selection where the user draws a loop while pressing down the right mouse button. The last alternative to select components is the **Select all** which selects all existing components in the layout. The last alternative under the Select command is the **Invert select**. This command selects all component in the layout that is not selected since before and vice versa.

Move

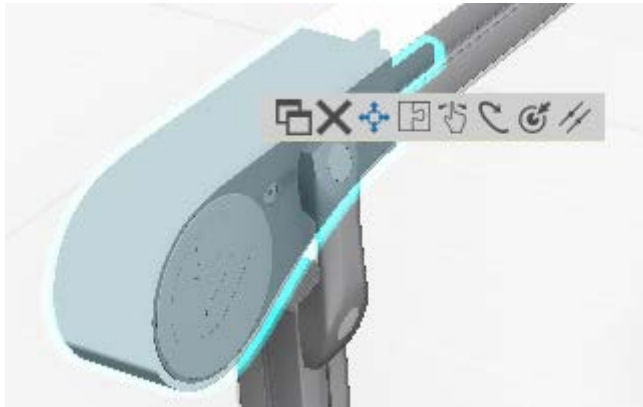


Picture: Move command (Manipulation ribbon group)

When the Move command is active and there is some selection of any kind the manipulator help shows up on the 3D-world (see picture above). The arrows are used for translating the selection along each axis. The rectangles are used for moving the selection in each 2D plane. Last, the arcs are used for rotating the selection around each axis. The X, Y, Z, A, B and C values are updated dynamically in the property window (see separate chapter).

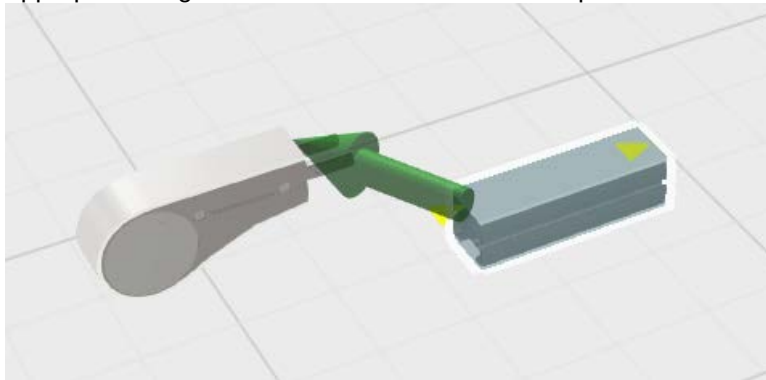
PnP

The building handles show only the most likely component that is possible to connect as the next component. If a component is needed which is not shown as a handle it could be possible to add it directly from the **eCatalog** instead. When the new component is in the layout it can be connected to the uncompleted conveyor using **PnP, Plug and play**. This option has to be turned on either from the **Manipulation** ribbon group (**Home** menu) or from the manipulation tool bar visualized when the component is selected.



Picture: PnP, Plug and play – Manipulate tool bar

When a component is close enough to another compatible component a green arrow appears indicating it is possible to connect in that location. If this component is moved even closer to the other component it is automatically connected to the open component end. It is also rotated to appropriate angle in order to connect to this component.



Picture: Connect via PnP, Plug and play

If no green arrow appears it could be caused by a number of different reasons:

- The plug and play option is not turned on
- The components are not compatible with each other
- It is the other end of the component that it is possible to connect
- The property **Inverted insert** must be changed

Interact

There are several ways to adjust a beam length. If **Interact** is turned (either from the **Home** menu or from the manipulate tool bar) it is possible to hold down the left mouse button on the beam and slide the length to required length. Connected components will still be connected at a new position. The **Beam length** can in the **Properties** window will be dynamically updated when using the **Interact** feature.



Picture: Interact – Manipulate tool bar

It is important to turn off the **Interact** mode after using it. Otherwise it will not be possible to move any components in the 3D world.

Interfaces

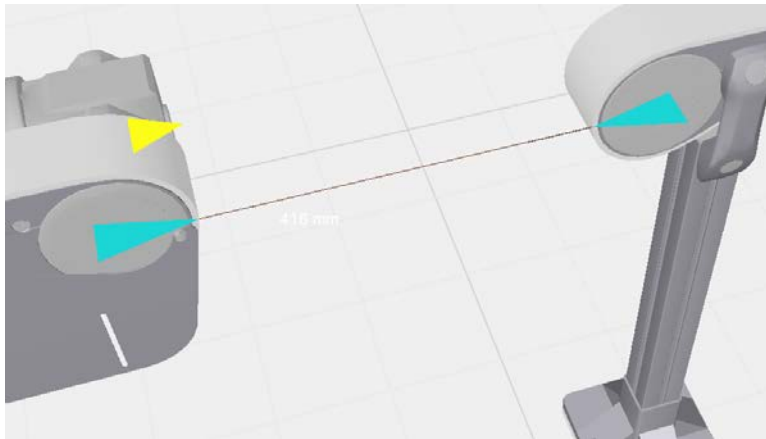
The Interface selection box enables interface connection and is for future use.

Tool (Home menu)

The tool ribbon group contains a number of misc component tools.

Measure (Home menu/Clipboard ribbon group)

The measuring feature calculates the distance between two user defined points. The Output window will also display the delta distance for X, Y and Z.

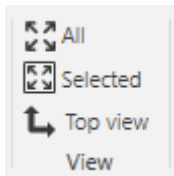


Picture: Measure, example

Align

The align tool is used to align two components. The alignment can be done on only the position, the orientation or both at the same time.

View (Home menu)



Picture: View ribbon group

The home menu contains a **View** ribbon group. These commands is used for adjusting the 3D screen.

All

The **Fill** command is adjusting the zoom and pan to fit all components in the layout.

Selected

Next command in this ribbon group is ***Selected***. It works like the command above but is only adjusting the view depending on the selected components.

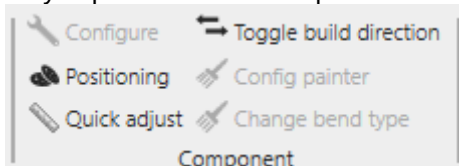
Top view

Short key (not implemented yet)

When the **Top View** command is triggered the 3D screen is automatically changed to a top view. This view is also set to orthogonal (very suitable when having a CAD-drawing as a floor object). This command button is a toggle function and is turned off by clicking the same button again. The **Top view**-mode is not possible to orbit, only pan and zoom.

Component

This chapter is only concerning all commands and settings that applies for a component. The majority of the location for these commands is in the Home menu but also the properties window is very important for the components.



Picture: Component – Home menu, Component ribbon group

Configure



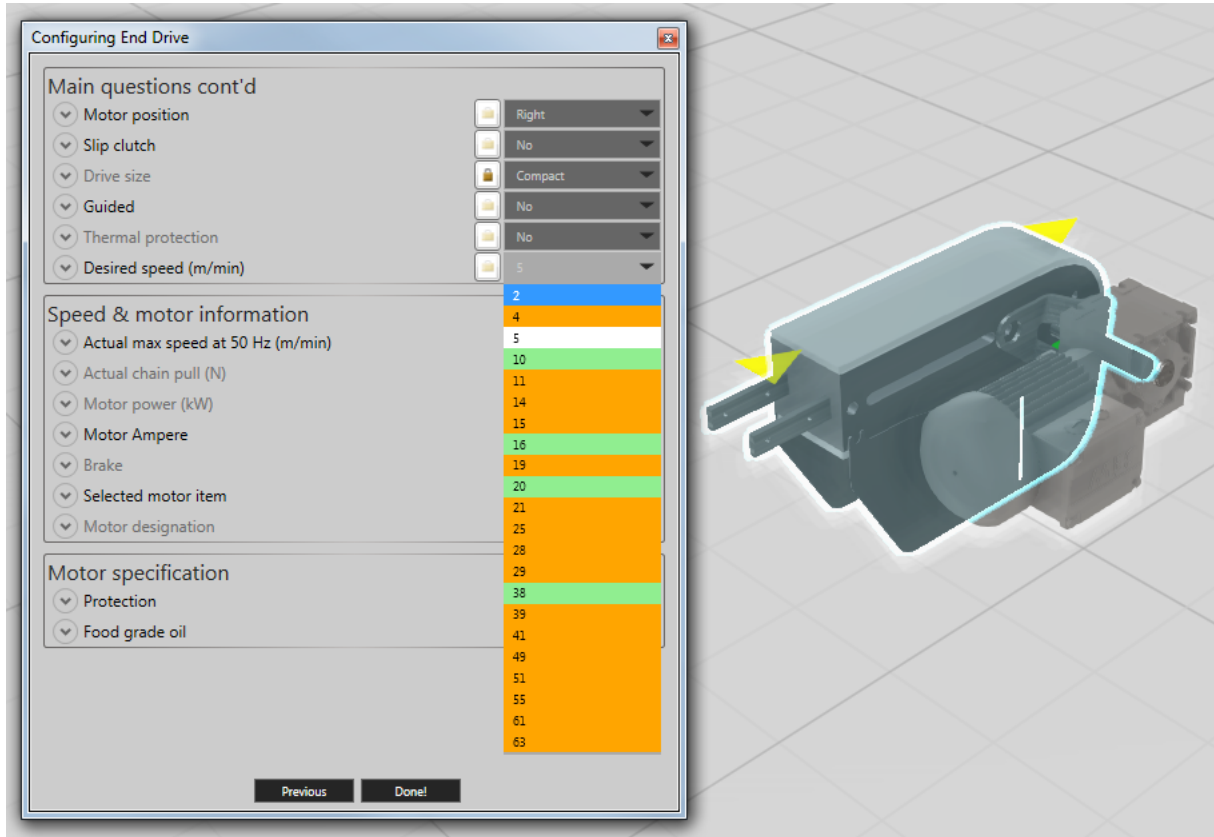
Short key: Ctrl + shift + C

There are a two ways to configure components in this tool. First the configure command could be triggered from the **Home** menu, **Component** ribbon group. Secondly the corresponding warning message could be double clicked on. The actual configuration differ a bit depending on the component type. This is covered in the sub chapter for each component below.

The most straight forward way to configure a component is by using the **Configure** command in the **Component** ribbon group. If more than one component are selected the **Configure** command will affect all these components.

Configuration window

When a component is configured a configuration window pops up containing a number of detailed questions about that specific component. Right next to the drop down list it is a lock symbol. This indicates if the value is explicitly set. In this case it won't be changed by any other change without a clear warning information (described below).



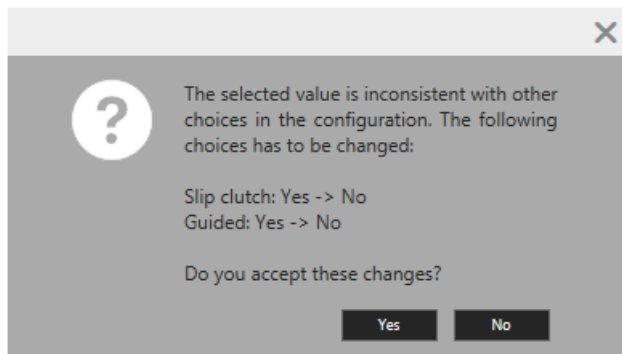
Picture: Configuration window, example with color coded alternatives

Values editable to the user has got a drop down list. All alternative are color coded according to the description below. The blue alternative in the example above is only indicating which is currently marked by the mouse pointer.

White:		The selected alternative
Green:		Possible alternative value. Doesn't require any other changes on a locked drop down list.
Orange:		Possible alternative but requires other changes on a locked drop down list
Red		Not allowed alternative

Table: Configuration window - color coding of configuration alternatives

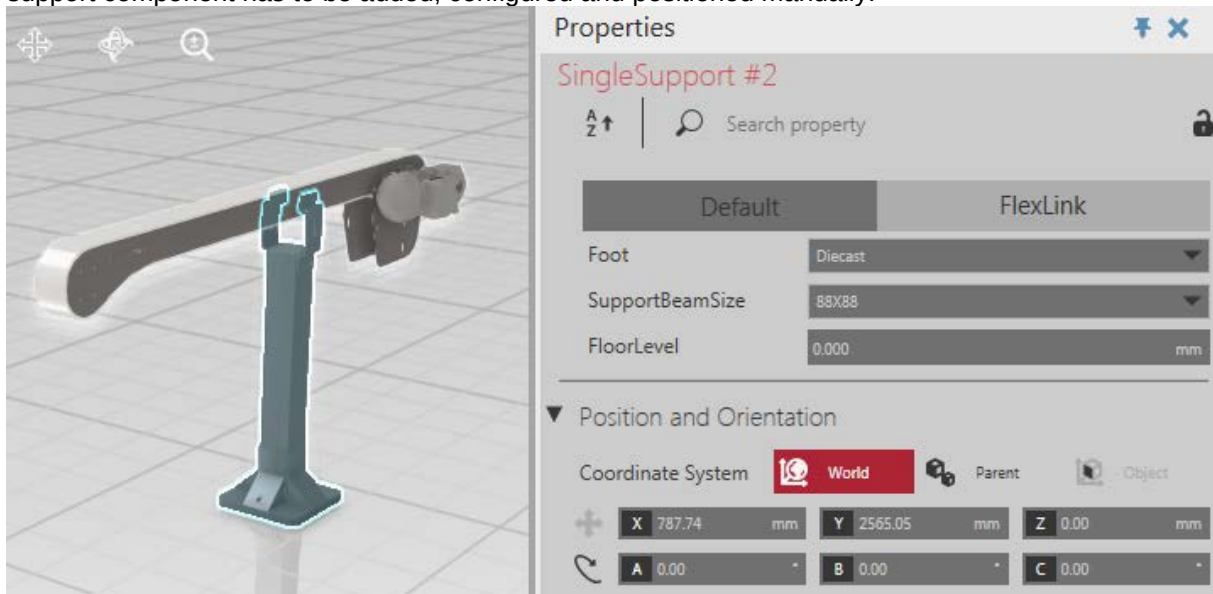
If the new value requires changes on any other check box value this is stated in a separate message window.



Picture: Configuration window - inconsistent configuration

Conveyor Support configuration

Each conveyor platform has several types of conveyor support components. When they are added into the layout they have a default configuration. These components doesn't have a separate configuration window. Instead the different choices are set through either the **Quick Property** window (double click) or the **Properties** window. The conveyor support component is added to the conveyor using **Plug and play**. When it connects to the conveyor it automatically adjusts its height to the height of the conveyor (**Top of chain/Top of pallet**). All conveyor support components has a property called **FloorLevel** (mm). This is a value which sets the individual floor level for each individual conveyor support component. It is defined as a Z-axis value relative the origin of the coordinate system. The component is also automatically adjusted to the floor level. Each conveyor support component has to be added, configured and positioned manually.

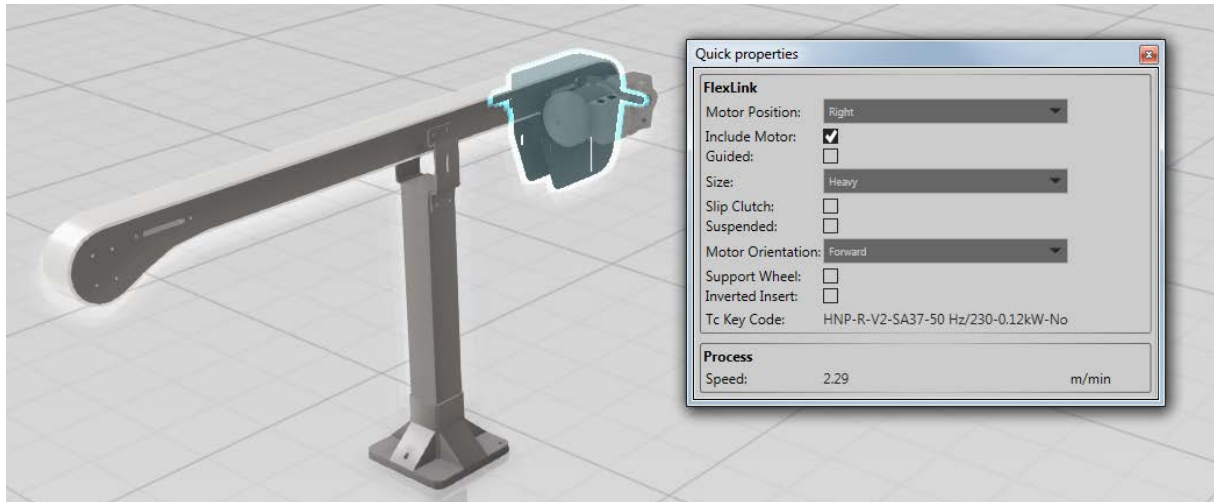


Picture: Configurations – conveyor support example

Drive unit configuration

Most of the drive units in the FlexLink assortment must be configured before they can be ordered. The configuration window isn't automatically triggered when a drive unit is added to the layout or even connected to a conveyor. It has to explicitly be triggered by either the **Configure** command in the **Component** ribbon group or double clicking on the corresponding warning message. The drive unit has also a lot of properties in the **Properties** window and in the **Quick Properties** window. If some property that is affecting the configuration is changed the color marking on the drive unit changes to yellow. This means the configuration is invalid and must be triggered again. In the **Quick Property** window it is also possible to read out the configuration string, **Tc Key Code**, see picture

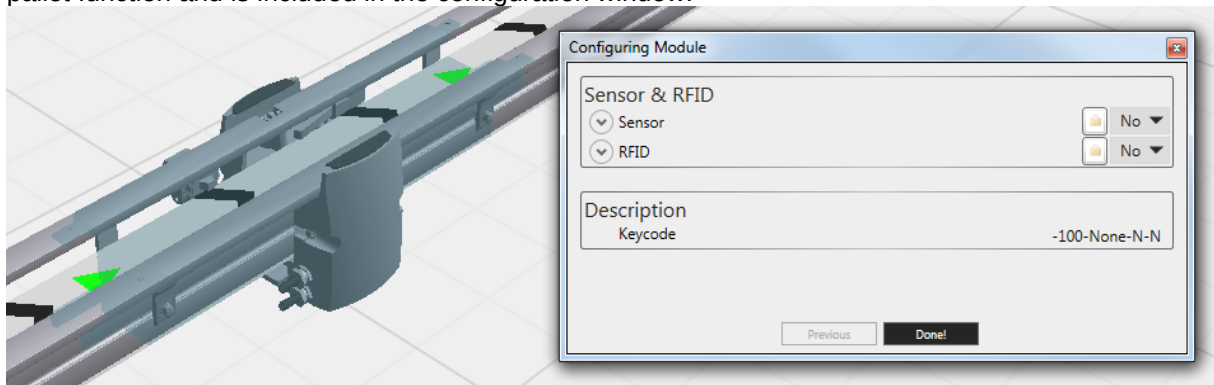
below.



Picture: Configurations – end drive unit example

Pallet modules configuration

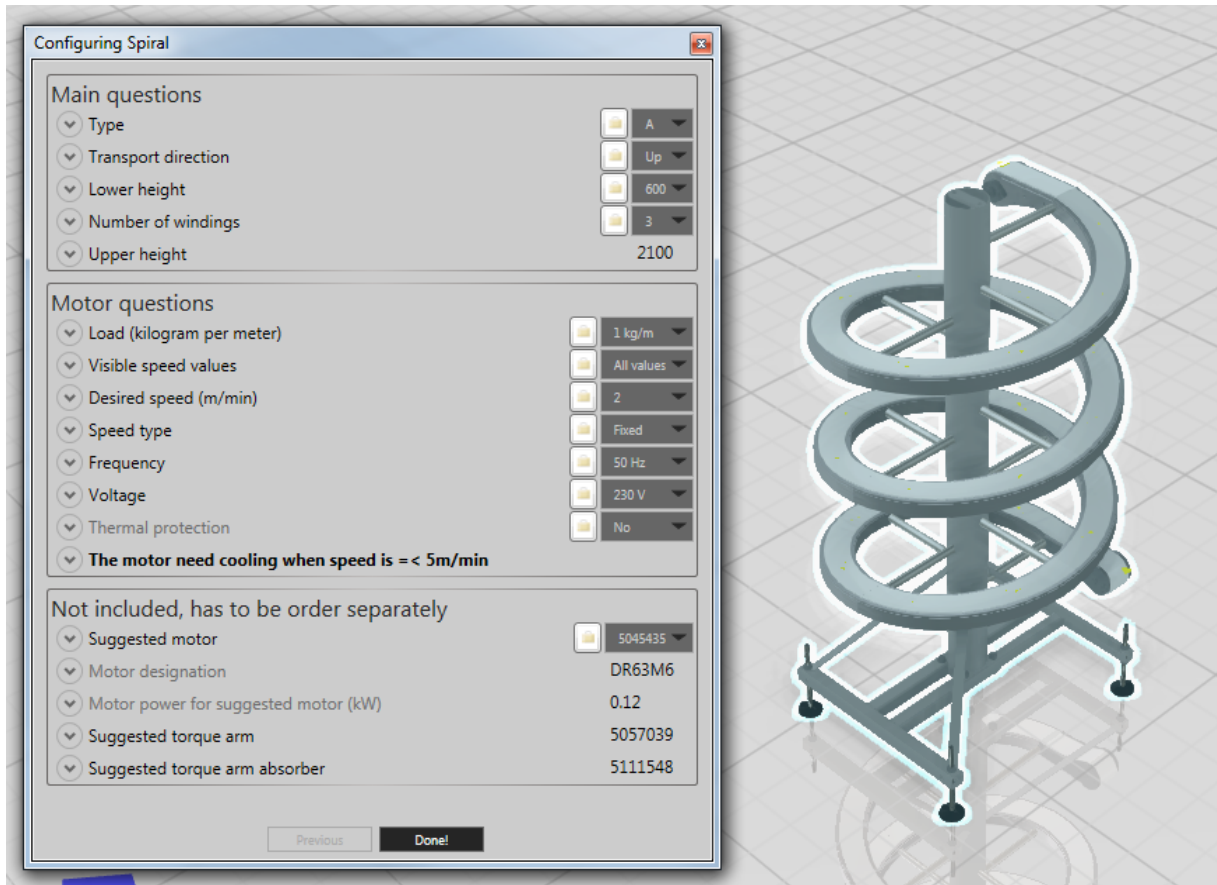
Pallet platforms (e.g. X85P) has functions which must be configured. The main question in these cases is the pallet size, **Pallet length (mm)**, and is concerning the complete conveyor. This property is set from the Conveyor menu (Pallet Ribbon group) and is described more in the **Conveyor** chapter. All other configuration settings (e.g. **Sensor** and **RFID**) can be individually set for each pallet function and is included in the configuration window.



Picture: Configurations – pallet function (basic locating module)

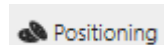
Spiral elevator configuration

The spiral can be configured via the **Configure** command in **Component** ribbon group. It is also a part of the **Config painter** command for both conveyor and component level. See the **Conveyor** chapter for more details about the conveyor **Config painter**.



Picture: Configuring Spiral

Positioning

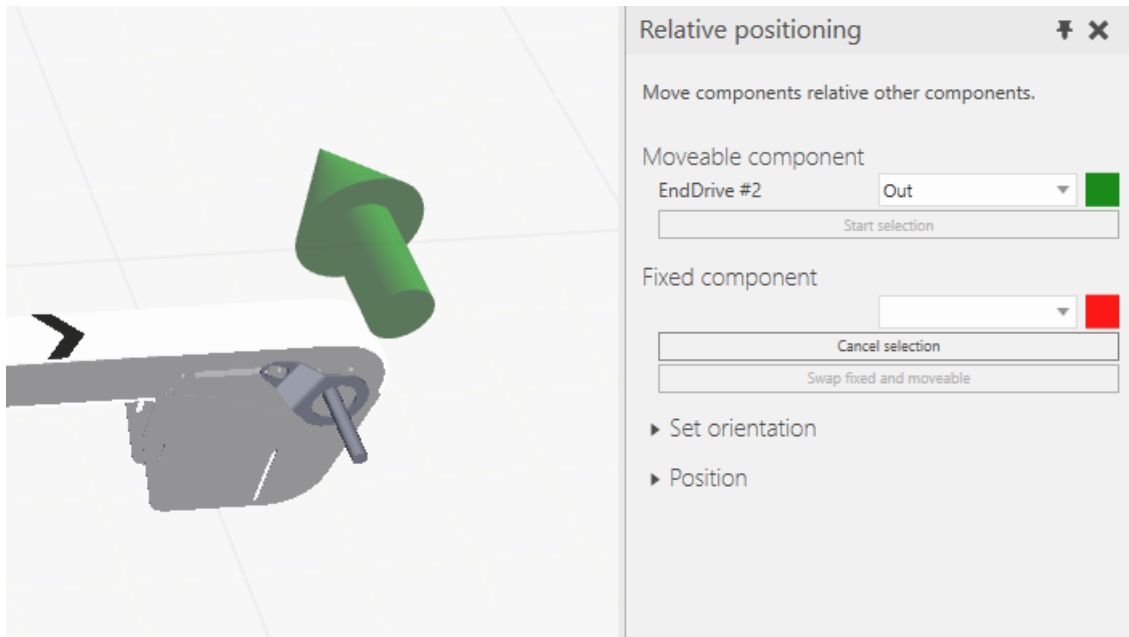


Short key: Ctrl + shift + R

This command is a useful tool to adjust a conveyor in relation to another already in the layout existing component. Even though the command is using the component as reference it adjusts the complete conveyor. It even adjusts other conveyors if they are connected using bridges. The command is divided into three parts.

Selection of component

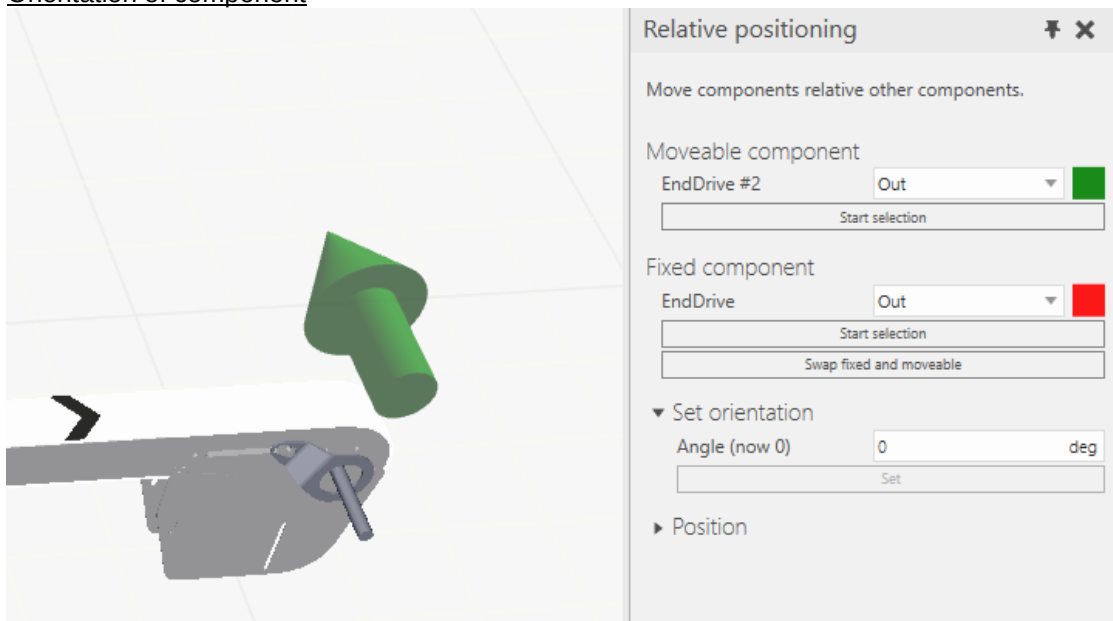
First the user has to select the component that should be moved to a new position or oriented to a new angle. When the **Positioning** command is triggered, the **Relative Positioning** command window appears and the movable component is marked with a green indicator arrow.



Picture: Positioning - Selection of component

The **Picker Tool** window informs the user to click on the component that shall remain in position. This fixed component will be marked with a red sphere. This component is the reference which all the adjustments are calculated from. The adjustment is calculated for one specific end of the component. As default the **In** end is selected but this can easily be changed to the **Out** end in the drop down list. Both component selections can be triggered again with the **Start selection** buttons. Also the two component points can be switch with the button **Swap fixed and moveable**.

Orientation of component



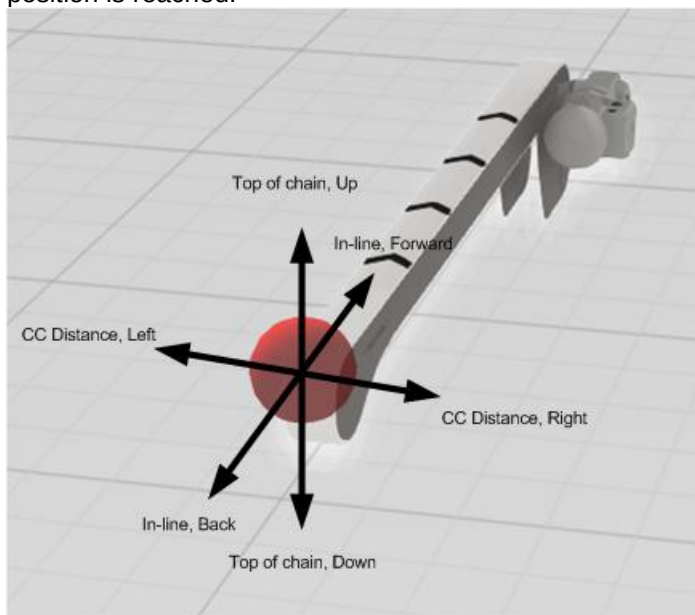
Picture: Positioning – Orientation of component

The next part of the command is the **Orientation of component**. This is used to rotate the whole conveyor that the moveable component is a part of compared to the fixed component. All three

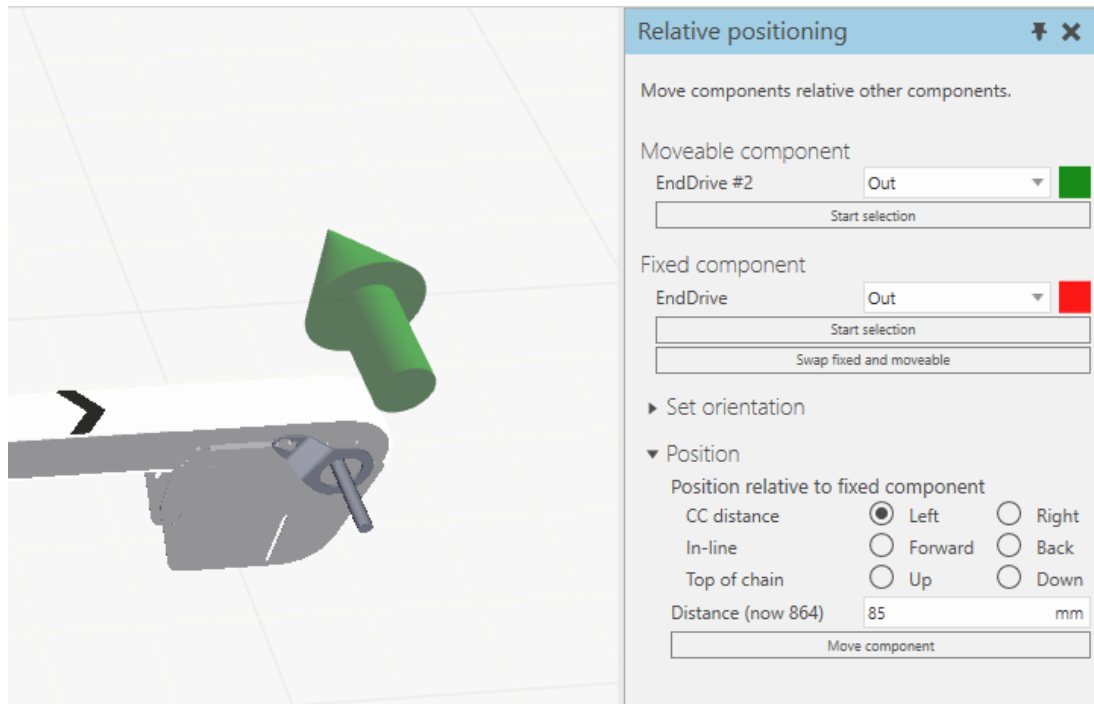
rotation values, A (X-), B (Y-) and C (Z-axis), are copied to the whole (movable) conveyor. If a value is set in the set orientation field then this is an offset to the C rotation value. As soon as the set orientation field is selected the red indicator will point in the zero angle for the static component and the green indicator will indicate the angle for the component that will be adjusted.

Positioning of component

The last part of the command is the **Position of component**. This command is divided into three different sub commands. Each sub command is translating the whole conveyor that the Movable component is a part of in a specific direction, see picture below. The **C-C distance** is sideways compared to the product path in the selected component point. The **In-line** is direction of the product path. The **Top of chain** is the height of the component. All these three sub commands can be executed with an offset written into **Distance to fixed component**. The component must be translated in only one direction simultaneously but the command could be repeated until required position is reached.



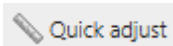
Picture: Positioning – Explanation of positioning directions



Picture: Positioning – Positioning of component

In the example above the positioning command has just been executed. The new position of the movable conveyor is 85 mm to the right side of the fixed conveyor. The green arrow indicates the direction where the offset is translating the conveyor further (positive direction). The indicator is showing this reference even before the movement is done.

Quick adjust

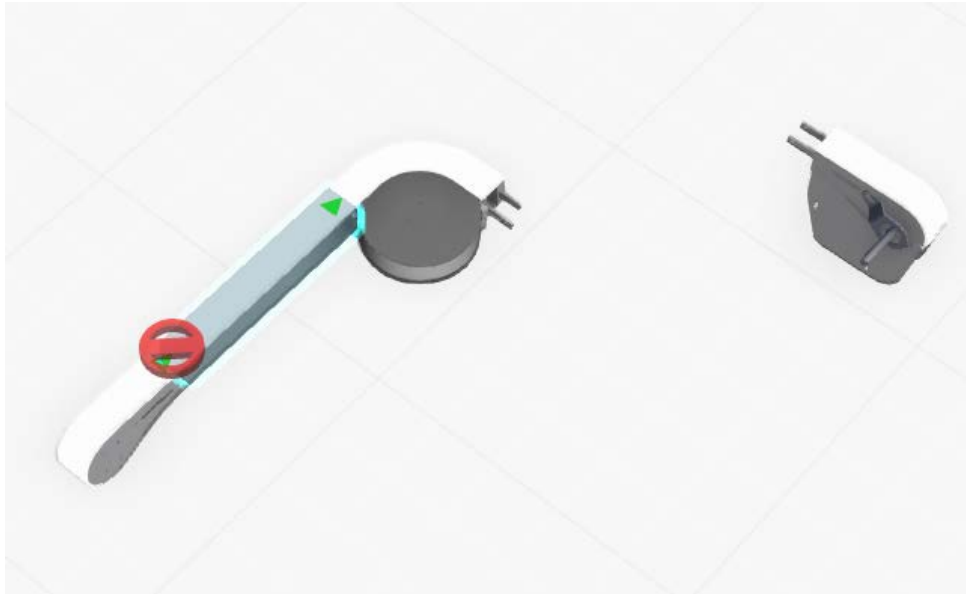


Short key: Ctrl + shift + Q

The **Quick adjust** command acts on either a beam or a bend. When the component that should be adjusted is selected the **Quick adjust** command is possible to run. Then the **Quick Adjust** command window informs the user to select the target component. After that has been done the original component is adjusted in order to reach the target component or as close as possible.

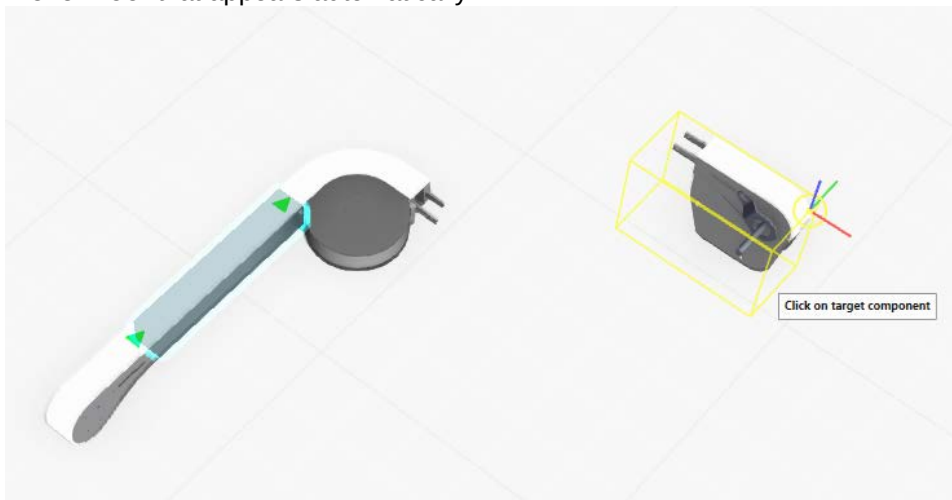
Quick adjust - Beam length

If a beam is selected the command tries to adjust the beam length so that the open conveyor end (not necessary the end of the beam) reaches the target component. It is very important that the build direction on the beam allows the beam to expand in the required direction (see picture below).



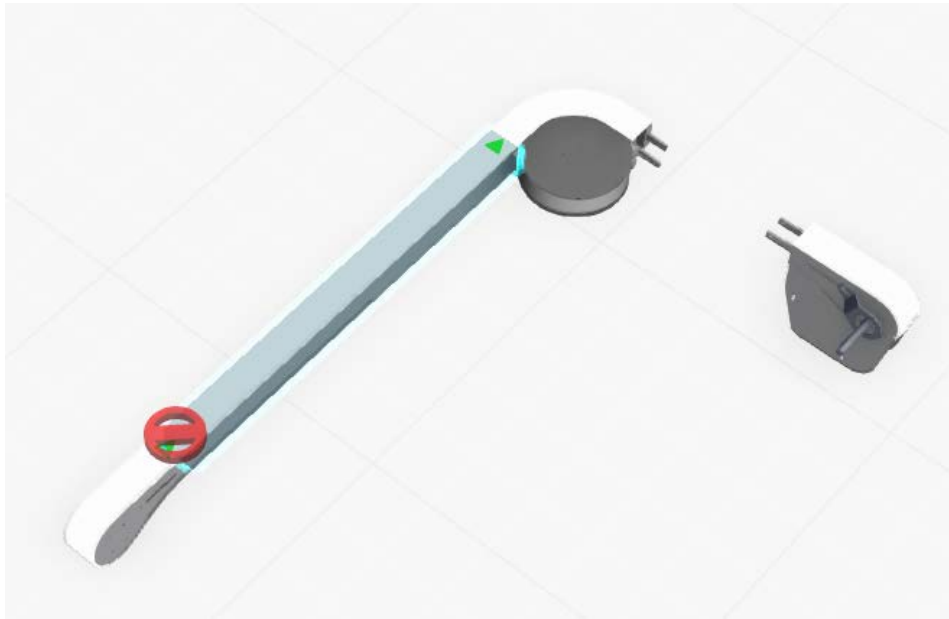
Picture: Quick adjust – beam, start condition

When the **Quick adjust** command is triggered the user has to select the target component with the **Picker Tool** that appears automatically.



Picture: Quick adjust – beam, select target component

The result of the quick adjust command is that the open end of the origin conveyor and the open end of the target conveyor is parallel. This is the optimal adjustment with respect of the beam. Next step would be to add a beam and use a **Quick adjust** on that length too.



Picture: Quick adjust – beam, command finished

Quick adjust – Bend angle

The other component that could be used for **Quick adjust** is bends. It acts in a similar manner as for a beam. The objective is to get the open conveyor end of the origin conveyor and the target conveyor parallel. Even for the bends it is important to verify that the build direction allows the conveyor end to be adjusted.



Picture: Quick adjust – bend, start condition

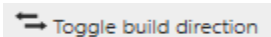
If a bend angle exists which makes the two conveyor ends parallel the bend angle is adjusted. Further steps would be to add and adjust beams before and after the bend.



Picture: Quick adjust – bend, command finished

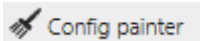
A complete example of how to work with the **Quick adjust** can be found in the appendix, under Best practices.

Toggle build direction

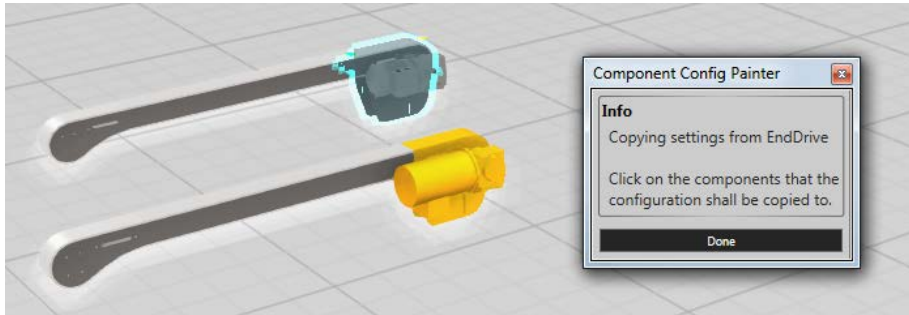


See Building concept chapter

Config painter

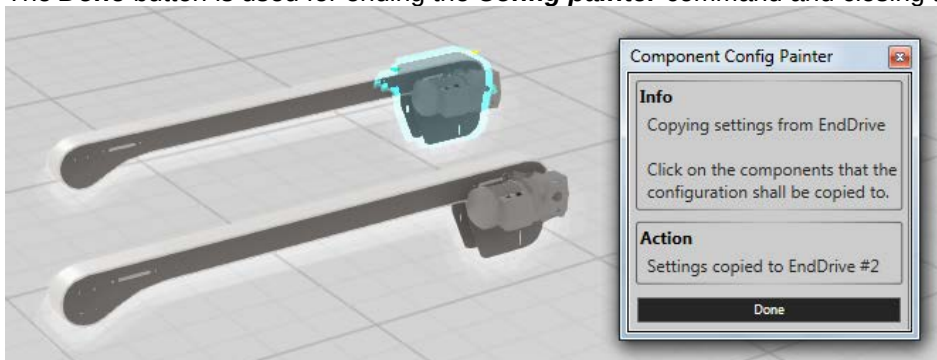


There are two different ways to copy a configuration from one place to another. This command focuses on the component level. The **Config painter** exists also for a complete conveyor. All component configurations can be copied by selecting the source component and then the **Config painter** symbol in the **Component** ribbon group. All components that have a configuration are possible to copy the settings. Examples of these are drive units, pallet functions and spirals. When the command is triggered a pop up information window appears, see picture below.



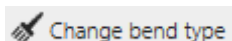
Picture: Component Config Painter – source component selected

The next step is for the user to select which component this configuration should be copied to, the target component. When this is done the user gets a feedback that the component configuration settings has been copied. This paint procedure can be continued to more components if required. The **Done** button is used for ending the **Config painter** command and closing the dialog window.

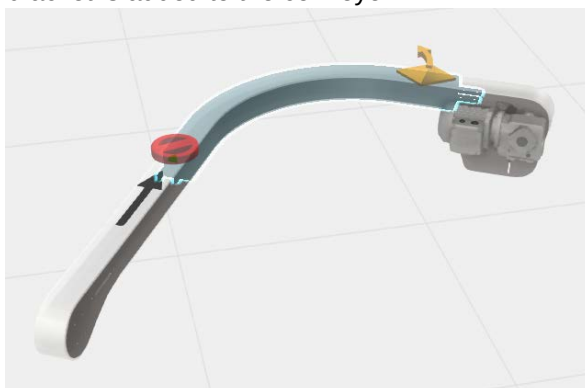


Picture: Component Config Painter – target component selected

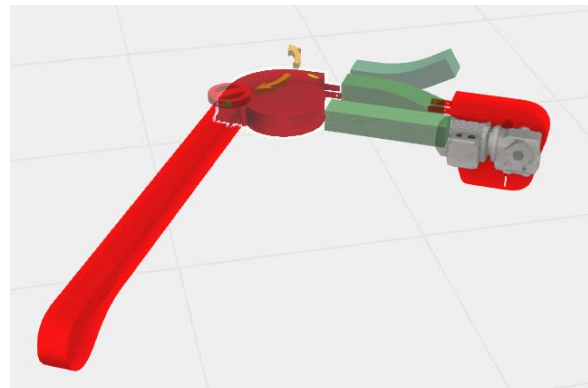
Change bend type



A lot of the platforms has both wheel bends and horizontal plain bends. In these cases there is a command for toggling between these two bend types in an existing conveyor. This command can be found in the **Component** ribbon group. The execution of the command is that it immediately changes the bend type for the selected bend component to the other bend type. The software is automatically trying to adjust the conveyor with regards to angles and beam lengths. If it doesn't find a solution it leaves this to the user. This means in some cases there is some manual work to fix the conveyor layout after this command. The **Change bend type** command requires that no guide rail bracket is added to the conveyor.



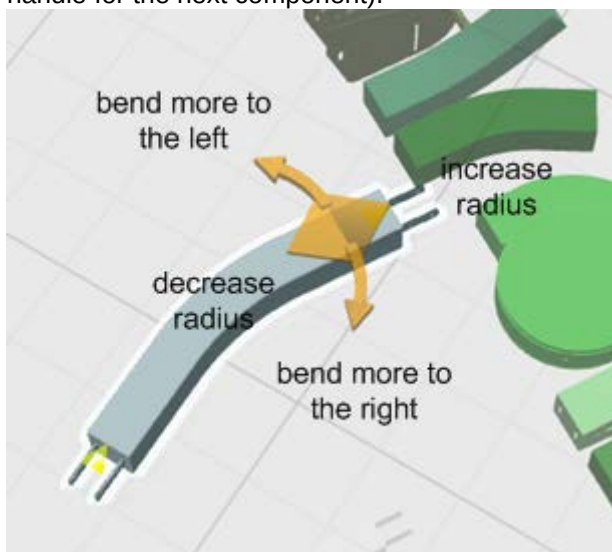
Picture: Change bend type – original conveyor



Picture: Change bend type – after change

Angle and radius handles

The horizontal bends and wheel bends have a specific type of building handles for adjusting the components. The larger orange arrows on the chain changes the radius of the bend. The smaller orange arrows changes the angle of the component. If limit is reached, on either the angle or radius, that handle is not visualized. These handles works even though the conveyor is continuing after the selected component. In this case all components beyond are moved (in the build direction). If CTRL is pressed when clicking on these handles a special feature is triggered. For the angle handle the first time this feature is triggered it tries to change the direction but keeping the angle value. If it already is bend in that direction it forces the angle to the maximum value. For the radius build handle the CTRL and click feature is only forcing the bend to reach the maximum value. These building handles is turned on and off with the show handle property in the **View** menu (same as the build handle for the next component).



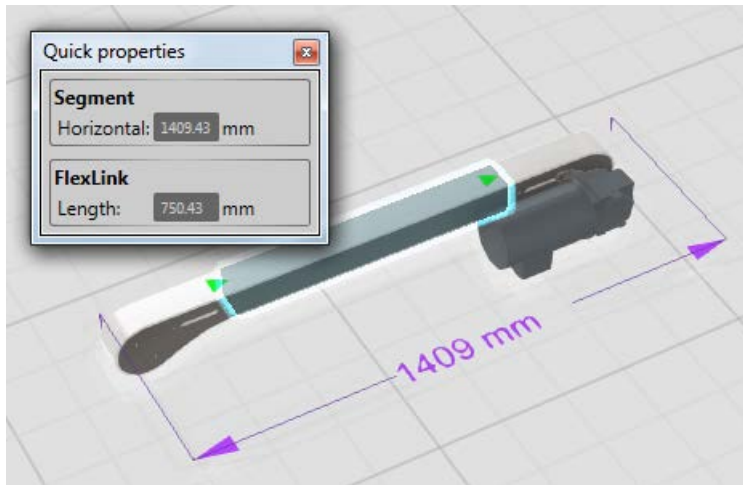
Picture: Adjust Bend angle and radius via handles

Quick properties window

By double clicking a component a **Quick properties** window appears. Here the user can change the most common properties for the selected component. Some of them could be reached from other windows also.

Beams example (Quick properties)

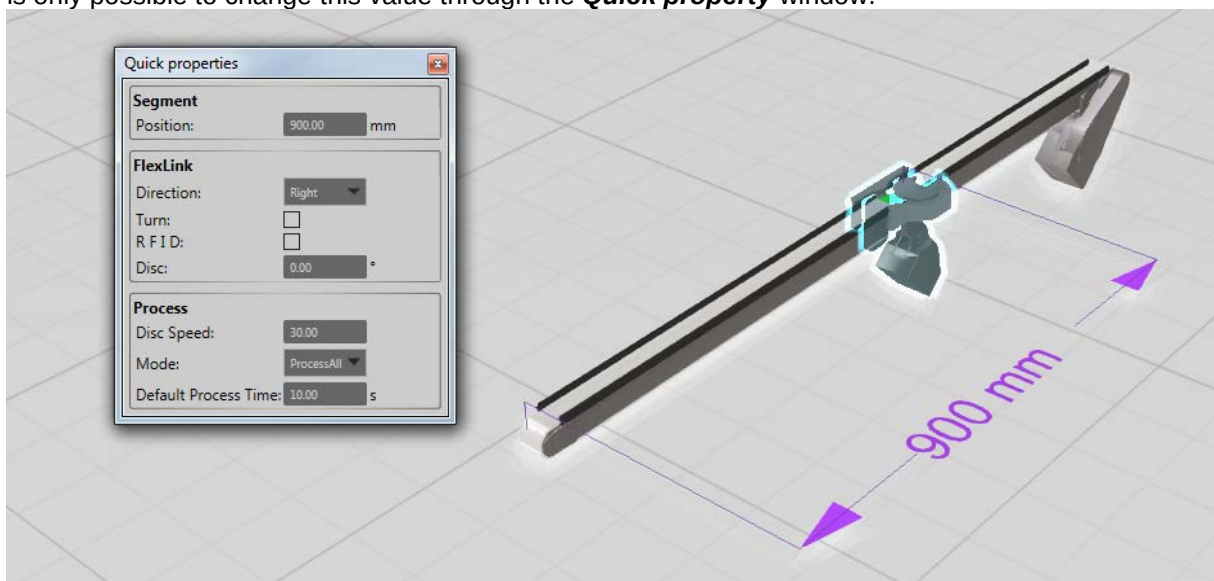
On a beam there are two properties in the **Quick properties** window that are controlling the length. The **Segment Horizontal** is the total conveyor length for that segment. This value is also visualized in the layout. Also there is a separate **Segment** command the **Conveyor** ribbon group where this value can be changed. The **FlexLink Length** is the actual length on the beam component. If one of those values is changed the other is automatically updated.



Picture: Quick properties – Beam

Function example (Quick properties)

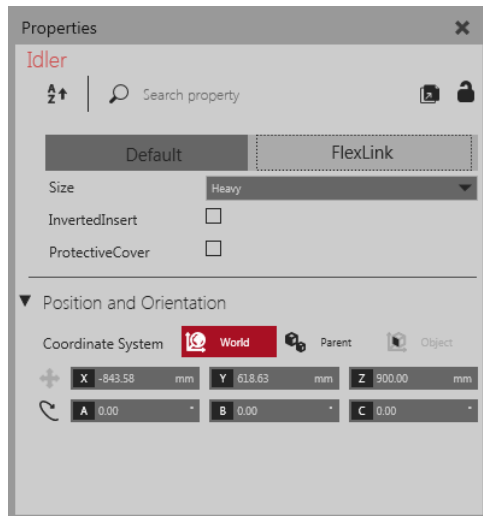
Functions that connect along a conveyor beam has a **Quick property** called **Segment Position**. This property can be useful for positioning the function. This value is also visualized in the layout. It is only possible to change this value through the **Quick property** window.



Picture: Quick properties – Function (X45e Locating unit)

Property window

When a component is selected the properties window is updated with data about the component. First of all the name of the component is declared at the top. This is the name of the selected instance of the component.



Picture: Properties example, X85 Idler



Lock symbol

All components can be locked and unlocked with the lock symbol. When a component is lock it can't be changed, both in terms of properties and its location.

Default tab

The default tab is containing some standard properties e.g. Name and Visibility.

FlexLink tab

The majority of the FlexLink specific properties are collected under the FlexLink Tab. But the individual properties are very different from component to component.

Product tab (only for Feeder components)

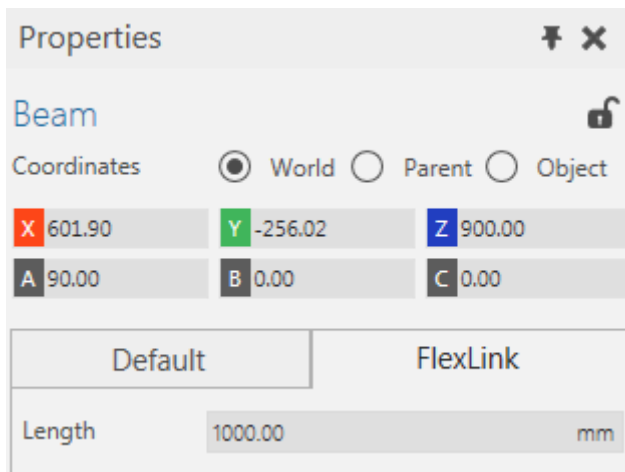
Components made for creating products during simulation has another tab called Product. The product specific data (e.g. size and shape) are collected in this tab.

Process tab (only for simulation components)

If the component has a need for properties controlling the simulation behavior they have another tab called Process.

Position & Orientation

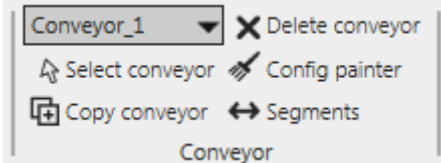
All components has a section for position and orientation in the properties window. The top row of this section contains a selection of coordinate systems. Further below is a table is describing the position and orientation of the selected component, relative the selected coordinate system. The X, Y and Z values can easily be changed by overwriting the new value in mm. If a rotation is required this can also be entered directly in the bottom row (in °). Rotation A is rotation around the Z-axis, B is around the Y-axis and C is around the X-axis. When the position and orientation letters (X,Y,Z,A,B or C) are clicked on the value is reset to zero (and also the corresponding position or orientation). See also chapter about the Move command (Manipulation ribbon group / Home menu).



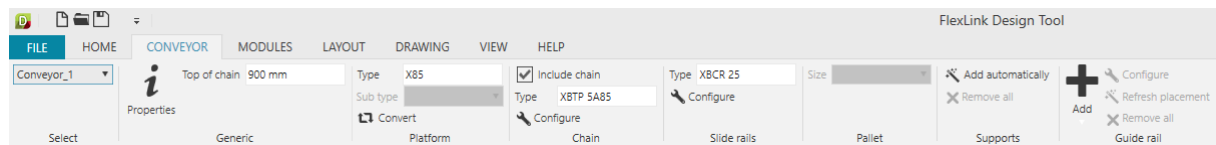
Picture: Properties window - Orientation

Conveyors

This chapter is focusing on features in the conveyor level. The most frequently used conveyor commands are gathered in the Home menu but there is also a dedicated conveyor menu.



Picture: Conveyors – Home menu, Conveyors ribbon group

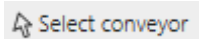


Picture: Conveyors – conveyor menu

Conveyor name (Home menu, Conveyor ribbon group)

The Conveyor ribbon group is always printing out the conveyor name of the selected component, Conveyor_1 in the example above. It is possible to use the drop down list to select any other conveyor in the current layout.

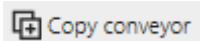
Select conveyor (Home menu, Conveyor ribbon group)



Short key: Ctrl + shift + A

A lot of the conveyor commands works on the complete conveyor even though only a conveyor component is selected. But if the command exists on both component and conveyor level it can be necessary to select the complete conveyor. It is done by selecting any conveyor component and the trigger then the select conveyor command. Typically this has to be done in order to translate a complete conveyor to a new position.

Copy conveyor (Home menu, Conveyor ribbon group)



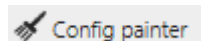
The copy conveyor command is both coping the selected conveyor and pasting it to the layout. The location of this new copy is the same as the copied conveyor but with a conveyor height of additional 500 mm.

Delete conveyor (Home menu, Conveyor ribbon group)

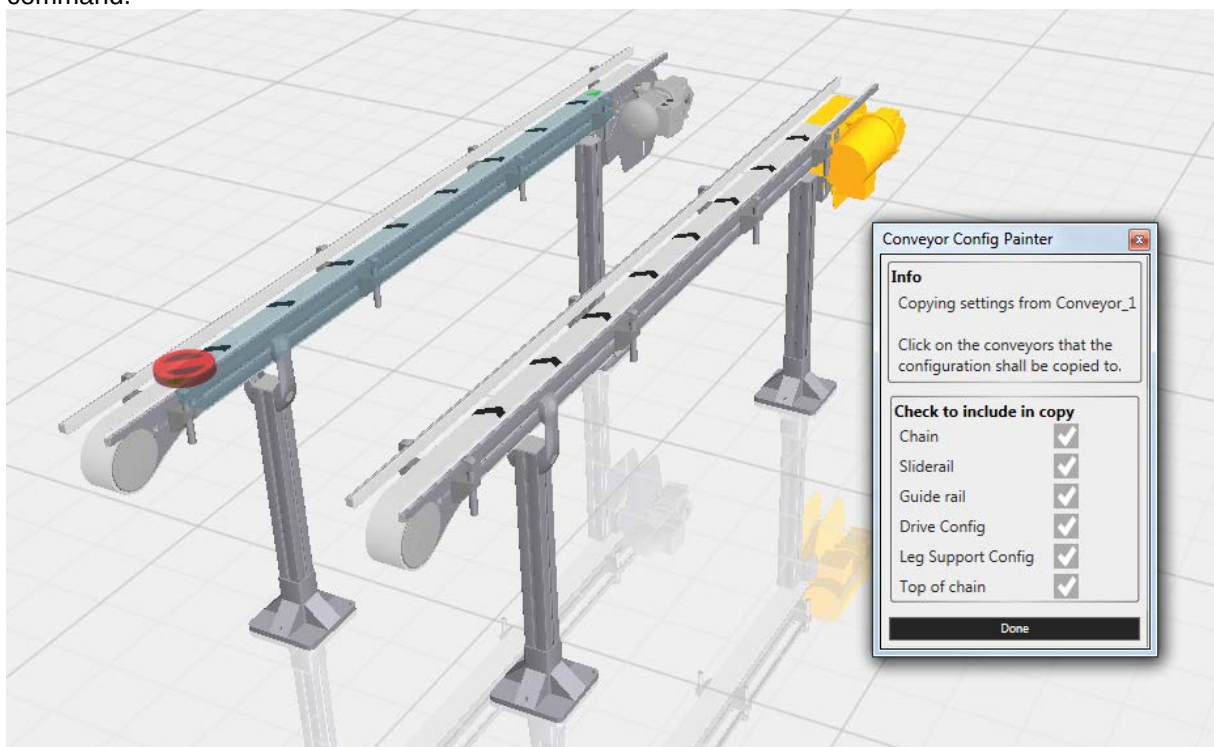


The delete conveyor command removes the selected conveyor completely. All components attached to this conveyor (e.g. conveyor supports and guide rail brackets) will also be removed. At the trigger of this command a confirmation window pops up with an option to cancel.

Config painter (Home menu, Conveyor ribbon group)



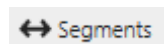
If one conveyor or a conveyor component is selected there is also a possibility to use the command **Config painter** from the conveyor ribbon group. Then a pop up window appears, letting the user decide what configuration parts in that selected conveyor that should be copied. If one of these options doesn't exist on the selected conveyor that option should be unselected as default, or even hidden in some cases. The **Guide rail** is only copied to the target conveyor if that conveyor already has a guide rail (bracket kit) of the same type. This is valid also for the **Drive Config**. The Top height is the Top of pallet/Top of chain parameter. For the **Leg support Config** the painter function is evaluating if both conveyor are containing the same type of conveyor support. In this case the configuration of one support of each type is copied to all supports of that type on the target conveyor. If the source conveyor is containing several supports of one specific type and they differ in the configuration it is recommended to not use the **Conveyor Config Painter** and to use the **Component Config Painter** instead. If the **Config painter** is used on a puck/pallet platform it also has the option **Pallet size**. Other configuration data for pallet functions is not included in this command.



Picture: Conveyor Config Painter – source conveyor selected

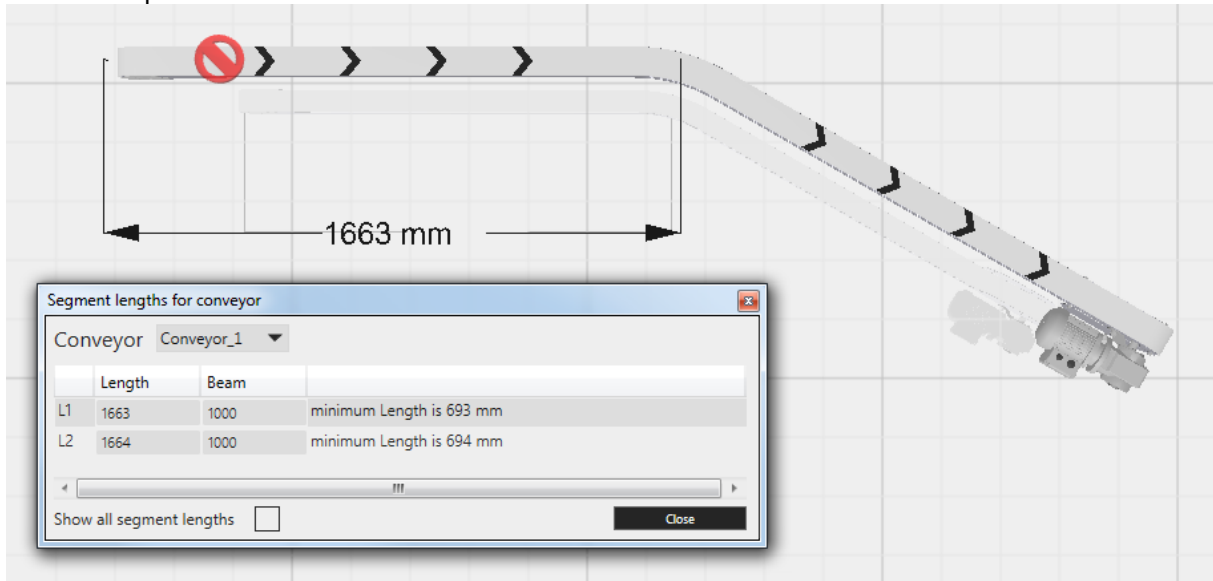
Then when the source is selected the user is requested to select the target conveyor or conveyors. The command is finished when the user clicks the **Done** button.

Segments (Home menu, Conveyor ribbon group)



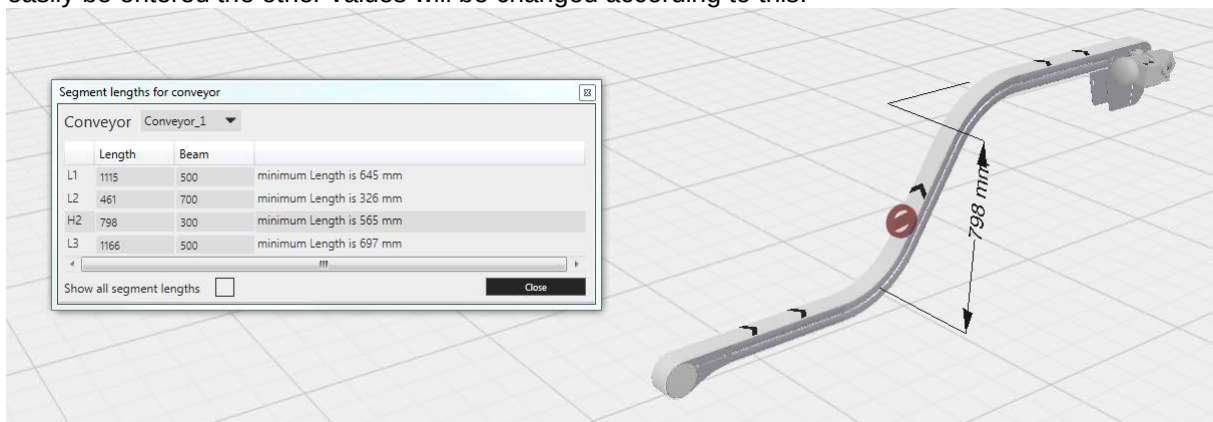
The **Segments** command is located in the Conveyor ribbon group. When the command is triggered the view is automatically changed to a top view and a table with all the segment distances for the selected conveyor pops up in the screen. The selected conveyor could be changed with the drop down list. One of the distances is showed in the top view as default but this could easily be changed by selecting another segment row in the pop up window. There is also a check box for showing all distances, **Show all segment lengths**. Every conveyor segment has two values, the **Distance** is the complete length and the **Beam** is the beam length. Either one of these could be changed by the user. If a value is entered that results in a distance shorter than the minimum distance the value will

not be accepted.



Picture: Segments, example

If one segment has got an incline or decline this segment has two distances. The L distance is the horizontal part and the H distance is the vertical part. If the height is the important value this could easily be entered the other values will be changed according to this.



Picture: Segments, incline example

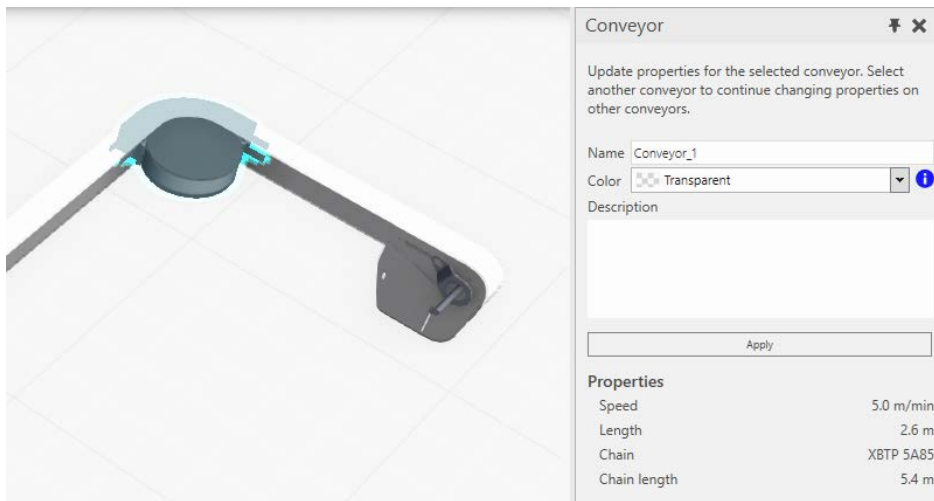
Generic

Properties



All conveyor can be named and described with the properties command. This information is used when the 3D-pdf is generated and exported to other FlexLink tools. To apply a change of the name, color or description the **Apply** button must be pressed, otherwise the information will not be updated.

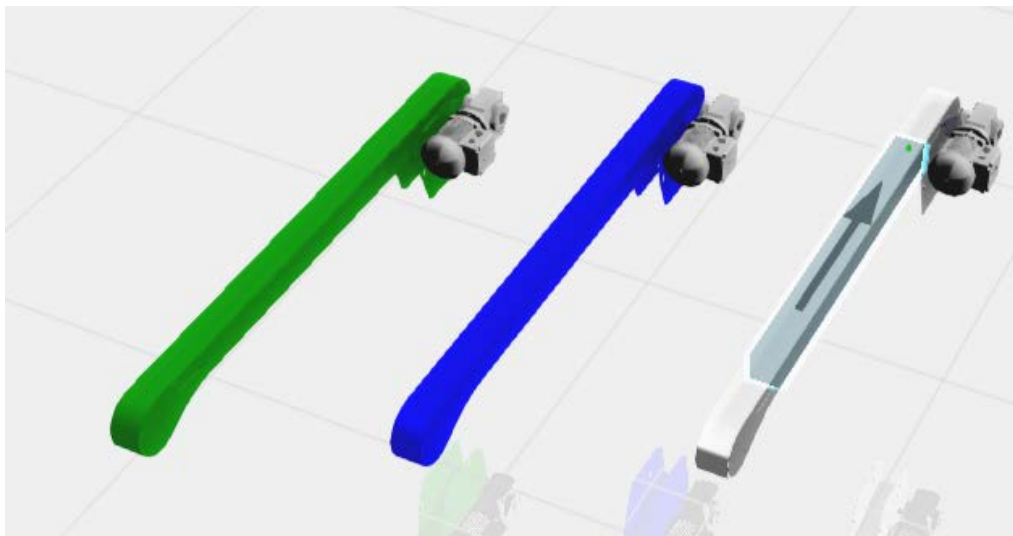
Properties on other conveyors can be easily changed, just click on one conveyor in the 3D view and the panel will show information about the selected conveyor.



Picture: Conveyor properties window

The user has also a possibility to do color marking on individual conveyors. This can be used for explanatory purposes in the layout. There are ten standard colors but also an advanced feature where a unique RGB-color could be set.

This command affects the selected conveyor only. All conveyors that should be colored must be selected and changed manually. The conveyor must not have any errors or warning that affects the component colors in order to show this coloring.



Picture: Color – palette

Top of chain

If the required top of chain value for the selected conveyor differ from the default top of chain set in the Layout menu this value is entered here. When this value is changed all conveyor supports are automatically adjusted.

Platform

Type

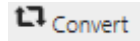
When a component is selected the platform type is written to the Type property. This is only for

presentation and can't be changed.

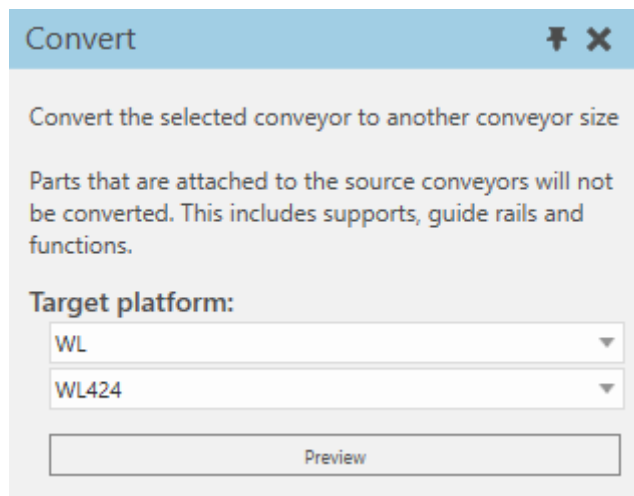
Subtype

If the platform of the selected component has variants the current sub platform is shown in the Sub platform property.

Convert



It is possible to automatically change all of the components in a conveyor to another platform. One component in the source conveyor must be selected before the Convert command can be run.

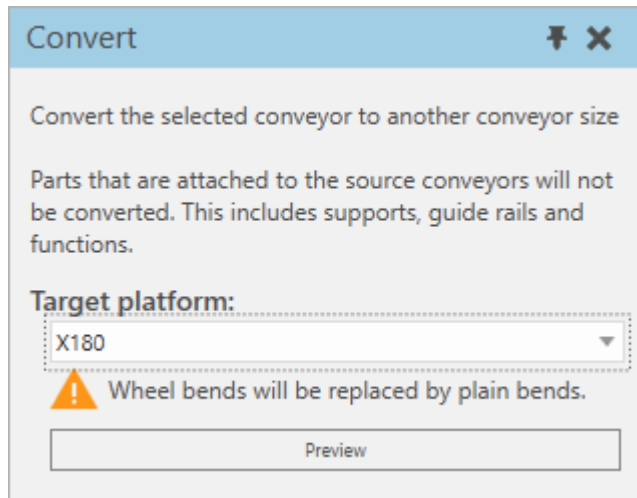


Picture: Convert: Source conveyor selected

The target platform is a drop down list containing all available platforms. Here it is possible to set which platform all components should be converted too. The user can also reselect the source again if needed.

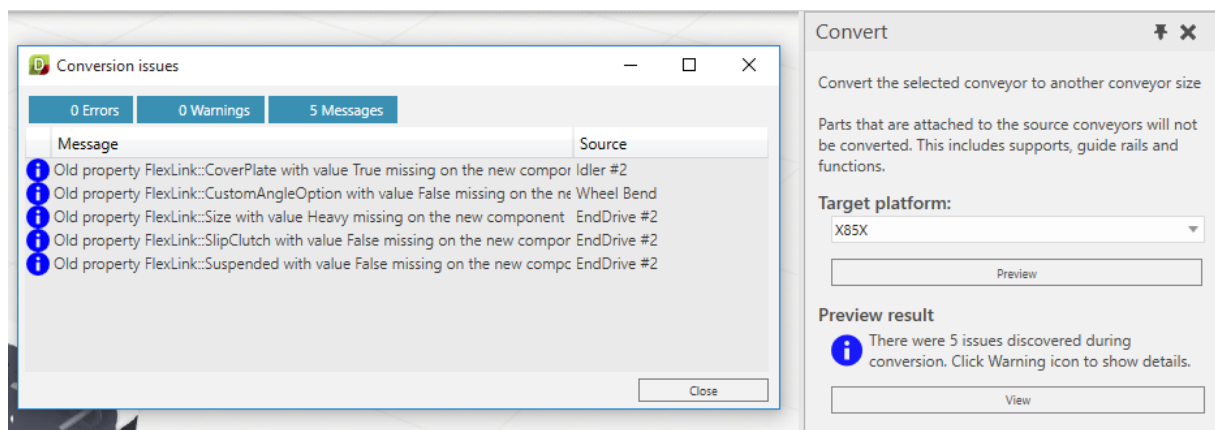
To preview the conversion, click **Preview** button and a new converted conveyor will be shown at the same position but 500 mm above the source conveyor. If this preview is acceptable, the new conveyor can replace the old by clicking **Replace**. If the preview is incorrect, cancel the conversion by clicking on **Cancel** and the conveyor preview will be removed.

If the converted conveyor cannot be a full copy of the original, then one or several warnings will be displayed under the platform selection.



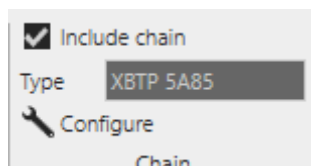
Picture: Convert: Preview may not be 100% similar to the original

After the preview, there may have been some options that could not be used due to the new platform. These warnings will be displayed below the **Preview** button and can be clicked to show a full list of the warnings.



Picture: Convert: Differences between the original and the converted conveyor

Chain



Picture: Chain ribbon group

The chain ribbon group contains some features concerning the chain for the complete conveyor.

Include chain (tick box)

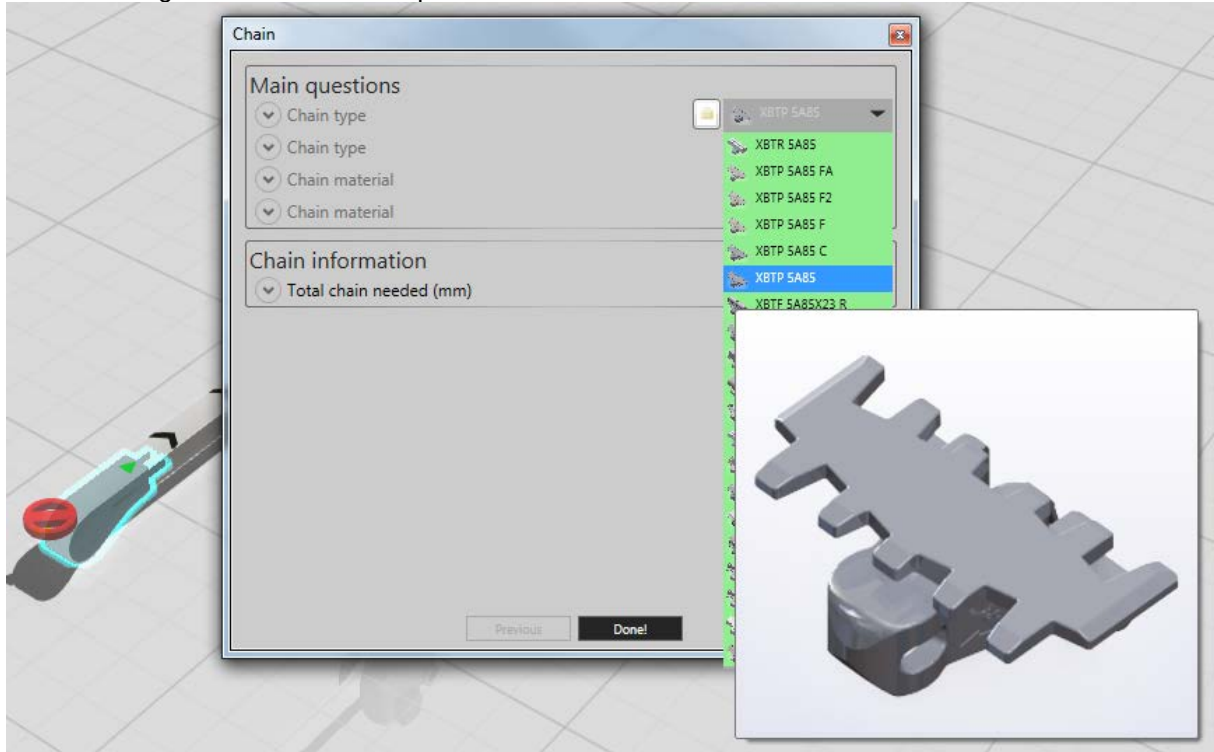
This option controls if the chain should be a part of the conveyor. This affects also the slide rail and the rivets. The BOM is only including these components if the include chain is selected. The chain visualization is also controlled by this option.

Type

The type property is written if a chain is included. This is the current chain type for the selected conveyor. This ribbon group contains a configuration button for the chain. See separate chapter about the chain configuration.

Configure (chain)

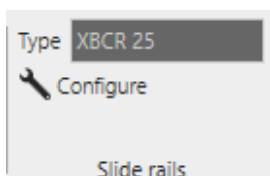
The chain configuration is reached from the chain ribbon group in the conveyor menu. It is also included in the conveyor **Config painter** command. If the chain configuration is not changed the default setting is a valid standard option.



Picture: Chain configuration window

In the chain configuration window the first chain type property is a drop down list with all chain types. When the mouse pointer is hovered above the small icon of each chain link type a bigger picture pops up on the screen. The rest of the properties is read only depending on this first property setting. At the bottom of the chain configuration window it is possible to read out the required chain length for the conveyor in focus.

Slide rails



Picture: Slide rails ribbon group

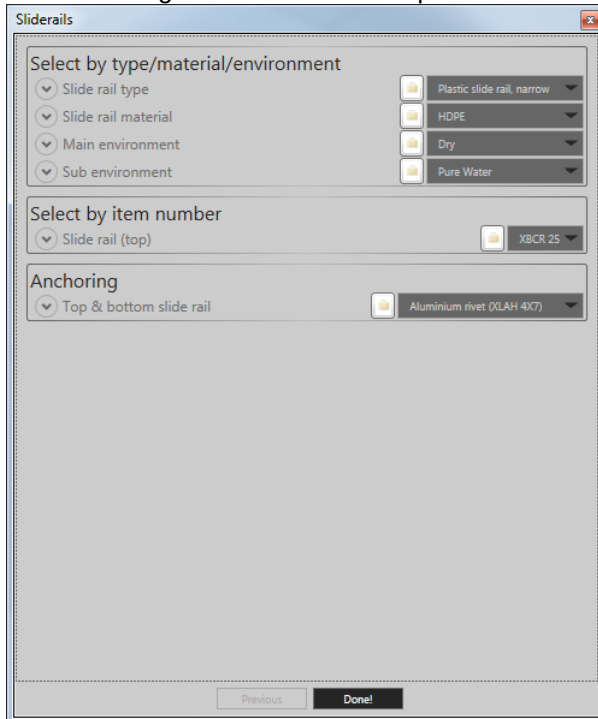
The slide rail commands for a conveyor is located in the **Slide rails** ribbon group.

Type

The type property describes the current slide rail type. This is a read only property and to change the slide rail configure must be used.

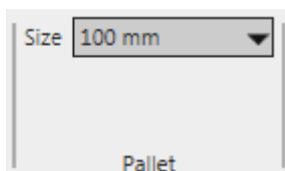
Configure (slide rail)

The slide rail configuration is reached from the slide rail ribbon group. This configuration is also included in the conveyor **Config painter** command. If the slide rail configuration is not changed the default setting is a valid standard option.



Picture: Slide rail configuration window

Pallet



Picture: Pallet ribbon group

The only property possible to set from the pallet ribbon group is the pallet size. This value is then used by all other data concerning the pallet configuration is set in the feeder component. All pallet conveyors get a default value on the pallet size. When one of the components in a pallet conveyor gets its pallet size changed all of the components in this conveyor will be updated with this setting.

Supports



Picture: Supports ribbon group

Add automatically

If a conveyor is complete but without conveyor supports it is possible to add them automatically. One of the conveyor components has to be selected to show the **Add automatically** command in the Support ribbon group. After the command is triggered the Add Leg Support command window pops requiring the user to enter the CC distance between the supports. The supports are added with the button **Apply**. First the software is trying to add supports at the beginning and end. Then it calculates the number of single supports and adds them with the required CC distance.



Picture: Add Leg support window – Add automatically

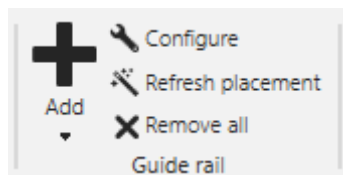
Refresh placement

When some conveyor supports have been added to the conveyor the commands in the ribbon group changes to include **Refresh placement** instead of **Add automatically**. This command opens up a similar pop up window as **Add automatically** but with the button **Update supports** instead of **Add supports**. The algorithm for the placement of supports is exactly the same.

Remove all

The Remove all command is removing all conveyor supports from the selected conveyor.

Guide rail



Picture: Guide rail ribbon group

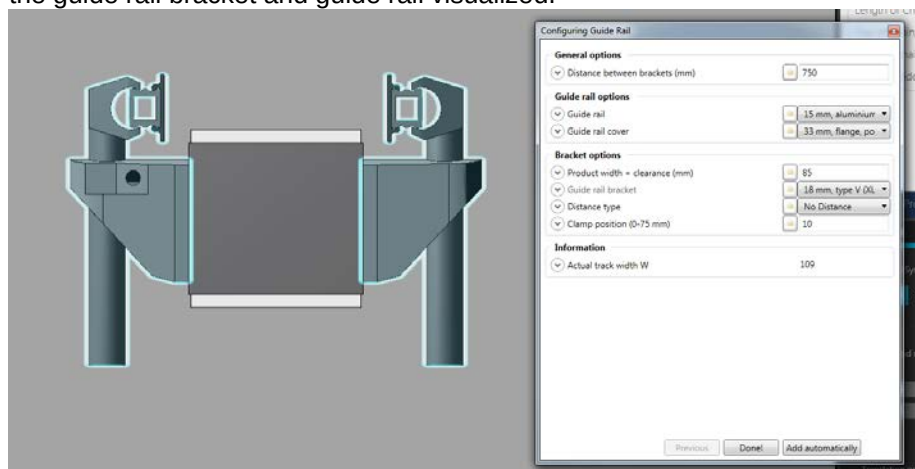
Every conveyor platform has a number of compatible Guide Rail bracket kit types. These are shown in the **eCatalog** under the **Guide rails** collection for each platform. In this manual section all information about the guide rails and how to work with these. All the guide rail ribbon commands is only possible to run when one conveyor component of that conveyor is selected (not the guide rail component itself). There is only one guide rail configuration for each conveyor. If a bridge component is used all guide rail features must be completed for each conveyor.

Add

The Add command contains a list with compatible guide rail kits. When any of these alternatives is chosen the configuration window of this guide rail kit appears automatically. If the conveyor already contains any guide rail kit a window appears asking if this should be replaced or not.

Configure

When a guide rail bracket component is added into the 3D world it is not connected to any other component. The guide rail bracket could then be connected to a conveyor component using plug and play. After being connected to a conveyor the guide rail configuration window is automatically appearing on the screen. At the same time the layout is changed to a cross section of the beam with the guide rail bracket and guide rail visualized.



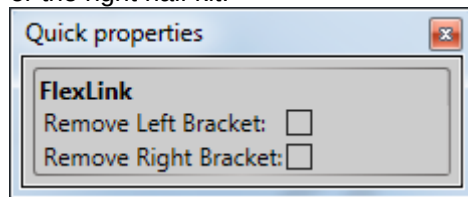
Picture: Guide rail configuration window

If the user changes any parameter in the configuration it is affecting also the graphical appearance in

the cross section view. When the configuration is completely filled in there are two choices to finish the configuration. With the done button the configuration window is closed and the guide rail bracket component selected in the 3D world is the only bracket for that conveyor. If the user wants this configuration to be used throughout the whole conveyor and that the software automatically calculates the number of brackets to use and the location of each the button **Add automatically** is used. In this case the guide rail itself is also visualized throughout the conveyor. The configuration is controlling the structure of one guide rail bracket half kit. The function with automatically adding of brackets is sometimes removing one half kit of the bracket pair depending on the conveyor layout.

Adjusting the guide rail bracket quantity and positions

If the user wants to manually remove more it is a property on every bracket to remove either the left or the right half kit.



Picture: Guide rail bracket – Quick properties window

These properties can be reached from either the quick properties window (picture above) or from the component properties window. In the BOM the quantity of guide rail bracket represent half kits. The user can also add more guide rail bracket kits by drag and dropping the same guide rail kit component from the **eCatalog** or just copy paste (CTRL + C and CTRL + V). To change a position of a guide rail kit the user can easily select it and drag it to a new position on the conveyor. It is also possible to delete single guide rail kits.

Refresh placement

If the user has changed the guide rail brackets and wants to trigger the automatically positioning of again this is done with the **Refresh placement** command. This overwrites all changes and recalculates the number of brackets and their positions.

Remove all

The **Remove all** command deletes all the guide rail bracket kits and the guide rails. After this action it is possible to add a new type of guide rail kit.

Modules

The Modules menu contains functionality for quickly creating conveyor modules. The new conveyor will be placed in the zero position in the layout. After the conveyor has been created, it is selected and can be moved directly around in the layout. Note that the created conveyor will be added as separate parts and is not to be confused with the Conveyor modules article in the Online Store.

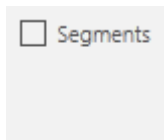


Platform



First the type of platform must be selected. To select a platform, click on the arrow below the platform name, and then choose the wanted platform. The selected platform name will be shown underneath the icon.

Options



Show segments

After the conveyor has been created, all beams are 1 meter long. If this checkbox is checked, then the Segments window will be shown where the length of the segments can be edited.

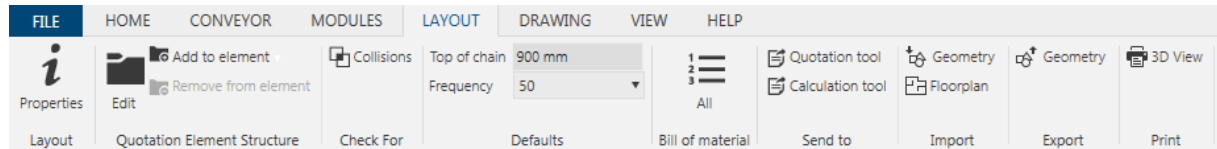
Add



To generate a conveyor of a requested shape, click the corresponding icon. Note that for some platforms it is not possible to create conveyors containing bends or similar, those shapes will be disabled and grayed out.

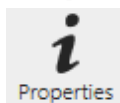
Layout

The main part of all layout feature can be found in the Layout menu. Especially the exporting to other tools is done from this menu.



Picture: Layout menu

Properties



For each layout there is a possibility to write and save data. The fields are fixed and showed in the picture below. This information is used when the 2D-drawing is generated. See the **Drawing** chapter for more details

Layout

Set information, customer name, quote number etc. to describe the layout

Customer name

Quote number

Revision

Description

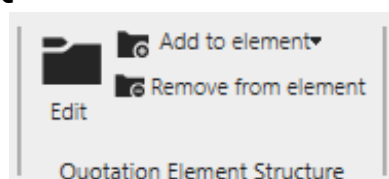
Properties

Created

14 December 2015 10:11

Picture: Layout properties window

Quotation Element Structure ribbon group



When the layout is exported to the FLQT software it is sometimes a need for grouping conveyors together. This can be done from the **Quotation Element Structure** ribbon group.

Add to element

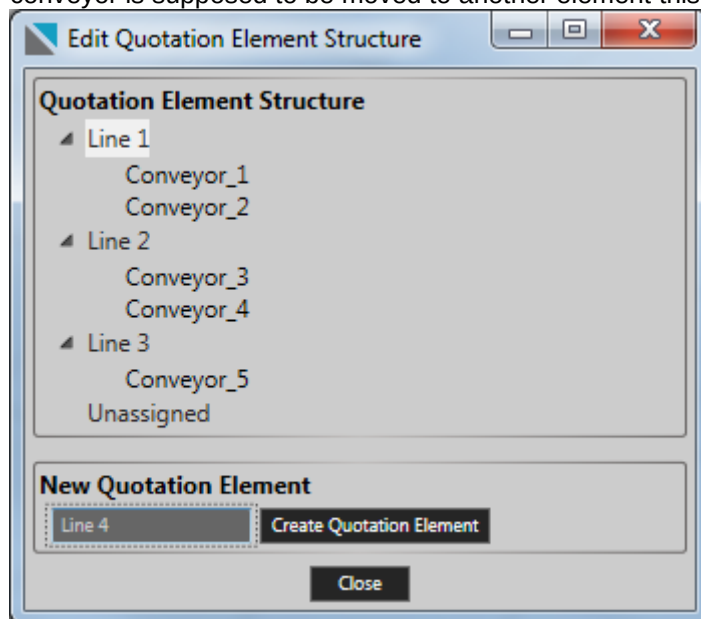
First of all the command **Add to element** can be selected. This is a drop down menu with some further options. If no groups have been created before the only possible choice is **Add to new quotation element**. Then the name of this new group must be specified. If no conveyor is selected there is still a feature for adding a new quotation element. Next time this command is selected the user has the choice of adding the conveyor to either one the old element or once again adding a new element.

Remove from element

If a conveyor is selected and that conveyor is already a part of an element the **Remove from element** command can be triggered. This removes this conveyor from the element. Later when the layout is exported further these groups are visible in the BOM structure. Conveyors that are not part of any group are categorized as not **Unassigned**.

Edit

The Edit command opens a window where the user modify the quotation element structure. New element could be created from the **New Quotation Element** section. Elements could be renamed or removed by right click commands. Conveyors could also be renamed by a right click command. If a conveyor is supposed to be moved to another element this is done by drag and drop.

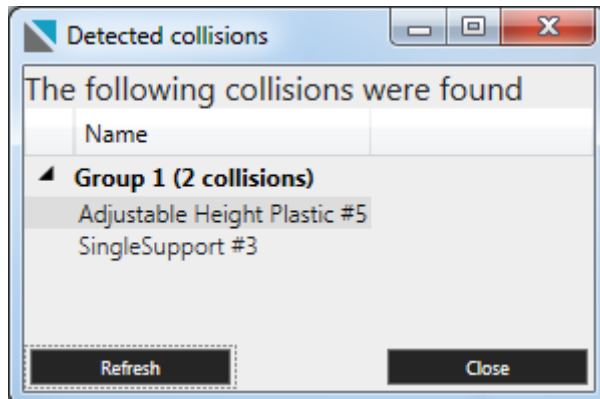


Picture: Edit Quotation element Structure

Check For

Collisions

This feature helps the user to find overlapping geometries regards to guide rails and conveyor supports. When the commands is run it returns with a list of fould collisions.



Picture: Collision detect, Detected collisions

Defaults

Top of chain

The 3D world of the layout has its own coordinate system. To help the user build conveyor there are a property in the Layout window called **Top of chain**. This is the level (Z-axis) where the components are located when they are added to the layout. This value is the layout default value and is stored for the complete layout. This means every conveyor component that is added to this layout is put to this height. The default (layout) top of chain value is 900 mm. This means when a new layout is created it automatically gets this default value as the layout top of chain value. This value can be adjusted for individual conveyors only. See the Conveyor chapter for more details.

Frequency

The frequency setting is giving the drive configuration the info of either 50 or 60 Hz to fit different geographical demands on the drive units. This setting is covering all of the motors in the opened layout. It is not possible to have motors with different frequency setting in the same layout.

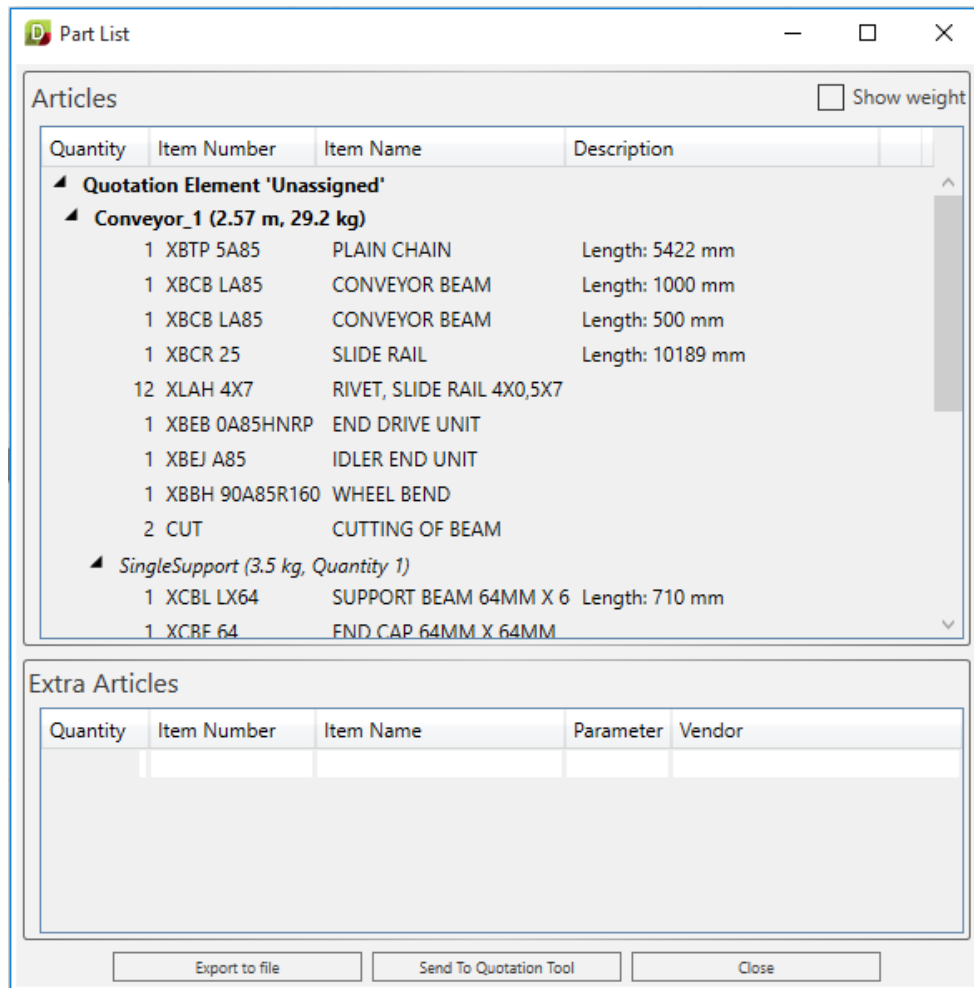
Default frequency (Hz) (File menu/Options/FlexLink Layout settings)

When a new layout is created this values is set by the default frequency in the Option FlexLink Layout settings, from the File menu.

Bill of material



After finishing the layout to required level it is sometimes needed to generate a Bill-Of-Material. This command is located in the Layout menu in a separate ribbon group and it opens up a Part List popup window. If any conveyor in the layout contains errors (e.g. not completed) the BOM can't be generated and there will be a warning note to the user.



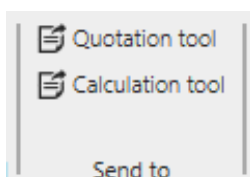
Picture: Part List window (Bill-of-Material)

The top level of the structure in the BOM is all the quotation elements. These corresponds to the groups defined by the user (**Quotation Element Structure** ribbon group). Each quotation element contains its conveyors. All conveyors contains a number of FlexLink articles but also conveyor supports and guide rails bracket kits.

Articles that isn't included in the layout can be manually added into the Extra Articles window. From this window it is possible to send this information further to the quotation or the calculation tool. This is the same actions as in the Send to ribbon group (below).

A weight is calculated for each conveyor, and to see exact weight for each separated article, check the **Show weight** check box. If there are any articles that does not have any weight, then no weight will be showed for the article and the conveyor that contains it.

Send to

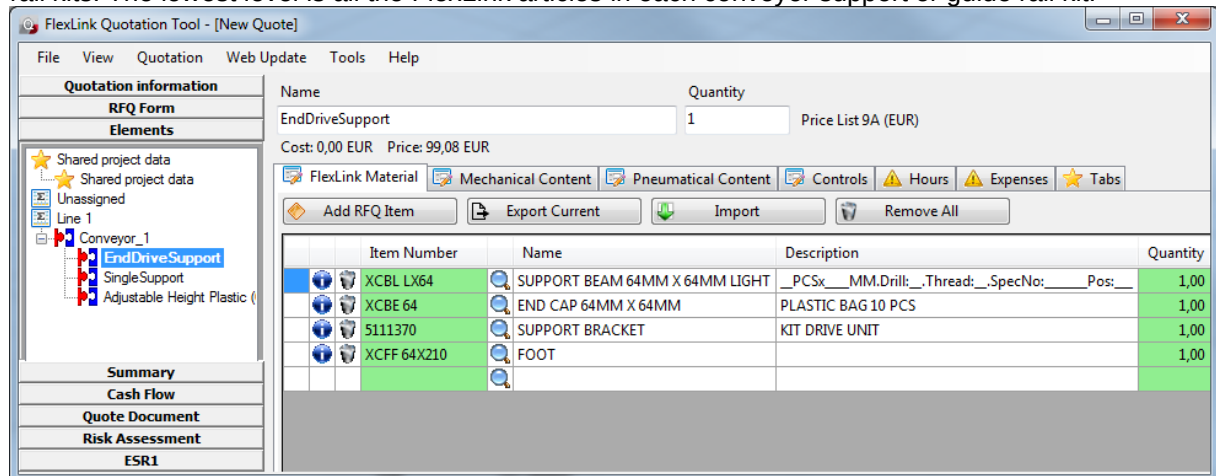


In the Send to ribbon group there are two buttons to export the layout to other tools in the FlexLink

engineering tools suite.

Quotation tool (FLQT)

The first alternative is to send it further into the FLQT, FlexLink Quotation Tool. This button is greyed out if the application FLQT isn't launched. In FLQT the articles will be sorted into three different sections. The top level is the quotation element structure. One step lower is all the conveyors. Beneath this are all FlexLink articles for each conveyor together with the conveyor support and guide rail kits. The lowest level is all the FlexLink articles in each conveyor support or guide rail kit.



Picture: FlexLink Quotation Tool, after import from FlexLink Design Tool

Calculation tool (FLCT)

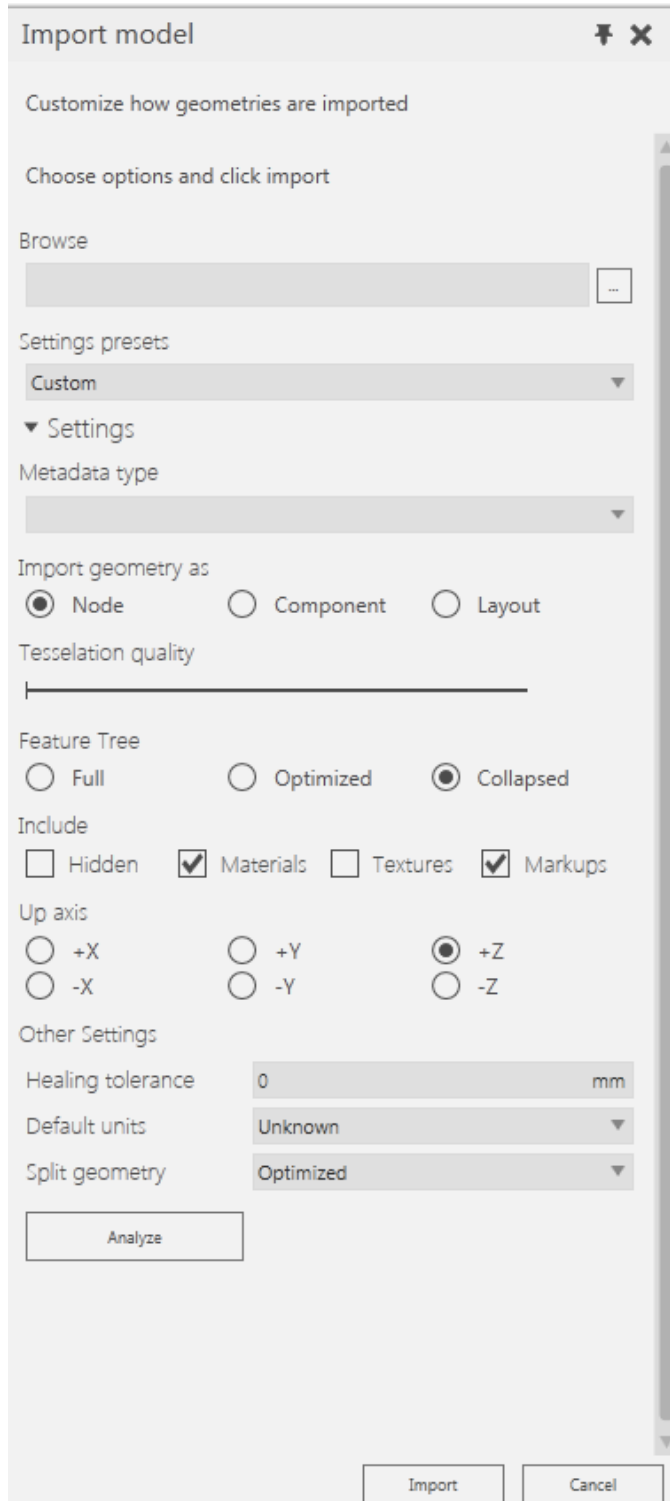
First alternative is to export it into an xml file for further use in the Quotation tool. All conveyor will be converted to FLCT components. Some components are not possible to export (e.g. pallet function).

Import

Geometry

This command is for importing geometries.

For information about supported formats check Appendix A.



Picture: Import Geometry setting window

Floorplan

In the software there is a concept of importing static pictures and showing them as a floor object. This object are referred to as **Floorplan**. The floorplan can be toggled on and off by switching **Show floorplan** in the view tab.

Add (Blueprint)

The **Add** command opens up a pop up window letting the user specify an image file to open. Compatible formats are dwg, dxf, jpg and bmp. It is possible to have multiple floor objects in the same layout.

Each blueprint object has a component property window which appears when the object is selected. In the Default tab there are three size properties, Scale, Width and Height. In the Advanced tab the blueprint object could be mirrored both horizontally and vertically and also be set to a degree of transparency.

There is a toggle command in the **View** menu (3D ribbon group), **Show Floorplan**, that controls the visibility of all blueprint objects. See View chapter for more details.

Export

Bill of material

This command exports the Bill of material into a text file that can be read by other tools, such as the FlexLink Online Store, FlexLink Calculation Tool.

The Bill of material can be exported in two different formats:

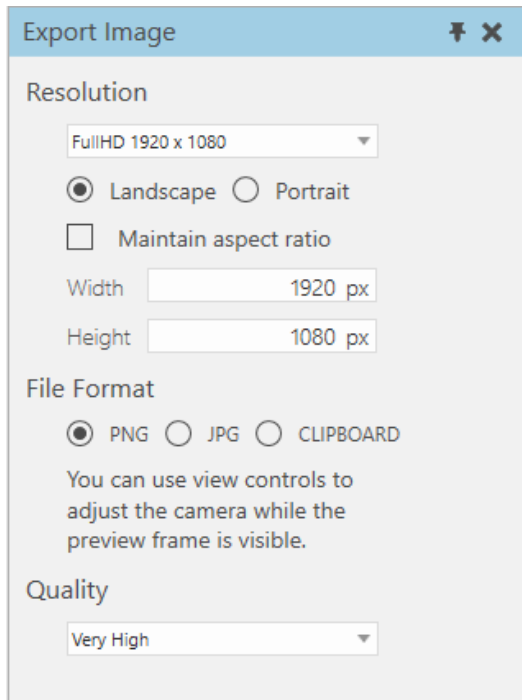
- FlexLink Format. Choose this format when exporting to other FlexLink Tools as it contains more information than the other format.
- Excel format, a common semi-colon separated list of items

Geometry

This command exports the 3D-world into multiple different formats. The user can choose to export all components or only the ones selected.

Image

The Export Image command is opening a command window for generating a bitmap picture of the 3D world. The **Export Image** button requires the user to fill in the file name and destination before the picture is generated. The **Clipboard** format is simply copying the same picture to the clipboard instead, for further use in other tools.

A screenshot of the "Export Image" dialog box. It has a title bar with a maximize icon and a close icon. The dialog is divided into sections: "Resolution" with a dropdown menu set to "FullHD 1920 x 1080", radio buttons for "Landscape" (selected) and "Portrait", a checkbox for "Maintain aspect ratio" which is unchecked, and input fields for "Width" (1920 px) and "Height" (1080 px). The "File Format" section has radio buttons for "PNG" (selected), "JPG", and "CLIPBOARD". Below this is a text instruction: "You can use view controls to adjust the camera while the preview frame is visible." The "Quality" section has a dropdown menu set to "Very High".

Export Image

Resolution

FullHD 1920 x 1080

☒ Landscape ☐ Portrait

☐ Maintain aspect ratio

Width 1920 px

Height 1080 px

File Format

☒ PNG ☐ JPG ☐ CLIPBOARD

You can use view controls to adjust the camera while the preview frame is visible.

Quality

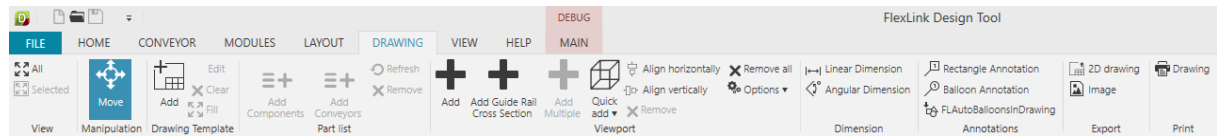
Very High

Picture: Export – Create image (bmp)

Drawing

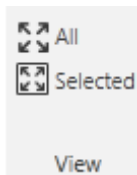
The drawing menu contains a lot of feature for the 2D drawing export. All actions done here are only affecting the 2D drawing area and not the rest of the tool. As soon as the drawing menu is selected the view of the tool changes to a 2D area instead.

Many of the features will show a Snapping window that makes it easier to snap the dimensions or balloons to certain points in the drawing. See the Snapping tool section above.



Picture: Drawing menu

View



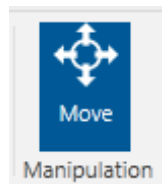
All

Fills the drawing window with the drawing template.

Selected

Fills the drawing window with the selected viewport(s).

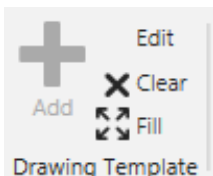
Manipulation



Move

When move is activated the selected viewport can be moved.

Drawing Template



When the drawing menu is selected the 2D area is complete blank. To add a drawing template the user must press **Add**.

Add

When working with the drawing features the first action has to be to add a default empty drawing.

This is done by the **Add** command. Before the drawing is created the user has the option to fill in some data in the **Add drawing** window.

Picture: Add drawing window

First section of the **Add drawing** window is the **Drawing type**. Here it is possible to set which drawing standard that should be used (ISO or ANSI), paper format (A0, A1, A2 or A3), projection and the scale. The alternatives in the projection are First angle (IOS) or third angle (ANSI). In the **Drawing properties** section there are a lot more field with information which is the data used as the text visible on the drawing export.

Edit

The **Edit** command is used for opening the **Add drawing** window again in order to change a value or just to check the content.

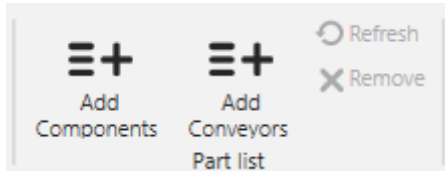
Clear

It is possible to remove the drawing completely by the **Clear** command.

Fill

The Fill command is exactly the same as the corresponding command for the 3D world. It adjust the 2D view in order to show all objects within the 2D area.

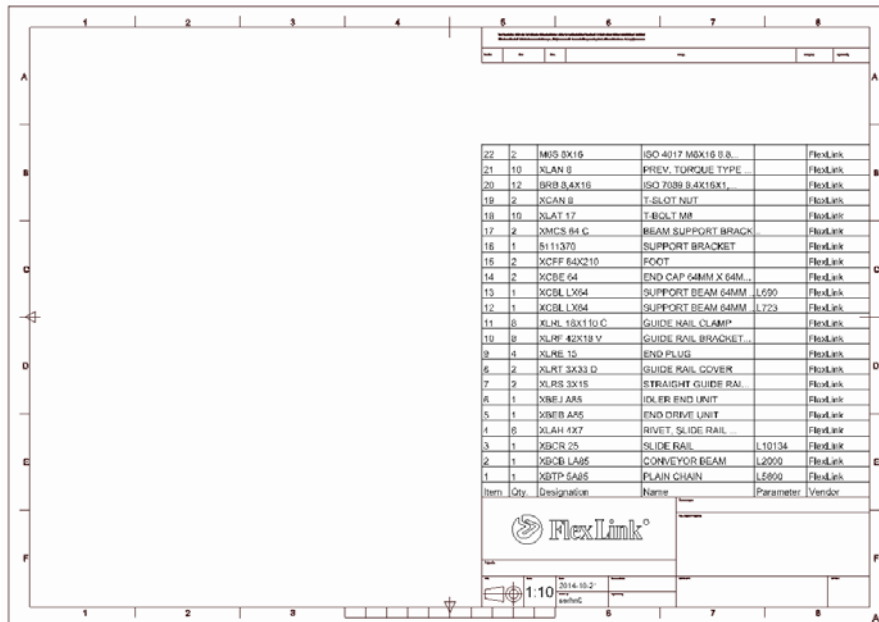
Part list



There are two types of part list, each described below. Only one part list can be added to one drawing.

Add Components

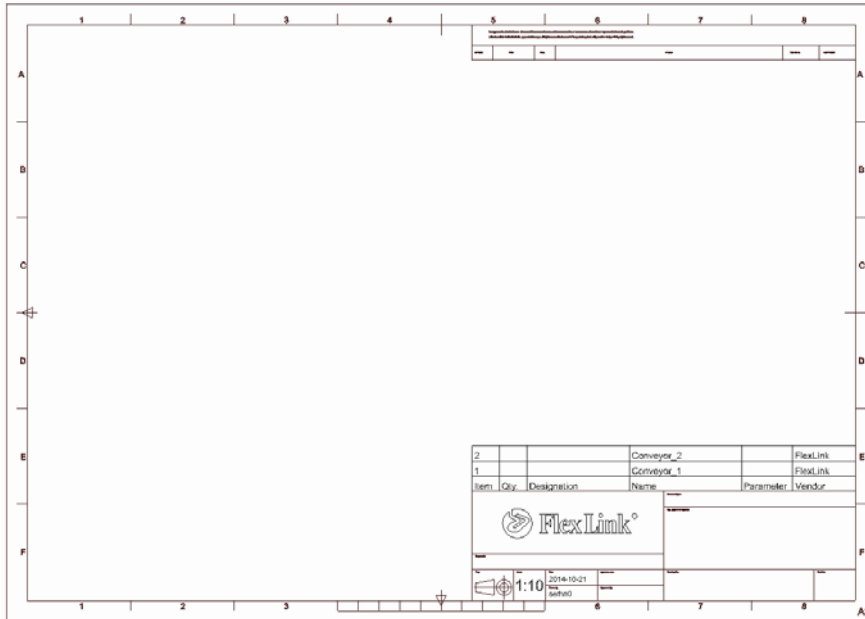
There are two different types of drawings depending on the part list. The first command to add a part list is the **Add components**. This command prints all of the components in the layout into a part list summary on component level.



Picture: Add Component – Part list

Add Conveyors

The other command for adding a part list is the **Add Conveyors** command. This creates a part list on the top level, conveyor level without any sub content.



Picture: Add Conveyors – Part list

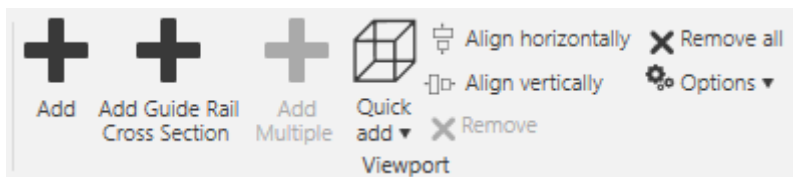
Refresh

The refresh command is rereading all info from the bill of material and updates the part list on the 2D drawing.

Remove

The remove command is completely removing the part list from the 2D drawing.

Viewport



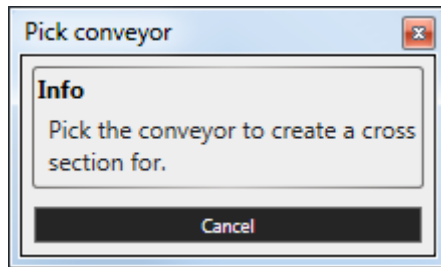
The Viewport ribbon group contains features for adding drawing snapshots from the 3D world into the 2D drawing.

Add

The **Add** command is used for creating a momentary snapshot from the 3D world and paste it to the drawing. When the command is triggered it automatically switches to the 3D world waiting for the user to define a snapshot area. It is possible to rotate and pan until required view before the area is selected using the left mouse button. Everything that is in the frame will be visible in the drawing. The snapshot view is then positioned somewhere in the 2D drawing and can be moved by just selecting it and drag it to correct position with the mouse.

Add Guide Rail Cross Section

There is also a function for adding cross section views of the guide rail bracket kits used in the layout. When this command is triggered the view is automatically changed to the 3D world with a pop up window informing the user to select which conveyor that should be used for the cross section.



Picture: Pick conveyor window – Add Guide Rail Cross section

When this is done the generated cross section is put on the 2D drawing and can be moved to required position.

Add Multiple

This command adds multiple view ports to the drawing template. The user is first asked to point out the top surface of the view port. Then the three different view port areas (top, front and side view) has to be defined. Finally these three view ports are created in the drawing area, located according to the projection setting (first angle or third angle).

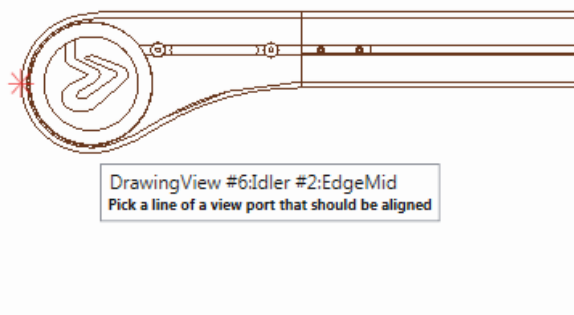
Quick add

The quick add gives the user the possibility to add a viewport for the whole 3D-view. Available options are:

- Top
- Bottom
- Left
- Right
- Front
- Back

Align horizontally

Two view ports (snapshots from the 3D layout) can be aligned both horizontally or vertically.



Picture: Align – Selection of a view port point

Two points has to be selected and the view point of the second point will be moved in order to align the two view points horizontally. The command symbol at ribbon group will be colored as long as the align function is started but not completed.

Align vertically

This command is exactly the same as the horizontally but align vertically instead.

Remove

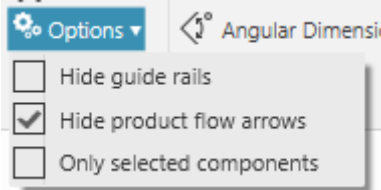
This command removes the view port from the drawing. This could also be done by selecting it and pressing delete on the keyboard.

Remove all

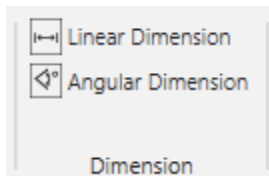
This command removes all viewports in the drawing

Options

In options the user can chose to hide the guiderails and the product flow arrows on the viewports. Please note that the setting will not affect viewports that already are in the drawing. The user can also change what should be added for **Quick add**. If **Only selected components** are selected only the selected components will be added when using quick add.

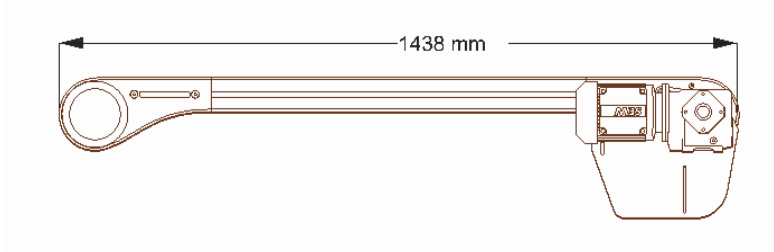


Dimension



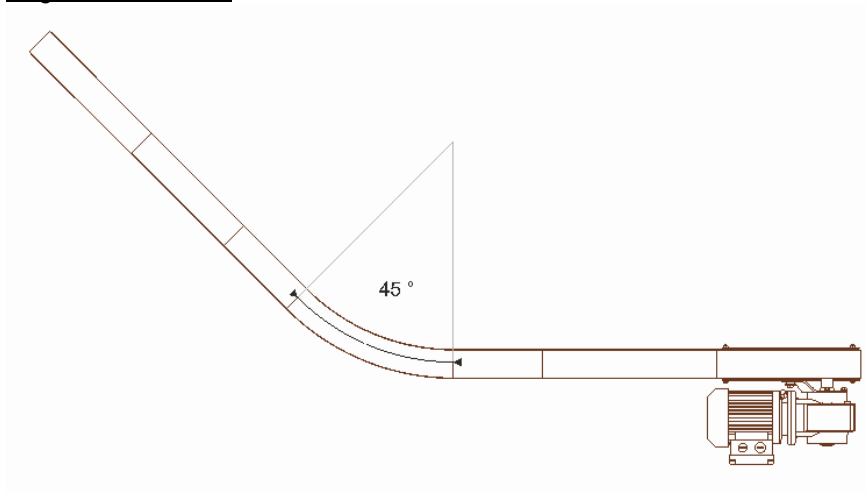
Linear Dimension

The dimension tools are also using the picker tool for selecting how the dimensions should be set.



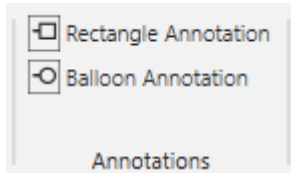
Picture: Linear Dimension, Example

Angular Dimension

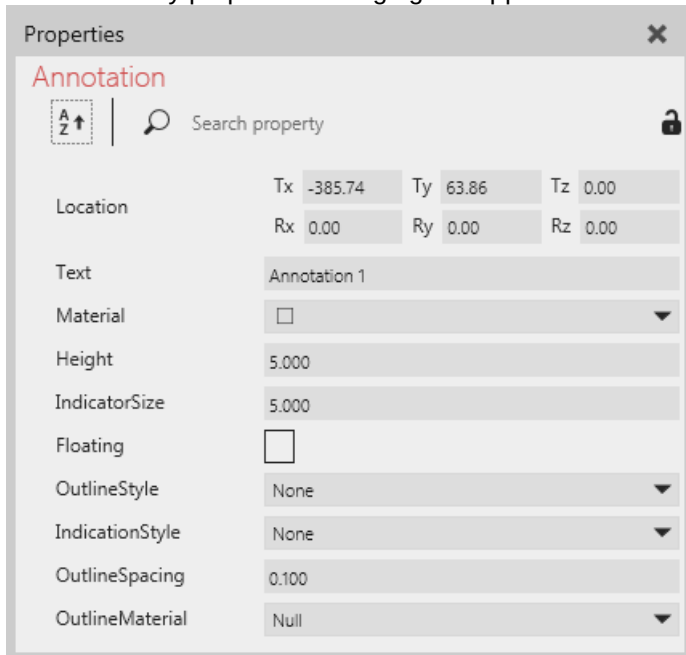


Picture: Angular Dimension, Example

Annotations

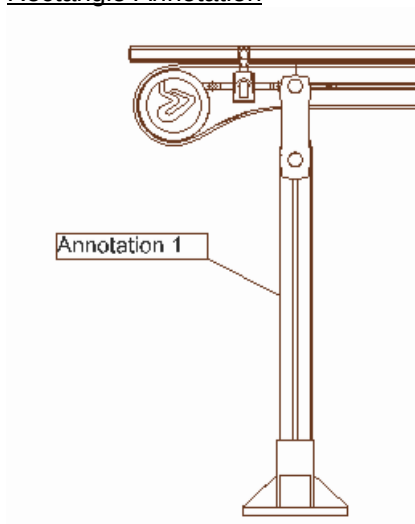


There are two types of annotations, rectangular or balloon. When they are added they both have a number of properties that could be changed through the Properties window. Besides the annotation text it is mainly properties changing the appearance of the annotation, see the picture below.



When one of the annotation commands is triggered the symbol in the ribbon gets colored indicating that the annotation function is active. The user can add multiple annotations of that type without triggering the command again but the function has to be turned off when all annotation have been added.

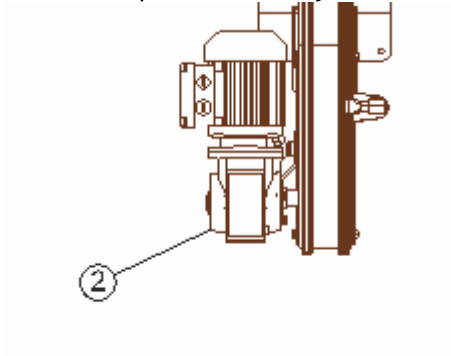
Rectangle Annotation



Picture: Rectangle Annotation, Example

Balloon Annotation

When the balloon annotation command is triggered all components in the view point will get small cross symbols. If any of these are selected with the mouse this component will automatically get a balloon annotation with referencing to the part list. There is also possible to click on any other object in the view port to manually add a balloon annotation.



Picture: Balloon Annotation, Example

Export

Geometry

This command exports the drawing into multiple different formats. The user can chose to export all components or only the ones selected.

Image

This command exports the drawing into a picture.

Print

Drawing

This command prints the drawing.

Simulation / visualization

The Design tool has a lot of features for simulating the conveyor lines. Even though the objective has been to reach as close to the real behavior as possible all simulations are an estimation of the reality. In many cases the objective with this part of the Design Tool is only to show briefly what can be done and give a 3D appearance of the running line. In this case it is very easy to create a simulation but then the term visualization is more proper to use. In this document the word simulation will be used even if sometimes the word visualization would better describe the feature. In order to get a proper simulation of the line a little more work has to be done by the user. If the line includes routing it is important to do an effort to make the simulation as accurate as possible. This routing concept and all its components will be described thoroughly in this chapter. Besides running the simulation in the software itself it could also be exported into a 3D-pdf.

Routing concept

In real applications there are sometimes requirements of route control. This often means that the products has individual ID numbers which defines where in the line this special product should be headed. In most cases this is implemented on pallet or puck systems but this is not mandatory. In FLDT there is a lot of virtual components for implementing route control. All of them are described separately in this chapter.

Process points

The basic idea behind the routing is to have a line with several destinations (position in the line) that processes the products in any way and every product has to reach these a certain predefined order. All these locations that can process the products are called process points and they can have one or more **ProcessIDs** connected to it. This means this process point can do all these operations. The **ProcessID** isn't necessary unique to one process point. Several process point can be set to the same **ProcessID** if all of them can do the same operation. In this way it is possible to create parallel production flow.

Route list

A route is a list of destinations (process points) that has to be reached in sequence. The individual products gets a route from the FlexLink Initialization component. Every individual products has also a dynamic pointer to the next destination of this predefined route. Some components in the **eCatalog** has a process point functionality. This means it has a **ProcessID** connected to it. When a product has its next destination matching with the process ID it is processed. When the product leaves the process point it has updated its own next destination from the predefined route. If the route is not changed and the product has reached the last destination the pointer is set to the first destination and the product is beginning a new cycle on the same route.

Decision points

In order to be able to implement the routing concept the line has to have some positions where the products can be separated from the main line and in to a satellite line. These positions are called decision points and is typically a diverter. The concept of a decision point is that it has two output conveyors. One of the output conveyor is thought of as the main conveyor and the other one as the satellite conveyor. The parameter **DivertID** can be filled in with arbitrary number of **ProcessIDs**. If the arriving product has a matching next destination to any of the **ProcessIDs** in the **DivertID** it will release it to the satellite conveyor. Otherwise it will release it to the main conveyor.

Simulation components

In the **eCatalog** there is a folder called **Simulation Toolbox**. This contains all of the simulation components and are placed in two sub folders, **Feeders** and **Generic**. They can also be found under the specific platform folders.

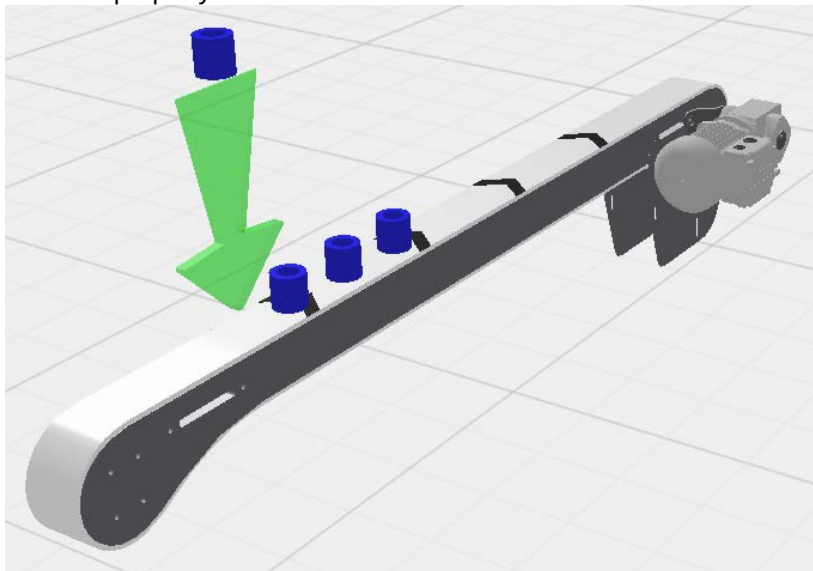
All of the arrow shaped components are only virtual components and they all have a parameter

IconSize where the size of the colored symbol can be changed. Also they have a visible option if it should be shown at all.

FlexLink Feeder

The feeder component is used for creating products onto a conveyor. In the Product property tab it is possible to change setting for the product size (height, width and length) and shape (box, cylinder, roll, bearing, bottle, pallet, container, cone or semisphere). It is also possible to set the behavior of the creation of products. In the Process property tab the first property, **Feeding Interval**, can be set to either **PartsPerMinute** or **Interval**. Depending on this first choice the value is then defined in the second property. The next property is called **Feeding Mode** and is by default set to continuously. If the feeder should have a max limit on the created products this property should be set to **MaxNumber**. In this case another property is shown, **MaxNumber**, which defines the total number of created products.

All of the created products will have the color blue. This is the default color (Null) and indicates that it hasn't got any route (see the **Initialization Point** component). The color could be changed with the **Material** property in the **Default** tab.



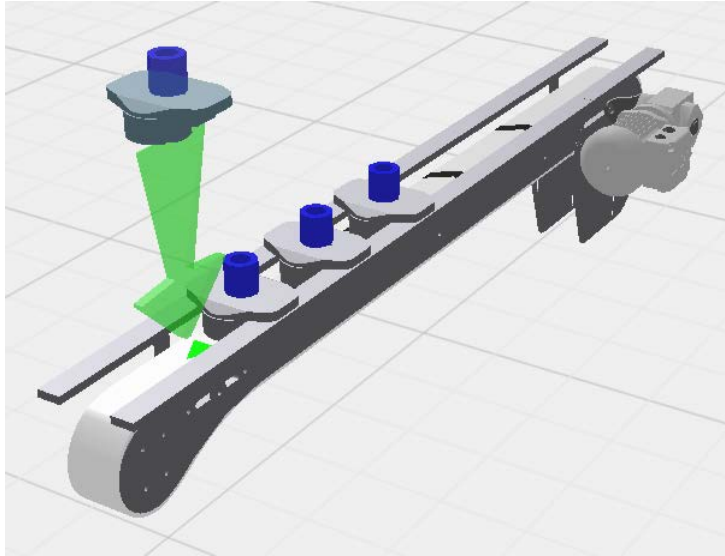
Picture: FlexLink feeder

FlexLink "platform" Pallet Feeder

All puck and pallet platforms has their own version of the feeder component. In the FlexLink property tab it is possible to set the **PalletType**. This specifies which type of puck/pallet that will be created. If RFID is required this option (**RFID tag**) should be selected. This has only an impact on the BOM. If the created pallets/pucks should be a part of the BOM this is fulfilled by selecting the property **IncludeBOM**. If this property is selected it is included in the BOM the **AmountOfPallets** sets the quantity. It is possible to select the product that is going to be transported on the puck/pallet in the same manner as for the feeder component.

For FlexLink X45e Pallet Feeder it is an option if the fixture should be visible (**IncludeFixture**). This is only a transparent generic top part of the puck. This part has no impact of the BOM.

All of the created products will have the color blue. This is the default color and indicates that it hasn't got any route (see FlexLink Initialization component).

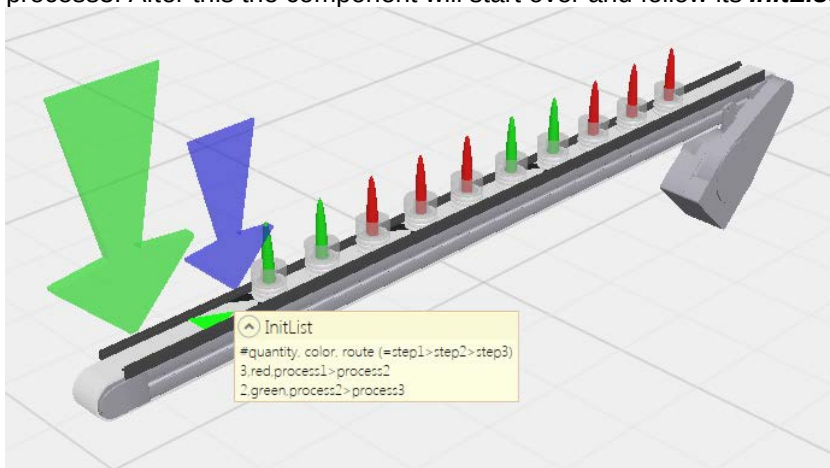


Picture: FlexLink X85 Pallet Feeder

FlexLink “platform” Initialization Point

All of the puck and pallet platforms has an initialization component. This is used for defining routes for each puck or pallet. The component is setting a route (sequence of destinations) to all passing pucks/pallets that doesn't already have a route. The implementation of this is done via an **InitList** where the user specifies the amount of pucks/pallets of a certain color and the route for this type. This can be repeated for other colors. The pattern is starting over after the component has received the total amount for the **InitList**. The **InitList** appears in the 3D layout when the **Initialization Point** component is selected. It is possible to change directly in this list (text file).

In the example below the first three pucks will be marked with the color red. If the upstream feeder component is set to not include products it will be the puck itself that will be color marked. These red pucks/products will get the process1 as their first destination process point. After process1 it will try to reach process2 and then start over. After these three red pucks the **Initialization Point** component will mark two pucks green which will have the destination route process2 and then process3. After this the component will start over and follow its **InitList** repeatedly.



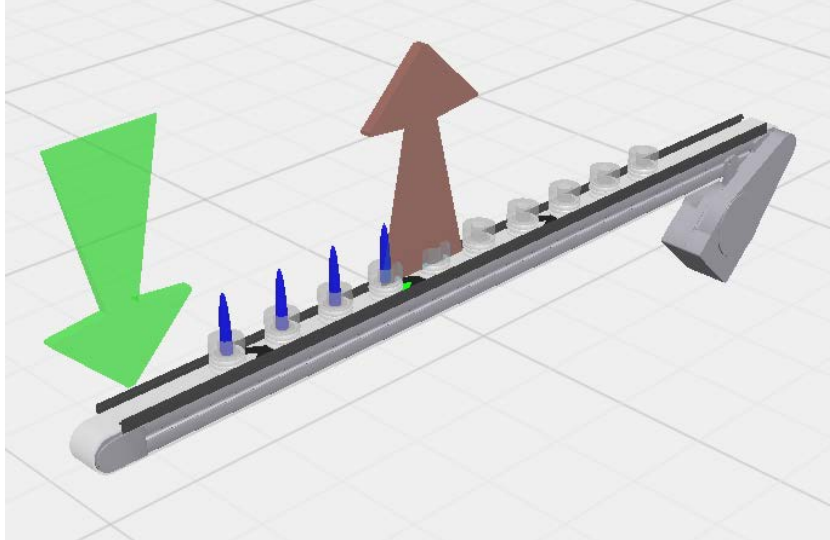
Picture: FlexLink X45e Initialization Point – InitList

The Initialization Point component has also got a **ProcessID** with the default value Init. This means all pucks with either no destination or the destination Init is going to get a route from the **InitList** of the component. If a route is ended with a **processID** corresponding to an Initialization Point component pucks with this route will complete all process steps and then go the Initialization Point

component to receive a new route. This route will then overwrite the previously executed route. All available colors for the **InitList** are listed in Appendix D

FlexLink Remover

The remover component deletes all products on a specific point of the conveyor. The property Remove can be set to either All or BasedOnID depending on the requirements. The BasedOnID option is only removing the units with matching destination compared to the parameter ProcessID. When selecting the RemovePallet property the complete pallet is removed from the line.



Picture: FlexLink Remover

FlexLink Changer

The FlexLink Changer component is used for modifying the products at a certain point in the line. In combination with this the changer component could also change the product shape and size.

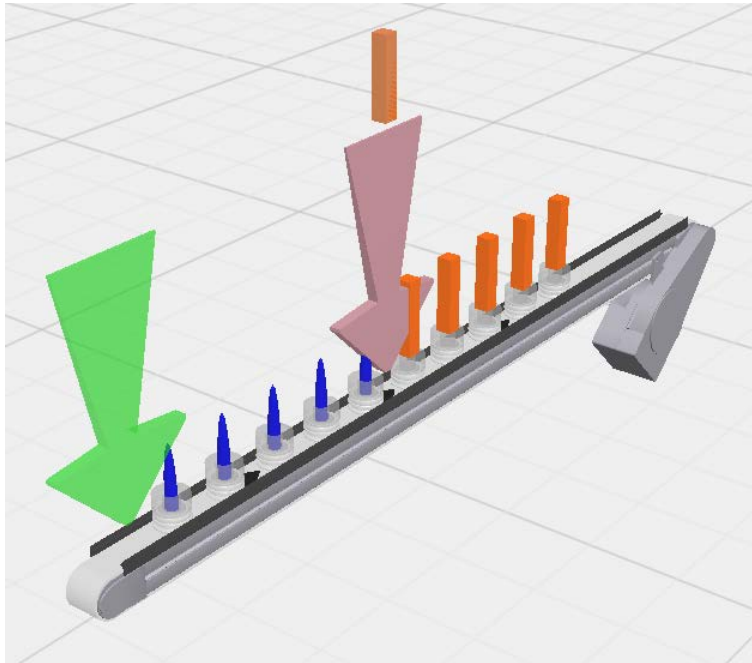
The **Change** property have two options, **All** or **BasedOnID**. If it is set to **BasedOnID** the process parameter **ProcessID** controls which products that is going to be a part of this changer behavior. Products without matching **ProcessID** as the next destination will just pass by the changer component. If all products is going to be processed the Change property should be set to **All**.

For the **Behavior** property there are three alternatives.

The simplest behavior is the **One-to-One** variant. When this variant is set the component doesn't change the number of products at the component. It could have a delay on the products before they are released from the component. This is set with the process parameter **ProcessTime** (s)

If the property instead is set to **Batching** the component waits until a certain number of products has reached the component before it releases one new. The number of products in the batch is set with the process parameter **BatchProducts**. If the **RemovePallet** checkbox is set all pallets that are not used will be removed.

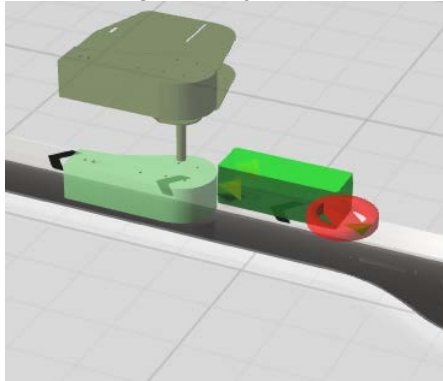
The splitter function is the opposite of the batcher. On every product that enters the component it releases a number of new products. The process parameter **AmountOfProducts** controls this number. The time between the new products can be set with the process parameter **ProcessTime** (s). The splitter variant is only possible for non-pallet/puck functions.



Picture: FlexLink Changer (One-to-One, Replace)

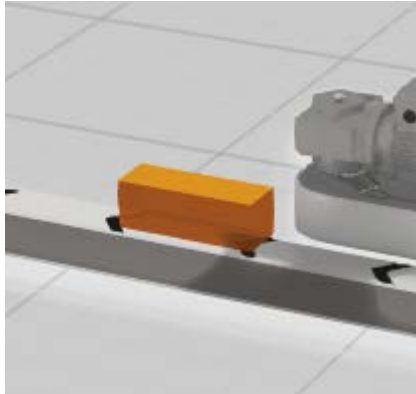
Wedge Connector

The Wedge Connector component is a help component used for building wedge conveyors. It has two main functions depending on the FlexLink tab property **ConnectMode**. If the component should grab products from a straight beam conveyor and put it onto the wedge conveyors (two opposite conveyors perpendicular to the in conveyor) the **ConnectMode** should be set to **ToWedge**. In this case the virtual **WedgeConnect** component will be green. Actually it is only transferring the products to one wedge conveyor but the user needs to draw two conveyors for the real application.



Picture: Wedge connector – ToWedge

The other option in the **ConnectMode** property is **FromWedge** and should be used when the products should be placed back from the wedge conveyors to the horizontal conveyor. This option is indicated by the orange color on the **WedgeConnect** component. The wedge connector could then be connected to a horizontal conveyor of any platform. The width of this platform becomes the default **ProductWidth** when it is connect. Then this value could be changed to appropriate value. This platform is also controlling what components will be added using the building handles. If another platform is to be used for the wedge (compared to the horizontal conveyor) the **WedgePlatform** property could manually be set to required platform. There are only certain platform that have the possibility to build wedges depending on if they have necessary cleated chain.



Picture: Wedge connector – FromWedge

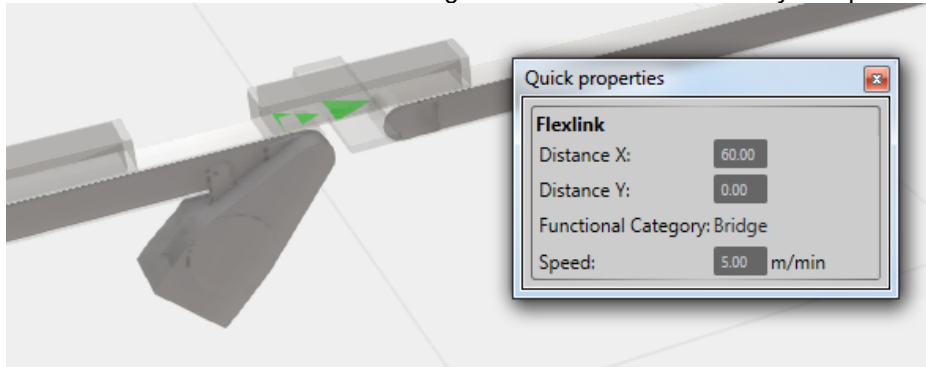
The wedge connector component is found in the accessories folder under all of the compatible platform folders.

The **HeightOffset** is controlling the space between the chain of the horizontal conveyor and the lowest part of the chain on the wedge conveyor. This is by default set to 10 mm.

There is a more detailed example of how to build a wedge conveyor in the appendix.

Virtual bridge

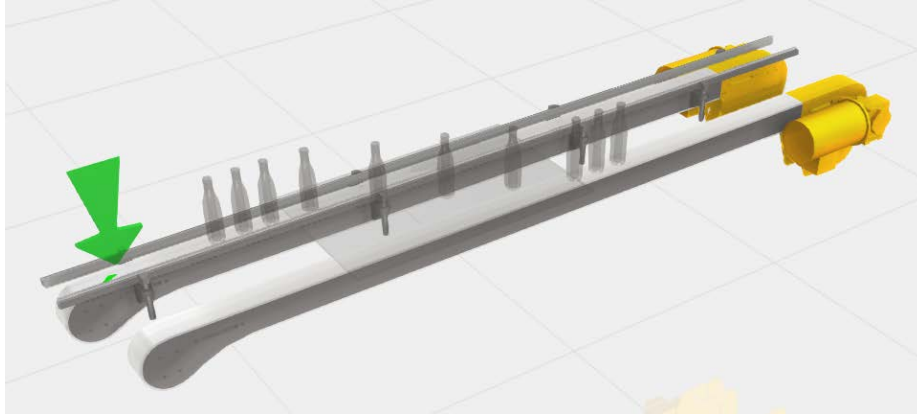
The virtual bridge component can be used for simulations where products needs to be moved from one conveyor to another. It is a virtual product in the sense that it has no geometry or BOM. The **Distance X** is the in-line space between the two conveyors. The **Distance Y** is the side transfer distance. It is set a zero as default. The Speed has to be set to a fixed value. It is not inherited from the connected drives. The virtual bridge can connect to almost any component end.



Picture: Virtual bridge, example

Virtual side transfer

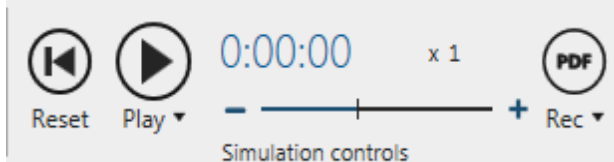
For transferring products in the middle of a conveyor beam to another parallel conveyor beam the side transfer can be used. It has also the same properties as the virtual bridge, **Distance X**, **Distance Y** and **Speed**. When guide rails are added to these conveyor it should follow the product path from one conveyor to the other.



Picture: Virtual side transfer, example

Simulation controls

The Simulation controls ribbon group is located in the **Help** menu. It contains commands, settings and information concerning the simulation of the layout.



Picture: Simulation controls ribbon group

Reset

The reset button is sets the playback time to zero.

Play

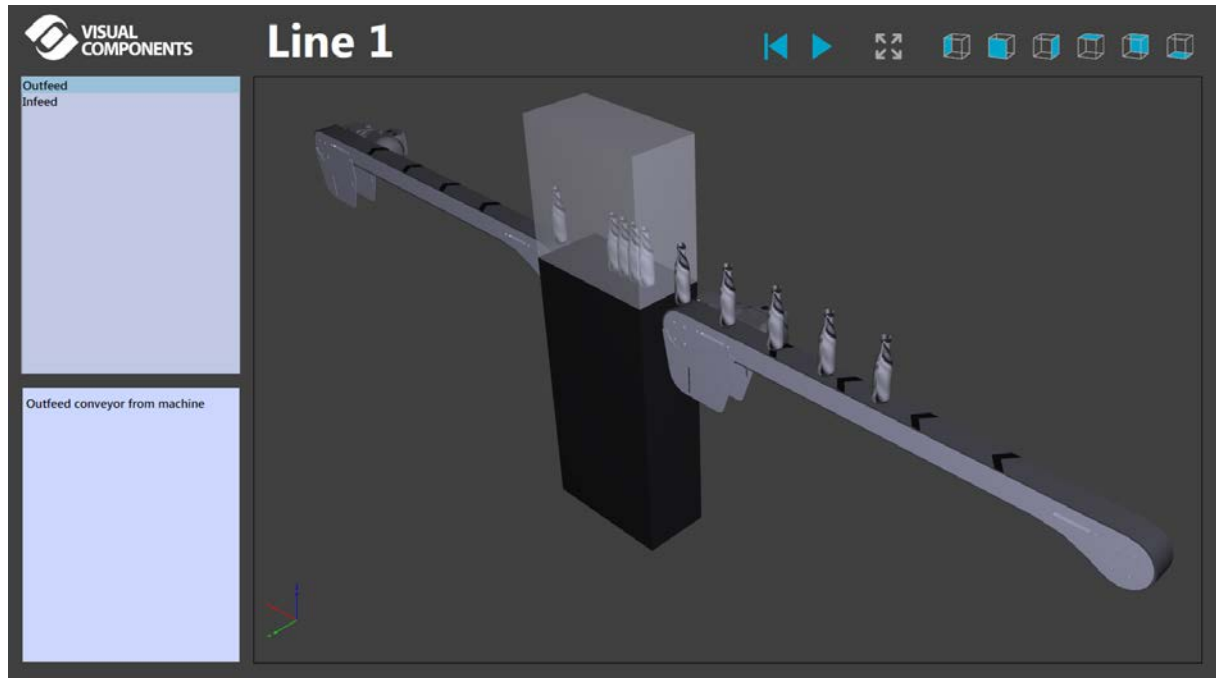
The simulation playback is started with the play button. One exception is when the playback should be recorded. Then the PDF button is used, see below.

Playback time

There is also an indicator showing the current time of the simulation playback. Next to this is an indicator which it is possible to change the playback simulation speed. The same simulation speed is also shown as a speed bar below the simulation time. This can be changed with the plus and minus symbols.

PDF

When the PDF button is pressed the user is required to specify the output file name and folder. After this is done the simulation starts automatically. The recording of this simulation is stopped by pressing the pause button or the PDF button once again. Then the 3D-pdf file is generated based on the recorded simulation.

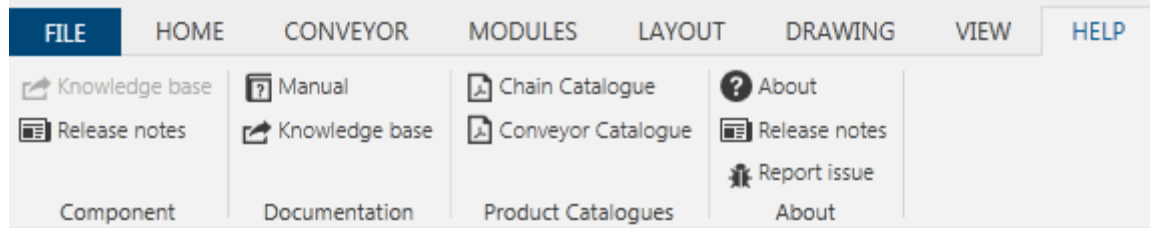


Picture: 3D-pdf, example

To open the 3Dpdf file it is recommended to use Acrobat software. When the file is opened it starts the simulation immediately and loops after completion. The 3D window could be changed with either the view icons at the top right part of the screen or by the mouse control. Left click is used for rotating, right click or wheel is used for zooming and both left and right buttons is used for panning. Beside the simulation view it is two information windows describing the conveyor names (above) and descriptions (below).

Help

This software tool covers a wide range of different features and the need of further information is often required. In these cases the Help menu can be very convenient. Here it is possible to find information about this software but also information concerning the FlexLink product assortment.



Picture: Help menu

Knowledge base components

If a knowledge base article is available for a component this button will be available when the component is selected.

Release notes

This button opens up the latest release notes for the components

Manual



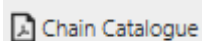
This document could always be reached from within the software. It is built in in the installation so it is possible to use even without an Internet connection.

Knowledge base



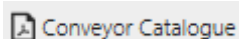
This is a link to the knowledge base system.

Chain Catalogue



The first link in the Product Catalogues ribbon group is the Chain Catalogue. This opens up the links and chain section for the platform of the selected conveyor component.

Conveyor Catalogue

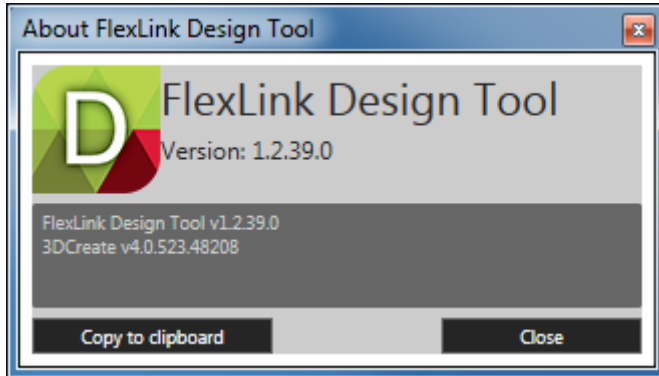


If a conveyor component is selected it is possible to open the catalog from the Platform ribbon group. It is only the catalog for the focused platform that is opened.

About

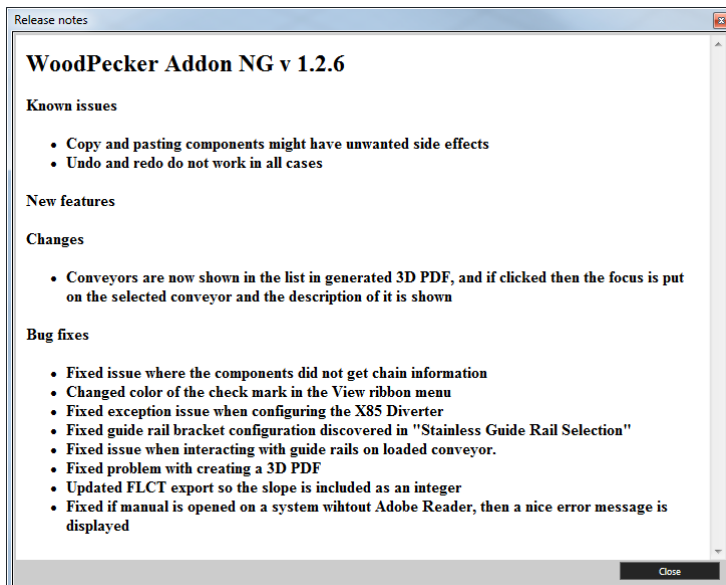
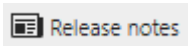


The About window is declaring which version of the Design Tool that is currently running. Also it contains the information about which version of the 3DCreate that this Tool is built on.



Picture: Help, About

Release notes



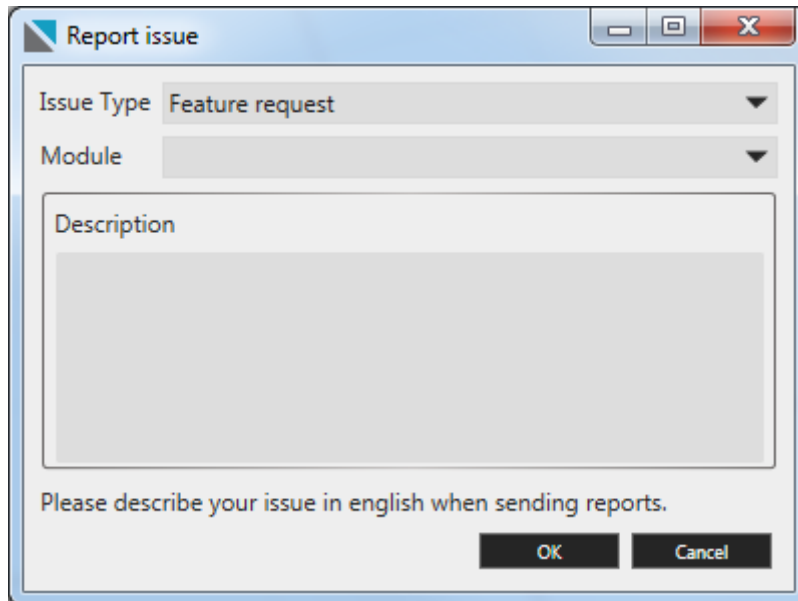
Picture: Help, Release notes – example

Reporting issue



In the **About** ribbon group there is a command for reporting found issues in the Design Tool. This information is then sent as an email for further action.

The first field is the **Issue Type** where it should be stated if the reported issue is a **Bug** or a **Feature request**. The next information to add is which module it concerns. The description area where the topic should be explained as detailed as possible. It is also recommended to add an email address for potential follow up question.



Picture: Report issue window

Visual Components Essential Help

If some topic is not covered or described too briefly in this document there might be more information available in documents (manuals, tutorials, etc.) from VC (www.visualcomponents.com). Also the question mark symbol in the top right corner of the screen (or F1) opens up a Visual component Essential Help for further reading.



Picture: Visual Component Essential Help (question mark symbol)

Abbreviations/terms

Design Tool	FlexLink Design Tool, FLDT
Quotation Tool	FlexLink Quotation Tool, FLQT
My FlexLink	Internet portal for external and internal users
Online store	FlexLink's webshop
VC	Visual Component, Company who develops the core software

Appendix A: Supported CAD file formats and versions

Native CAD readers

Format	File extension	Version (1.1.2014)	Tessellation	B-rep
3DStudio	3ds		Yes	Yes(*)
Autocad	dwg,dxf,dgn	-2013	Yes	Yes
Deneb part files		R8-R14	Yes	No
Robface	rf	R4	Yes	No
Sketchup	skp		Yes	No
STL (ascii and binary)	stl		Yes	No
VRML	vrml, wrl	1 & 2	Yes	No
(*) B-rep supported through proprietary extension to the format				

Native writers

Format	File extension	Version (1.1.2014)	Tessellation	B-rep
3DStudio	3ds		Yes	No
Autocad	dxf	R9	Yes	No
Deneb part files		R9	Yes	No
Robface	rf	R4	Yes	No
STL (ascii)	stl		Yes	No

CAD Readers based on Tech Soft 3D technology

Format	File extension	Version (1.1.2014)	Tessellation	B-rep
ACIS®	sat	-R21	Yes	Yes
Autodesk Inventor®	iam, ipt	-2011	Yes	No
CATIA® V4	model, dlw, exp, session	-4.2.5	Yes	Yes
CATIA V5	CATPart, CATProduct, cgr	R10-R22	Yes	Yes
CATIA V6	3DXML	2011-2013	Yes	No
Creo™ Parametric / Pro/Engineer®	prt, asm, neu	R13 – Creo2	Yes	Yes
I-deas / Siemens PLM Software's NX™	arc,	-NX5	Yes	Yes
IGES	igs, iges	3 & 5	Yes	Yes
JT	jt	7 – 9.5	Yes	Yes
Parasolid®	x_t	-20	Yes	Yes
Solid Edge®	psm, par, asm		Yes	Yes
SolidWorks®	sldprt, sldasm	-2012	Yes	Yes
STEP	stp	203 & 214	Yes	Yes
Unigraphics	ugprt, ugasm	Unigraphics 11 – NX8	Yes	Yes
VDA-FS	vda		Yes	Yes

Appendix B: Short cut keys

Ctrl + H:	Toggle build direction
Ctrl + Shift + A:	Select conveyor
Ctrl + shift + F:	Fill
Ctrl + shift + I:	L-distance
Ctrl + shift + R:	Positioning
Ctrl + shift + P:	Conveyor properties
Ctrl + shift + C	Configure
Ctrl + shift + Q	Quick adjust
F11	Toggle full screen mode

Appendix C: Best practice

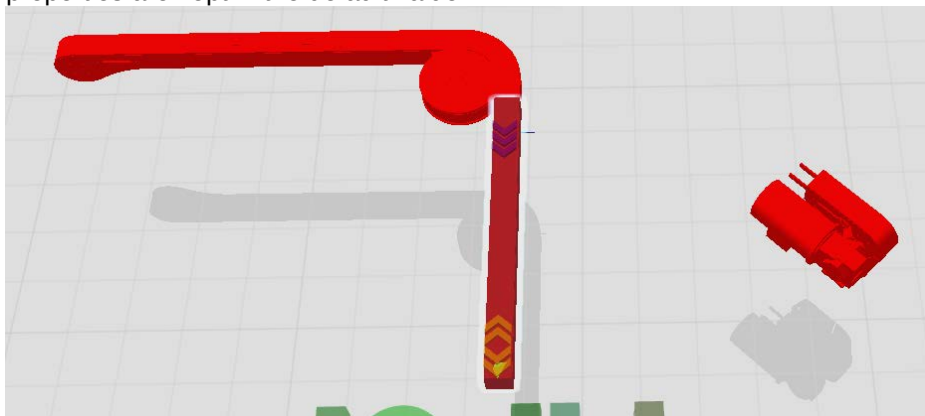
Quick adjust example

This is an example of how to use the purple triple arrow. At start the position of an Idler end and a Drive unit is known.



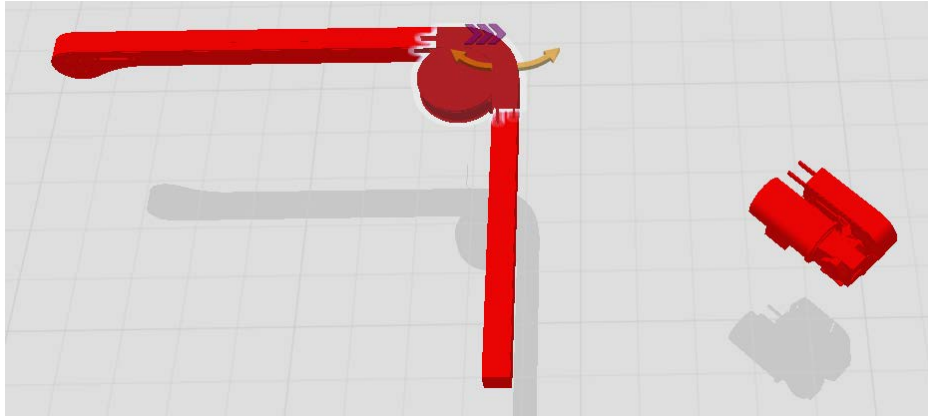
Picture: Start condition

The first step is to quickly draw the component type and order using the build handles. In this case the required order, following the Idler end, is a Beam, a Wheel bend and then another Beam. All properties are kept in the default value.



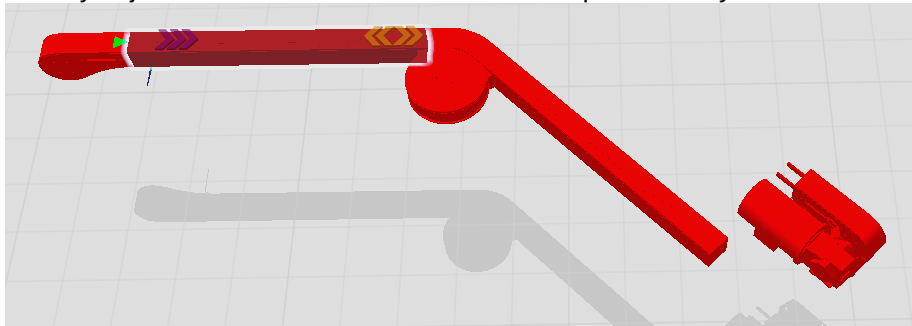
Picture: Quickly add the required components

The first component to adjust using the triple arrow is the Wheel bend, because this affects the angle. Select the Wheel bend. Make sure the handles shows on the correct end of the component. Otherwise change it with the short command CTRL+H. Select the triple arrow and then when the FlexLink Picker Tool appears select the drive unit as the end point.



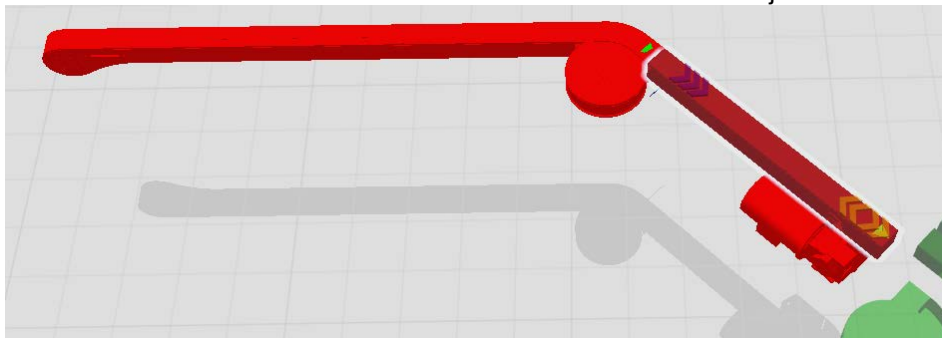
Picture: Adjust the Wheel bend using the triple arrow.

After the angle is correct it is a good solution to adjust the Beam length from the Idler and forward on. It is possible to make this in any order. Sometimes this command has to repeat once more to an already adjusted beam in order to achieve a complete conveyor.



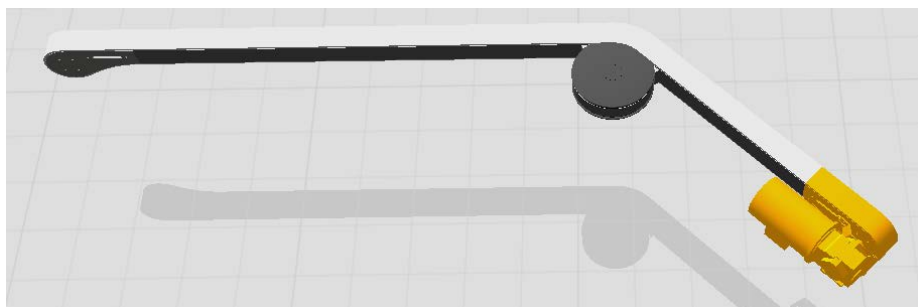
Picture: Adjust the first Beam using the triple arrow.

The final beam is not connected to the Drive unit and must be adjusted to correct length.



Picture: Adjust the second Beam using the triple arrow.

Then all components are connected, indicated by the color change.

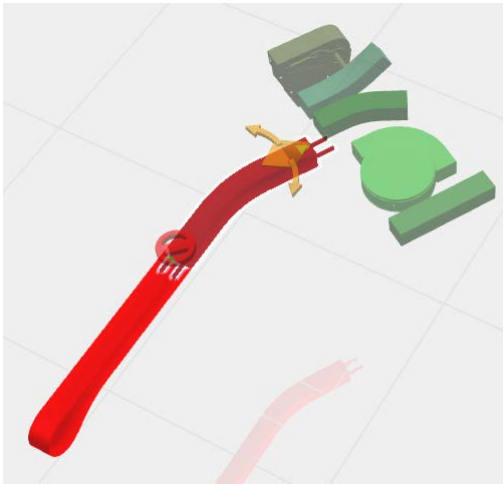


Picture: Complete conveyor

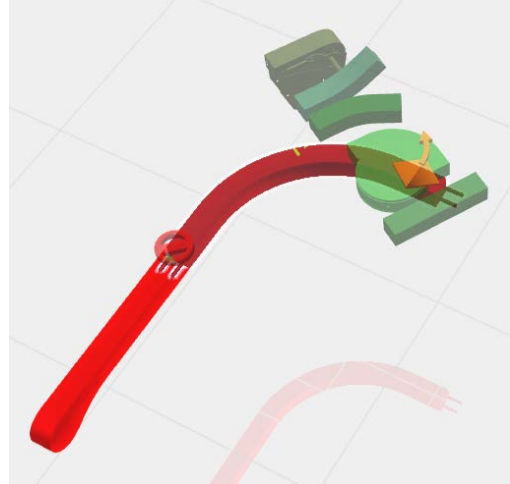
Toggle build direction examples

Example 1

First example of the Toggle build direction is a conveyor built from an Idler end to a Horizontal bend. When the bend is added the static symbol is automatically set to the end of the bend that is connected to the existing conveyor. This is the most intuitive and straight forward approach. Then when the bend angle is changed (to 90° in this example) only the open end is affected. The static component end and all components connected beyond this is not affected.



Picture: Toggle Build direction, 30°



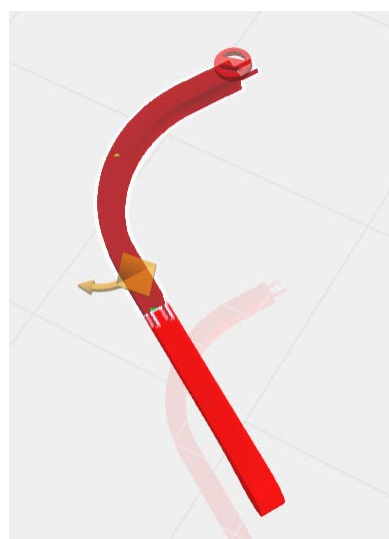
Picture: Toggle Build direction, 90°

Example 2

The other example is showing the same change on the bend but with the Toggle build direction set to the opposite component end. When the bend angle is changed to 90° the open component end of the horizontal bend is not changed due to the build direction. On the other hand all the bend end connected to the rest of the conveyor is affected to the angle change. Further all components connected beyond this end is also affected.



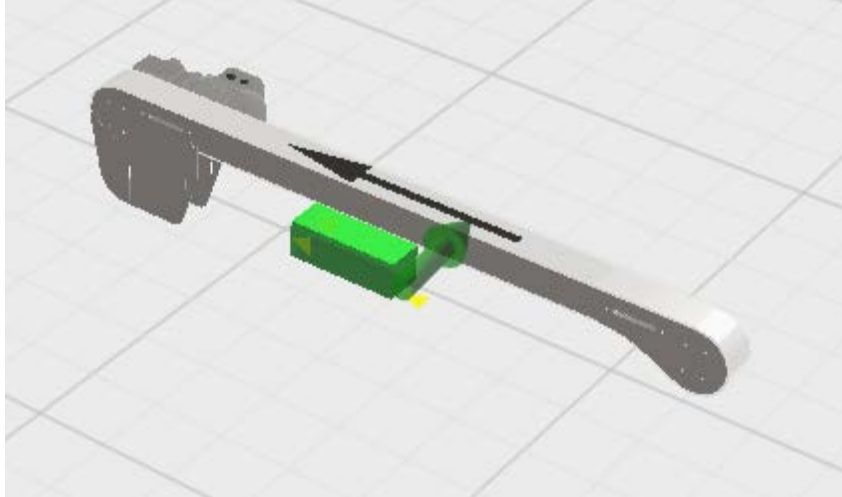
Picture: Toggle Build direction, 30°



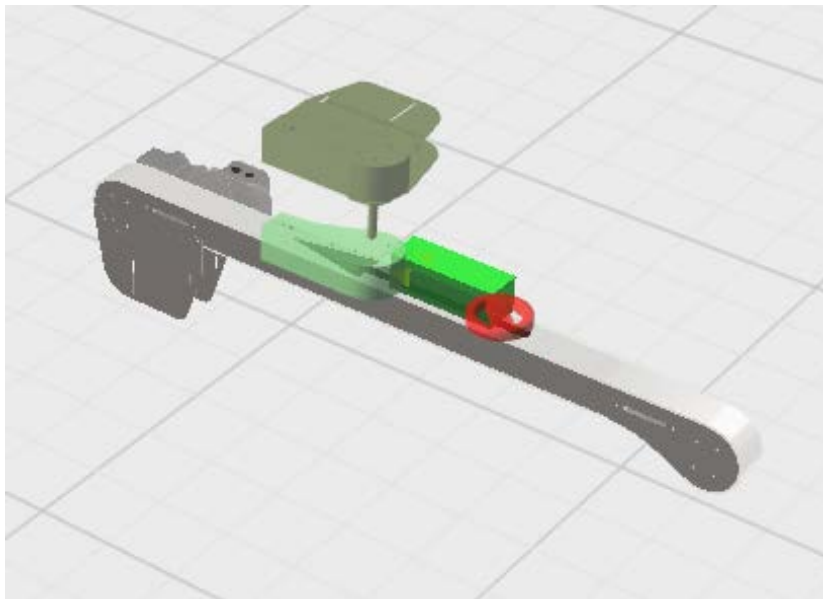
Picture: Toggle Build direction, 90°

X85 Wedge example

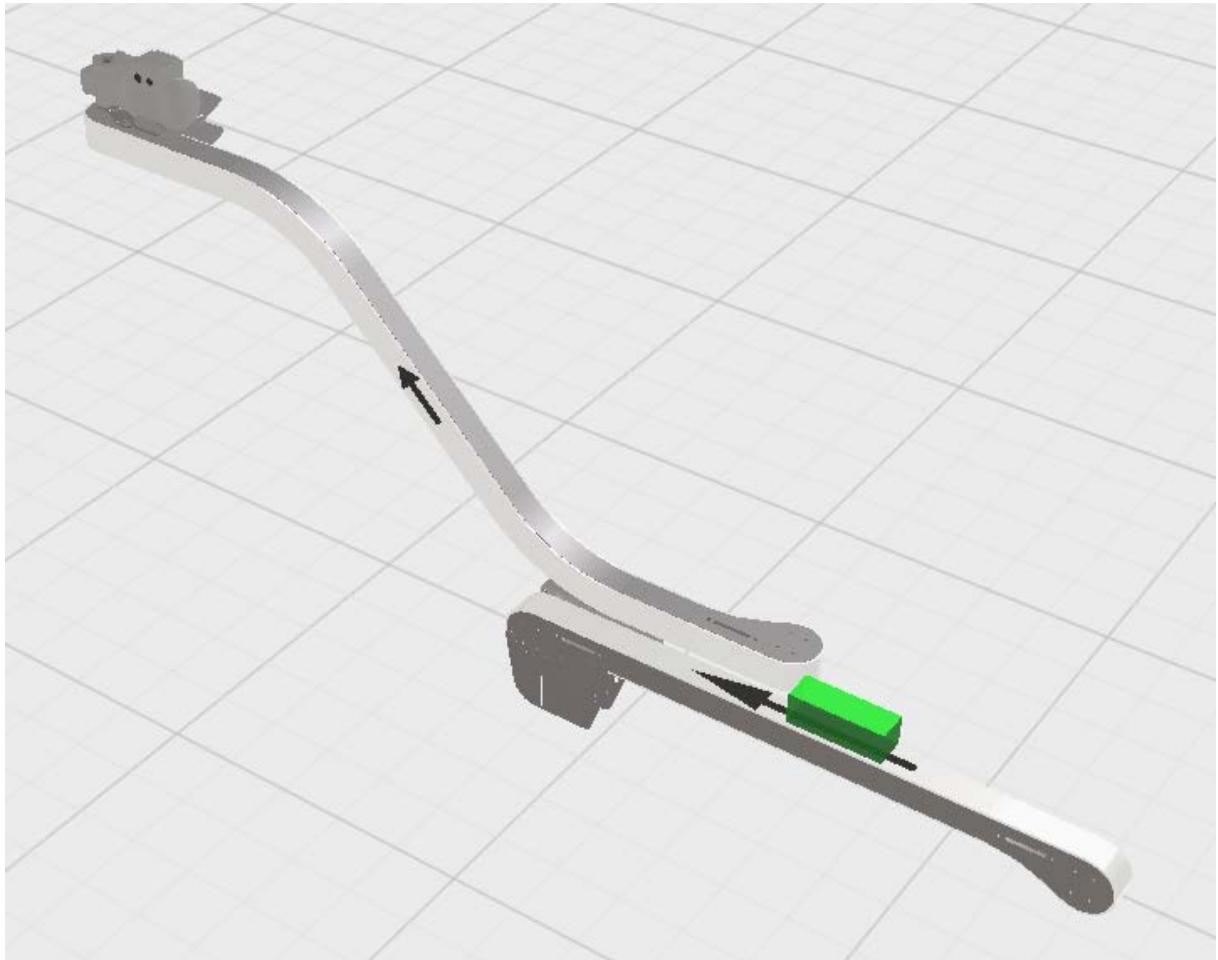
Below is an example of how to build a wedge conveyor. In this example the platform X85 is used but there are several more platforms that is available when building wedges.



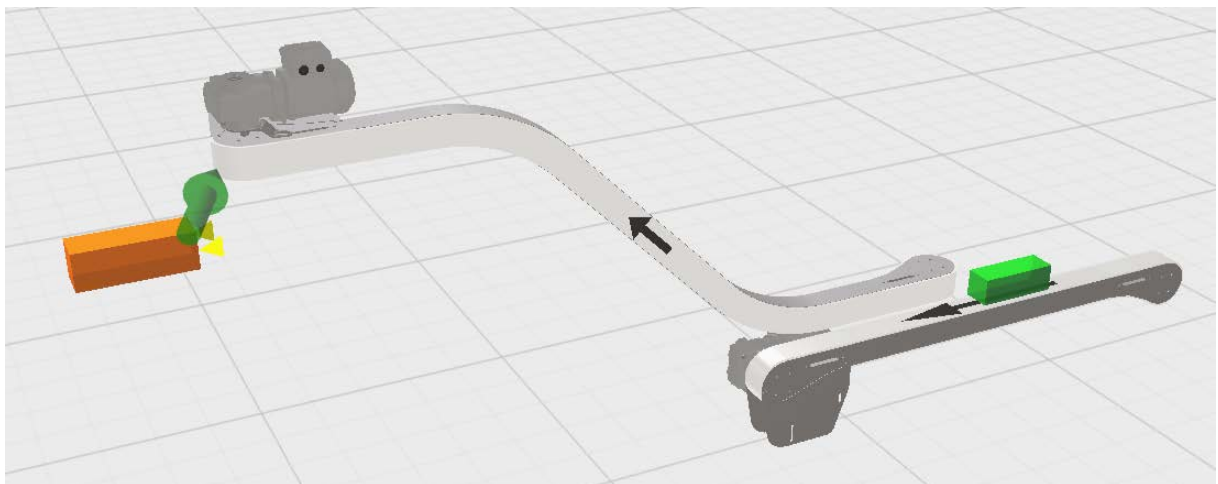
Picture: Wedge example, Add Virtual Wedge component, ToWedge



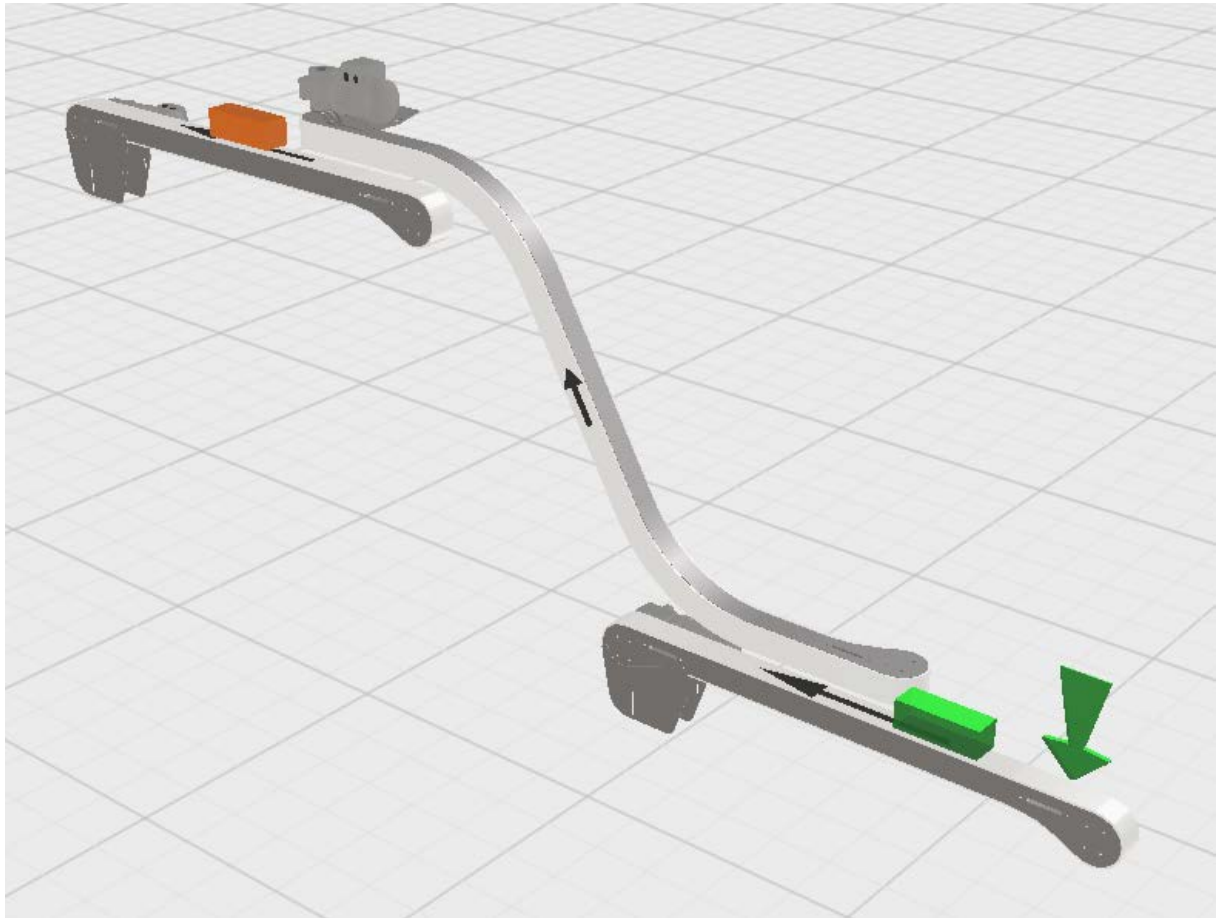
Picture: Wedge example, Building handles



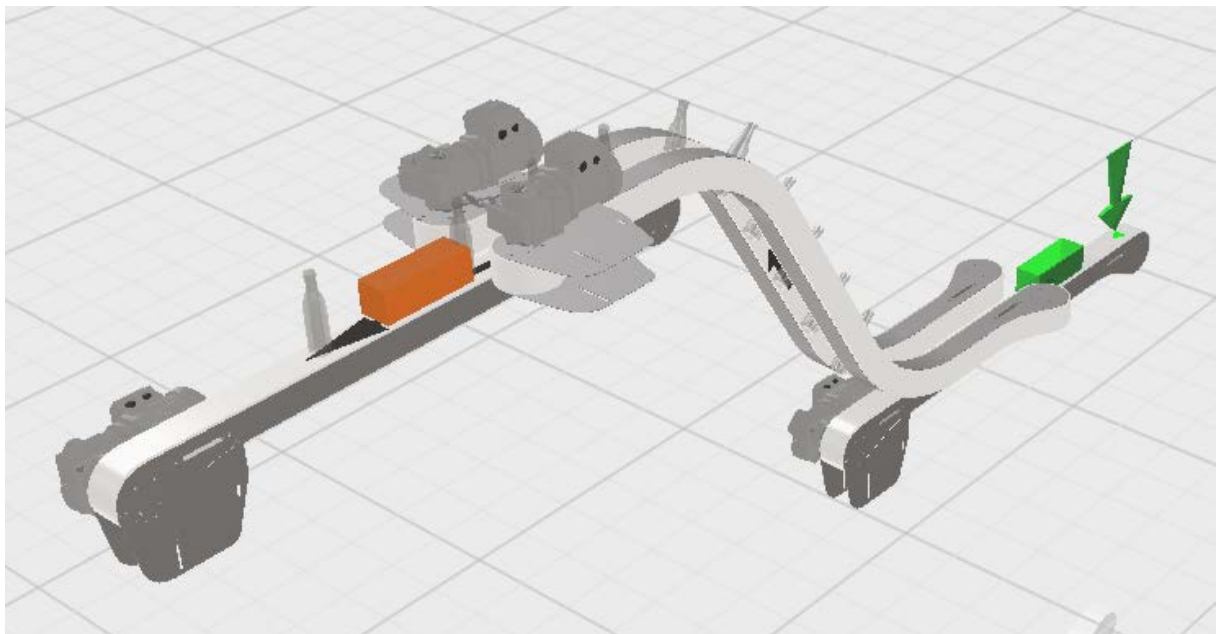
Picture: Wedge example, Complete first Wedge conveyor



Picture: Wedge example, Add Virtual Wedge component, FromWedge



Picture: Wedge example, connect a horizontal conveyor to the second wedge component



Picture: Wedge example, Add the parallel wedge conveyor

Appendix D: InitList colors

The following materials can be used wherever in the Design Tool where it is possible to set Materials. These Materials (colors) can also be used in the ***InitList*** of the FlexLink Initialization point component: The material names are not case sensitive)

- aluminium
- beige
- black
- blue
- brown
- chrome
- clear_glass
- cyan
- green
- grey
- orange
- pink
- purple
- red
- reflective_glass
- steel
- white
- yellow

Appendix E: Platform dependent building concept

X45/X45P/X45H/X45HP

For the X45 platform families there are four different platforms; X45, X45P (X45 for puck), X45H, X45HP (X45H for puck).

X45 is the platform using the low profile beam (41 x 45 mm) and the slide rails for non puck-applications. This platform can be added with guide rail and guide rail bracket kits.

X45P uses the same beam but with a combined slide rail / guide rail adjusted for puck handling.

X45H is the platform with the higher profile beam (69 x 45 mm). As for X45 this platform can be added with guide rail and guide rail bracket kits. To use X45H with pucks the X45H platform should be used which has guided slide rail and guide rail for pucks.

Puck applications

For puck applications there are a number of different functions. These will connect to either X45P or X45HP conveyor components.

The X45P-functions are components added to the conveyor components. A function can have up to three different conveyors that it interacts with. In FLDT it is only one of them which it belongs to, one main conveyor for each function. This is not necessary equal to which conveyor that function is mounted to in reality. When that main conveyor is selected and moved all of its functions are moved along with it. Most of the functions has a quick property, Segment position. This is deciding the position on this main conveyor which it belongs to. The only exception from this is the locating satellite function which is a bit different. It snaps (connects) to its in conveyor but it doesn't belong to it. This means both its interacting conveyors could be moved without moving the locating satellite unit.

When a function snaps to a conveyor beam it automatically detects if this conveyor is an X45P or an X45HP conveyor and adjusts its bracket to the corresponding height. If there is another conveyor beam interacting with this function and that conveyor could be of either X45P or X45HP.

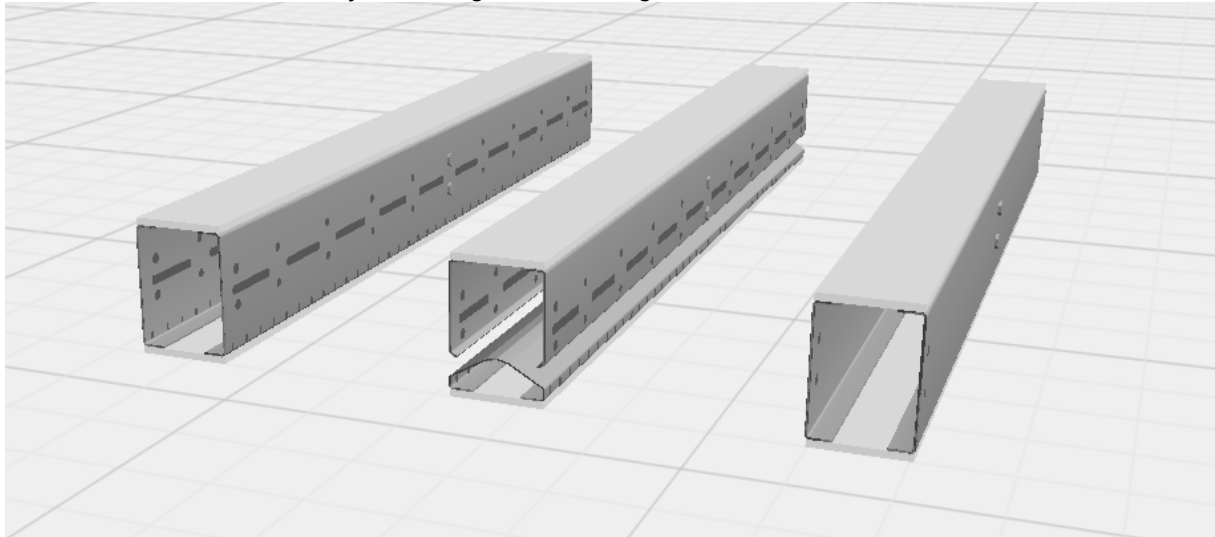
The functions has got some properties that is important to know about. Most of them can be found under the FlexLink property tab. All functions has the Direction property. This decides if it should be a right or a left version. In some cases the CAD geometry is updated at the connection to a conveyor beam. Some function has also got the RFID option. This replaces one photocell with an RFID reader instead. The change is both in the geometry and in the BOM. The last property in the FlexLink tab is the Disc value. This is only a setting of which angle should be showed in the static layout. As soon as the simulation starts it overrides this angle.

In the Process tab there is the disc speed property. This is the rotation speed of the guide disc movement. The rest of the process properties are described in the simulation chapter.

Often in puck layouts there are two parallel conveyors with the distance of 60 mm in between. To achieve this the Positioning command, CC distance is very useful. The CC value 60 is prefilled as a default for all X45 beams.

X85X

The stainless platform X85X has predefined hole configurations. These are the only available positions for attaching guide rail brackets and conveyor supports. The beam component exists in three different version, exactly according to the catalogue.



Picture: X85X Beams: X, Y and C

Pallet systems X85P/X65P/XKP

All of the conveyor components for these platforms has guide rails included when they are added to the layout. The guide rail bracket components has to be added separately. As for all guide rail brackets these could be automatically located when the first bracket is configured. If no guide rail bracket is included in a pallet line like these there will be a warning to the user. As soon as one bracket is added to warning disappears. This way of building differs from all other platforms because all other platforms doesn't show the guide rail until the bracket is located and configured.

If pallets should be included in the BOM the pallet feeder component should be used and BOM included option should be turned on. But if pallets should be positioned only for visual appearance in the print layout the pallet component should be added to the layout at required positions. These components will not be included in the BOM. The component exists in one version for each platform and can be set to the appropriate pallet size.



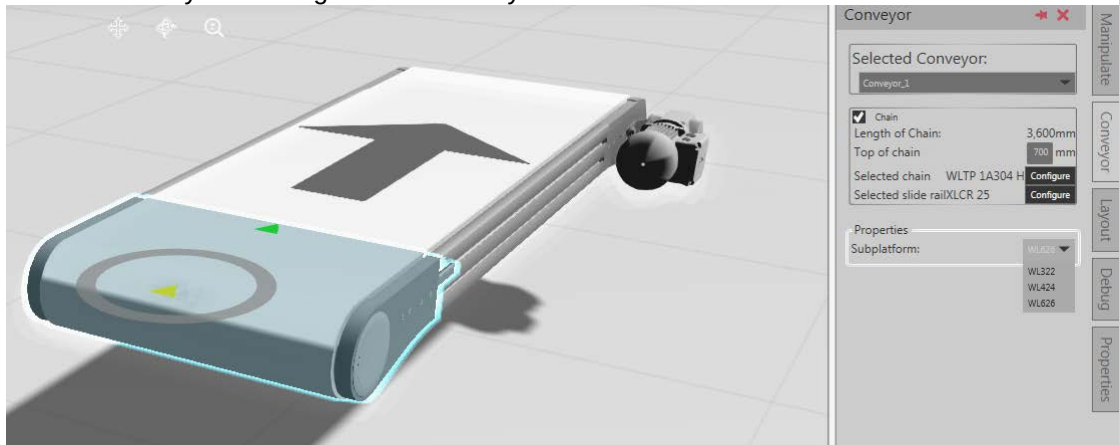
Picture: Pallets manually positioned

In pallet system like these the pallet size could be set on either one of the pallet functions or at the conveyor level. Then all of the other pallet functions and also the manually positioned pallets are

changed to that size. Changes made to the pallets are only affecting that pallet and will be overwritten if one of the pallet functions is changed.

WL

The WL platform has three sub platforms. These could be selected in the **Conveyor** property tab using the dropdown list shown in the picture below. When changing this all of the components in the selected conveyor is changed automatically.



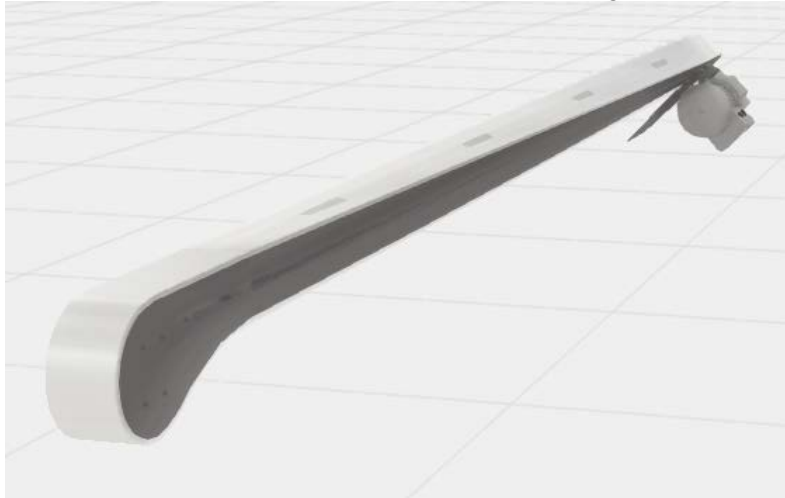
Picture: WL – Conveyor property Sub platform

Appendix F: Special component

This chapter describes some standard component types that need some more explanation about the features they have.

Twisted beam

The twisted beam consists of the half beams which are then bend to desired angle. The length of the twisted beam can't be shorter than 2500 mm or longer than 3000mm. It can be bend

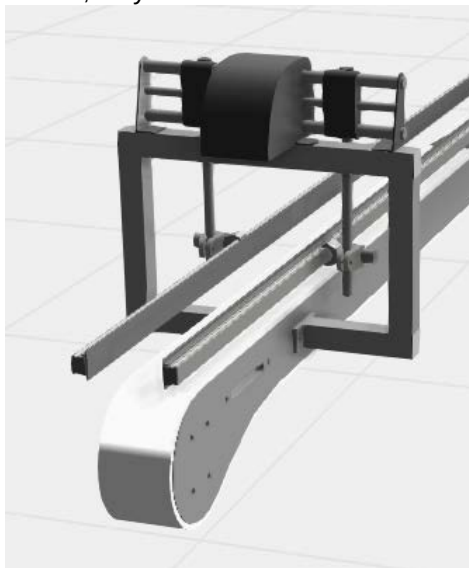


Picture: Twisted beam

AGS

The AGS (automatic guiding system) has one component in the **eCatalog**. This is only the guide rail and brackets kits. For finalizing the system in real applications some more components has to be considered. These are the control box and the junction box. Description of these can be found in the catalogue in the GR chapter.

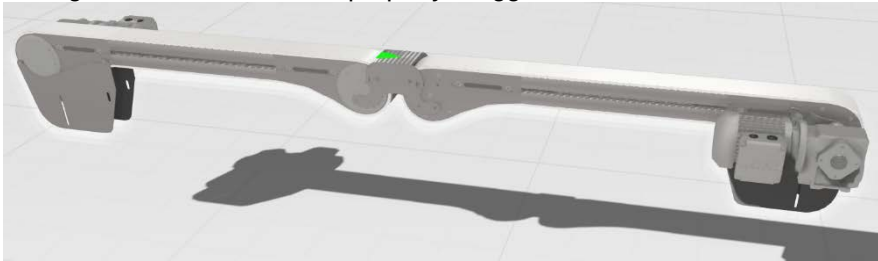
The AGS kit is also a little bit different to many other guide rail kits because it can't be mounted as half kit, only on one side.



Picture: AGS example

Bridge

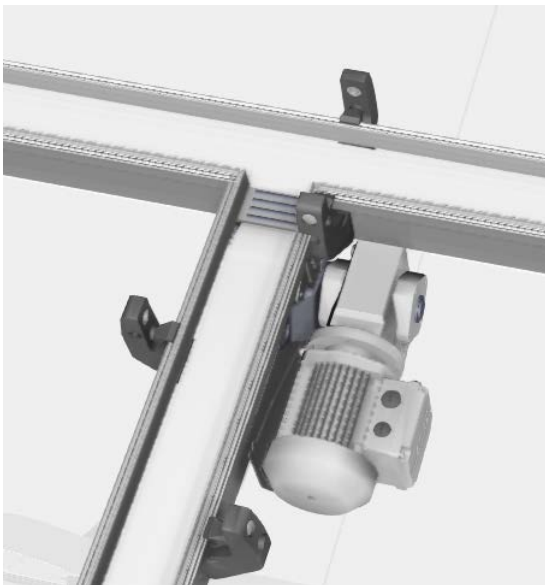
There are some standard bridges in the assortment. These can connect to both Idlers and End drive units. Also there can be variants in size (medium/compact or heavy). When the bridge connects to a component it automatically detects which component it is and adjusts itself. If an Idler end or End drive is not connecting using plug and play the bridge might expect another setting on the Inverted insert property. Toggle this can solve this.



Picture: Bridge (X85)

Bridge to beam

The bridge to beam component is used for merging product flows from a 90 degree conveyor. It can't be used for the reverse direction, i.e. diverting onto the 90 degree conveyor. When the guide rails are added to the straight conveyor it automatically stop the visualization at the bridge. The component is automatically adjusting its geometry and BOM depending on what component is connected on the 90 degree side.



Picture: Bridge to beam example

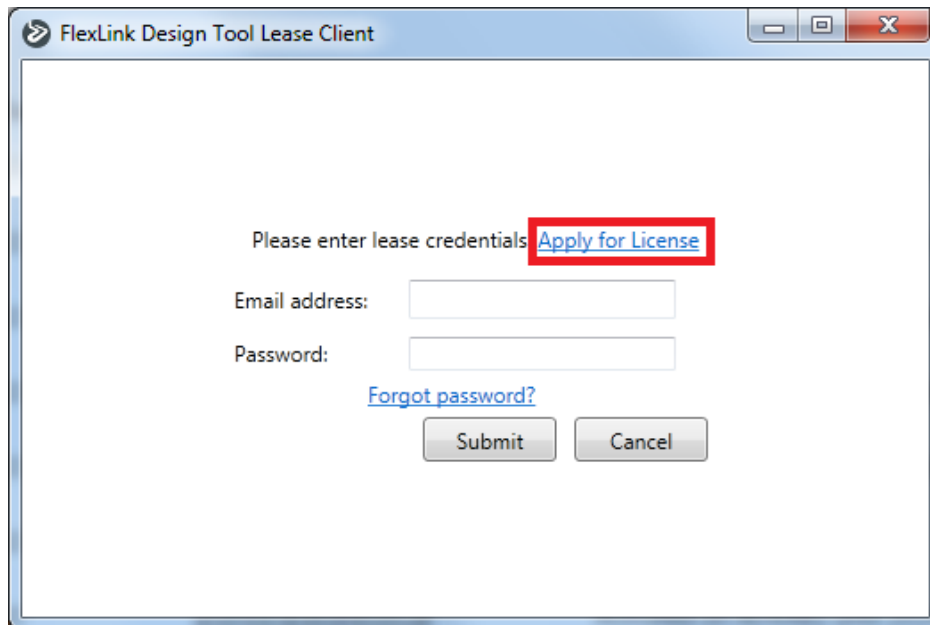
Appendix G: Lease Management System

To be able to use the Design tool every user needs a lease. All leases are handled by a local administrator. A user that want to use the Design tool needs to apply for a lease.

FlexLink as a company has many licenses for the design tool and the local administrator can lease these licenses to users for a specified time. The lease will be connected to the users email so there will be no need to keep track of the license code.

Applying for a lease

1. Open the Design Tool  (Start menu>All Programs>FlexLink Design Tool>FlexLink Design Tool.exe)
2. Click on "Apply for License". This will take you to the application form for the Design Tool.



FlexLink Design Tool Lease Client

Please enter lease credentials [Apply for License](#)

Email address:

Password:

[Forgot password?](#)

3. Fill in the form. An email will be sent to the local administrator for approval of the account.

Apply for FlexLink Design Tool lease

Please enter your contact details in order to apply for a FlexLink Design Tool lease. After you have submitted your application, you will be contacted by a FlexLink representative with more information.

E-Mail *

Given Name

Family Name

Company *

Country *

* = Field is required

Submit

Reset

4. Once the local administrator has accepted the application you will get an email from noreply@flexlink.com. Follow the link to activate your account.

Activate your lease account

1 message

noreply@flexlink.com <noreply@flexlink.com>

To:

Dear

An account has been created for you at the FlexLink Design Tool lease web site:
<http://flexlink.visualcomponents.net/>

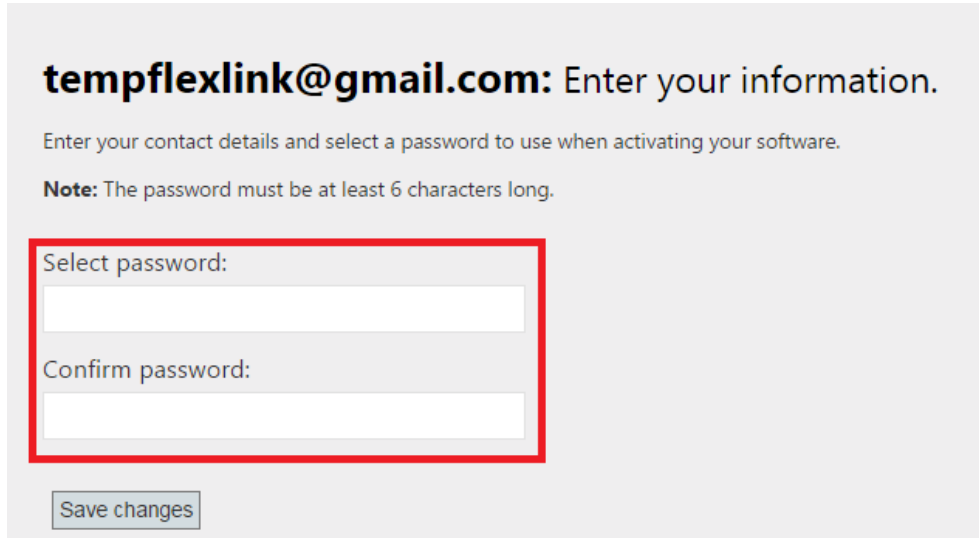
Follow the link below to activate your account:

<http://flexlink.visualcomponents.net:80/Account/CollectInfo.aspx?enrollCode=GJPBTPbC..pAHsXfLPUTmPaYN6iY.1aw.QFikLuRVjG>

Best regards,
FlexLink Engineering Tools team

PS: This is an automatically generated email message, please do not reply directly to this message. If you have any questions or concerns, please contact softwaresupport@flexlink.com

5. Fill in a password for the account and press “Save changes”.



tempflexlink@gmail.com: Enter your information.

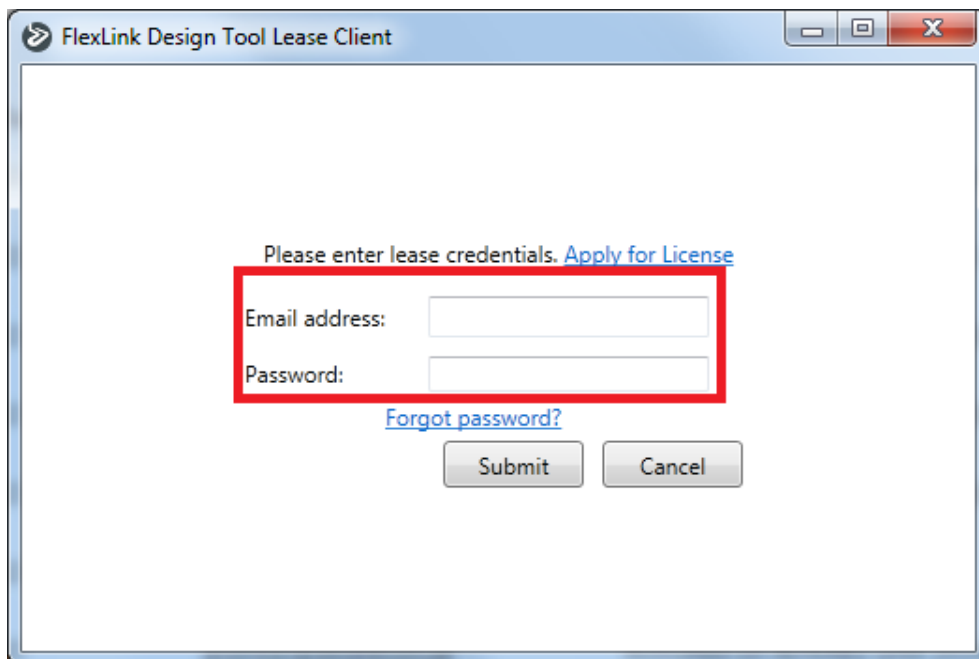
Enter your contact details and select a password to use when activating your software.

Note: The password must be at least 6 characters long.

Select password:

Confirm password:

6. Your account has now been activated. Open the design tool and sign in with your email and password.



FlexLink Design Tool Lease Client

Please enter lease credentials. [Apply for License](#)

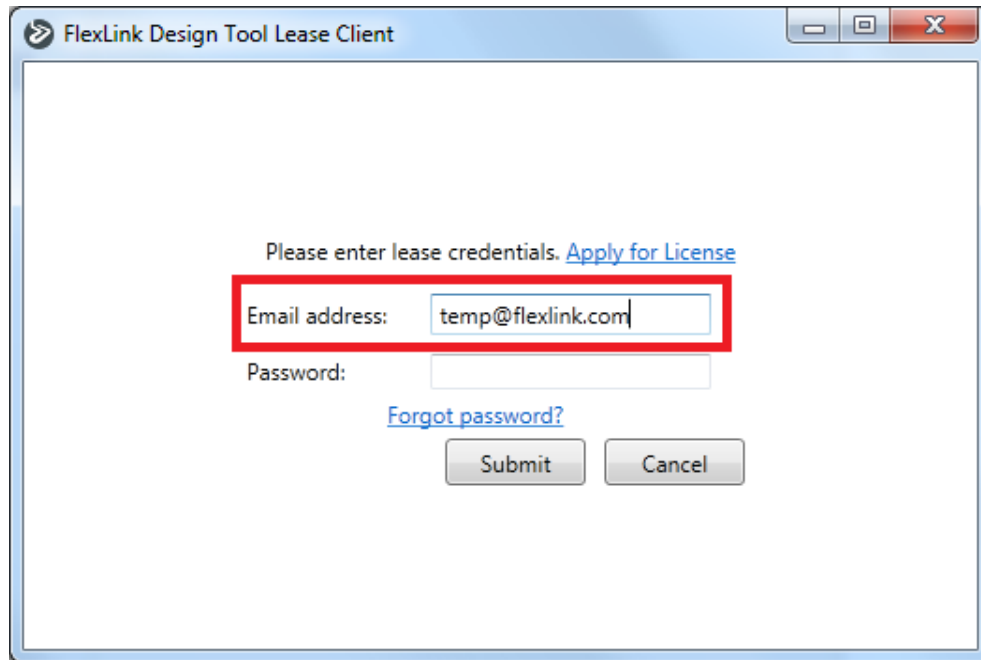
Email address:

Password:

[Forgot password?](#)

Reset password

1. Open the Design Tool  (Start menu>All Programs>FlexLink Design Tool>FlexLink Design Tool.exe)
2. Fill in your email address



FlexLink Design Tool Lease Client

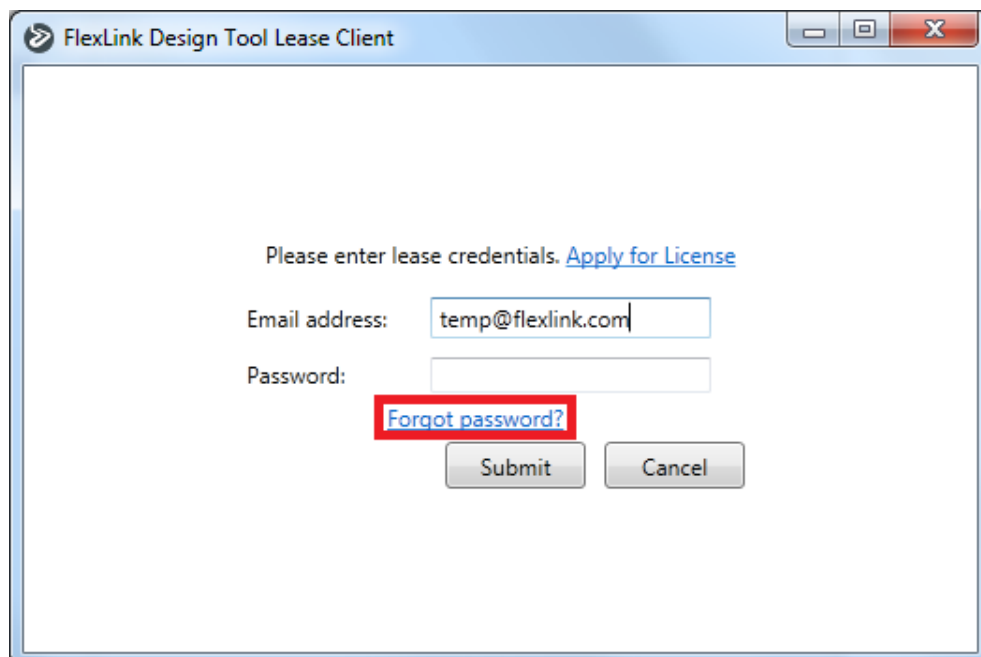
Please enter lease credentials. [Apply for License](#)

Email address:

Password:

[Forgot password?](#)

3. Click on "Forgot password?"



FlexLink Design Tool Lease Client

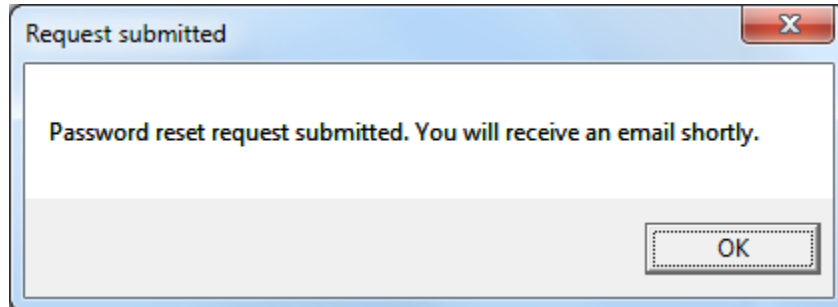
Please enter lease credentials. [Apply for License](#)

Email address:

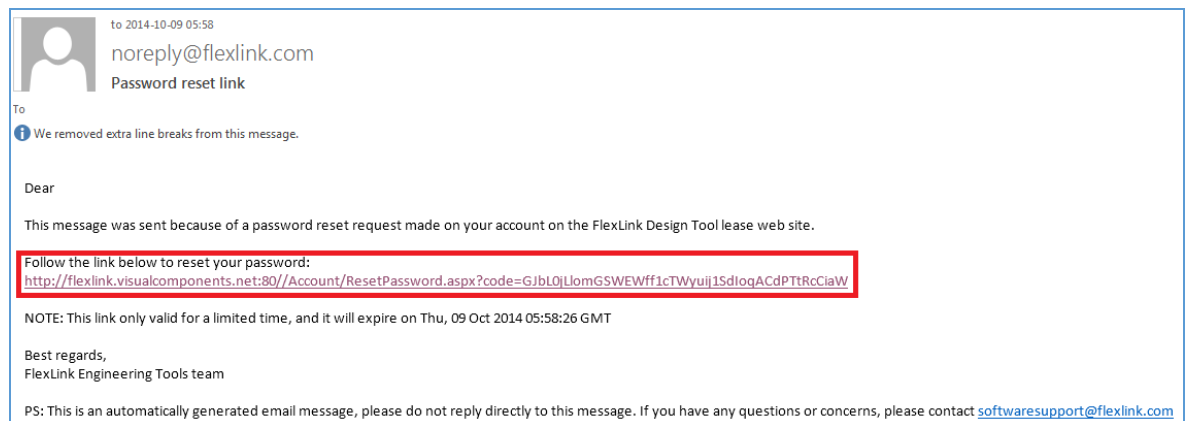
Password:

[Forgot password?](#)

4. Your request for password reset will be sent and an email with instructions of how to reset the password will be sent to your email.



5. Open the email and click on the link to reset your password.



6. Fill in your new password and press "Reset password"



Reset password. Enter a new password.

New password

Confirm new password

Reset password

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Switch computer

When using a lease on a computer it is activated on that specific computer. Therefore the lease needs to be deactivated before it can be used on another computer. Please note that if this is not done it will take 30 days before the lease is available to use on another computer.

1. Open the "FlexLink Design Tool Utility" (Start menu>All Programs>FlexLink Design Tool>FlexLink Design Tool Utility.exe)
2. Click on "Deactivate"

FlexLink Design Tool Utility

Lease

Deactivating a lease will make it possible to use the lease on another computer. Leases are, after activation, locked to a computer and will remain so until the lease has been de-activated. The de-activation of a lease can only be done on the computer that it was activated on.

Switching user will deactivate the lease and reset the user credentials so a new user can activate a lease on the computer.

Clean eCatalogue

Empty the eCatalogue settings and remove the local folders. This is good to do if you would like to download all component blocks again. Please, note that you need to enable Local copy of some of the eCatalogues as it will be reset as Online by default. To enable Local copy of an eCatalogue is described in the FlexLink Design Tool manual.

Reset configuration

Resets configuration and removes configuration files. This can be used if you would like to start with a clean slate.

Collect and send log files

Collecting and sending log files to Software support will help the engineering tools team to investigate and solve installation or other file issues.

A file explorer will be opened containing the packed file, this packed file should be attached to the email window that is opened at the same time. Make sure to fill out a short description about the problem and then send the email to the softwaresupport@flexlink.com. This may take up to 5 minutes.

Version: 2.0.10

